Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```python
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```
    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
```

```python
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)



    return df,dict_features
```

```python
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8028

Nombre de lignes df clean	8028
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5619
Taille dataset test	2408
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_4h_mean, r_NH3_st_4h_std, r_NH3_st_4h_rms, r_NH3_st_4h_median, r_NH3_st_4h_min, r_NH3_st_4h_max, r_NH3_st_4h_diff_last, r_NH3_st_4h_slope, r_temperature_st_4h_mean, r_temperature_st_4h_std, r_temperature_st_4h_rms, r_temperature_st_4h_median, r_temperature_st_4h_min, r_temperature_st_4h_max, r_temperature_st_4h_diff_last, r_temperature_st_4h_slope, r_umidity_st_4h_mean, r_umidity_st_4h_std, r_umidity_st_4h_rms, r_umidity_st_4h_median, r_umidity_st_4h_min, r_umidity_st_4h_max, r_umidity_st_4h_diff_last, r_umidity_st_4h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

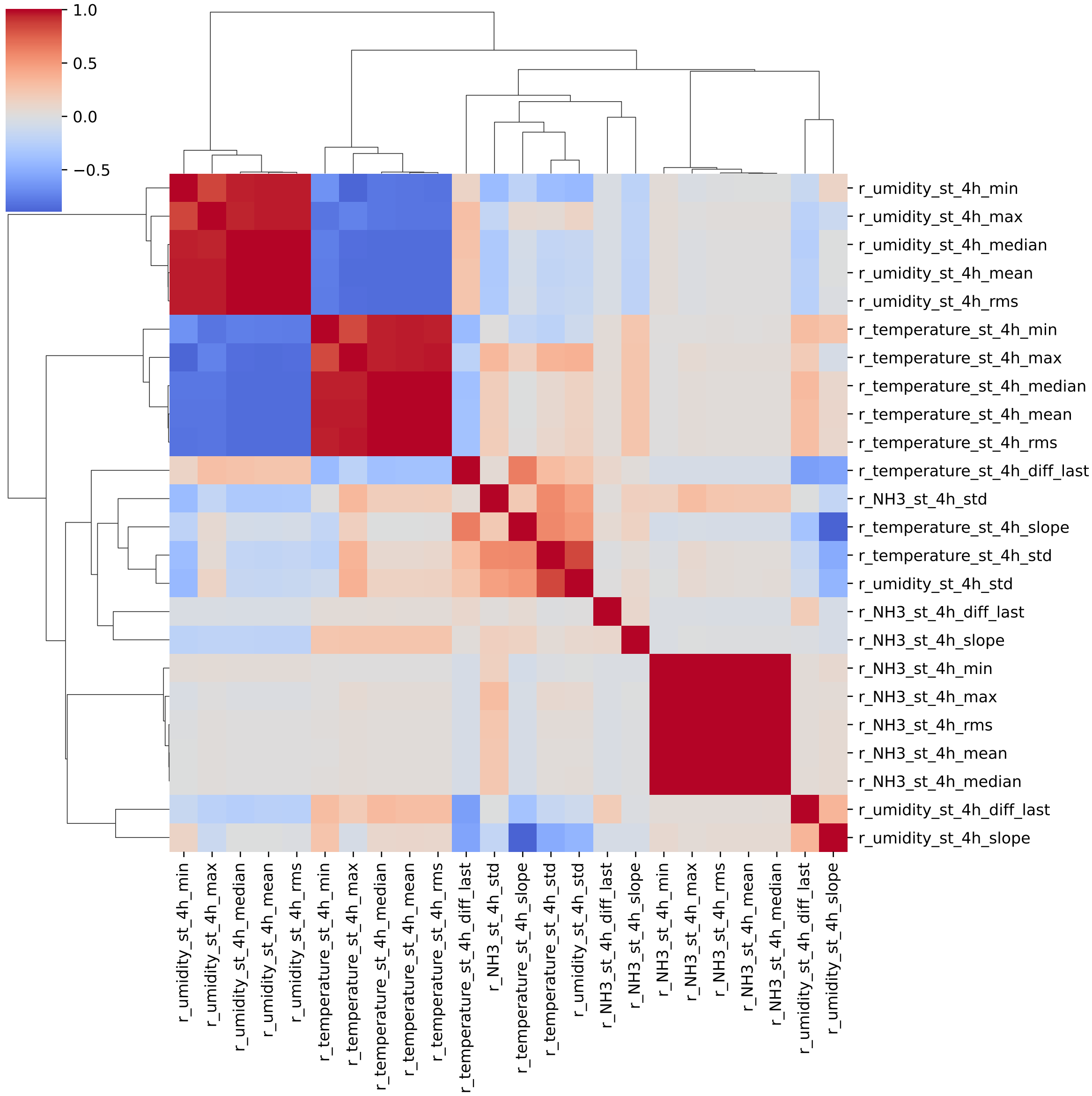
cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_4h_max, r_NH3_st_4h_min, r_NH3_st_4h_rms, r_NH3_st_4h_mean, r_NH3_st_4h_median ; **cluster 7:** r_NH3_st_4h_std ; **cluster 8:** r_NH3_st_4h_diff_last ; **cluster 9:** r_NH3_st_4h_slope ; **cluster 10:** r_temperature_st_4h_min, r_umidity_st_4h_max, r_temperature_st_4h_mean, r_umidity_st_4h_min, r_temperature_st_4h_median, r_umidity_st_4h_mean, r_temperature_st_4h_max, r_temperature_st_4h_rms, r_umidity_st_4h_median, r_umidity_st_4h_rms ; **cluster 11:** r_umidity_st_4h_std, r_temperature_st_4h_std ; **cluster 12:** r_temperature_st_4h_diff_last ; **cluster**

13: r_temperature_st_4h_slope, r_umidity_st_4h_slope ; **cluster 14:** r_umidity_st_4h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_4h_max, r_NH3_st_4h_std, r_NH3_st_4h_diff_last, r_NH3_st_4h_slope, r_temperature_st_4h_min, r_umidity_st_4h_std, r_temperature_st_4h_diff_last, r_temperature_st_4h_slope, r_umidity_st_4h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_4h_max, r_NH3_st_4h_std, r_NH3_st_4h_diff_last, r_NH3_st_4h_slope, r_temperature_st_4h_min, r_umidity_st_4h_std, r_temperature_st_4h_diff_last, r_temperature_st_4h_slope, r_umidity_st_4h_diff_last

Len(Features_X): 14

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

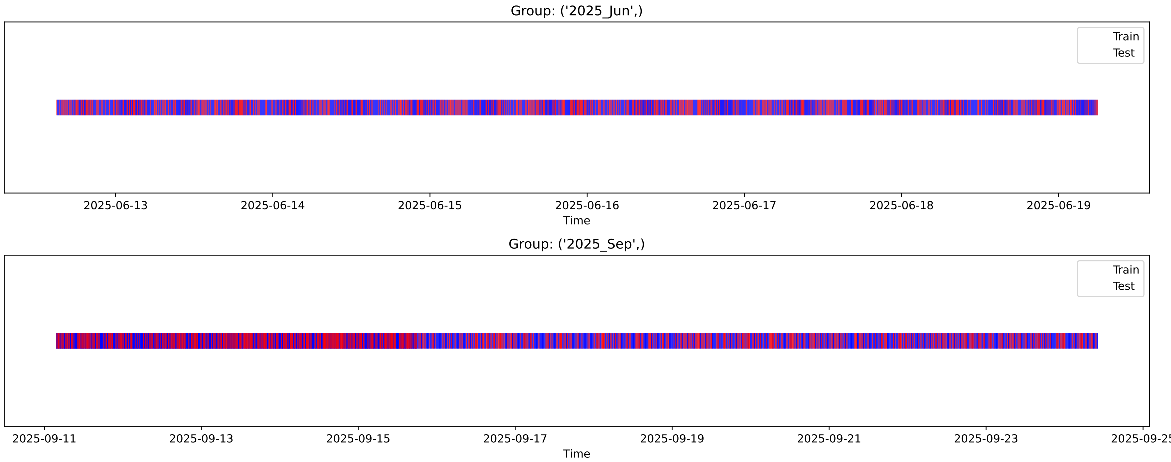
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

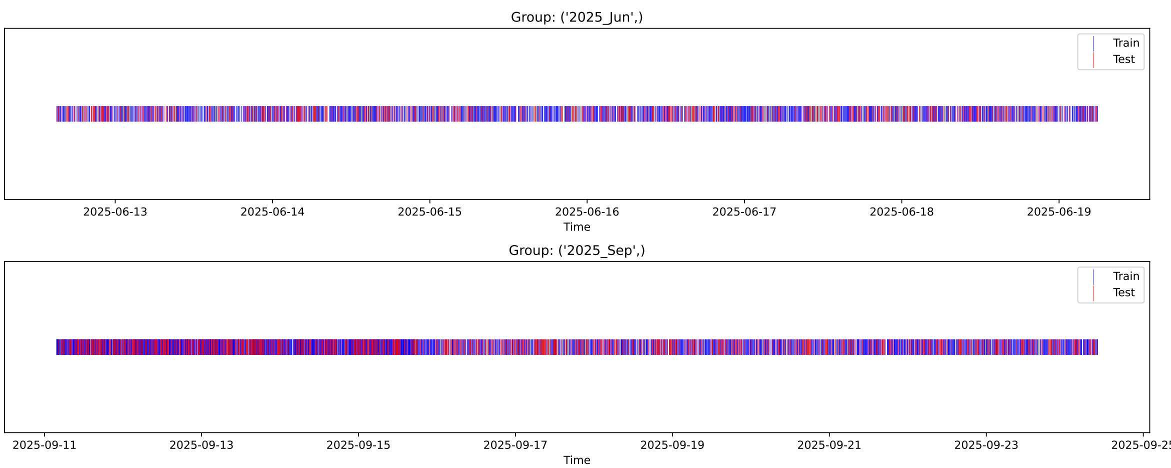
External Split: Train: 5619 samples Test: 2409 samples



3.3.2 Internal CV Folds

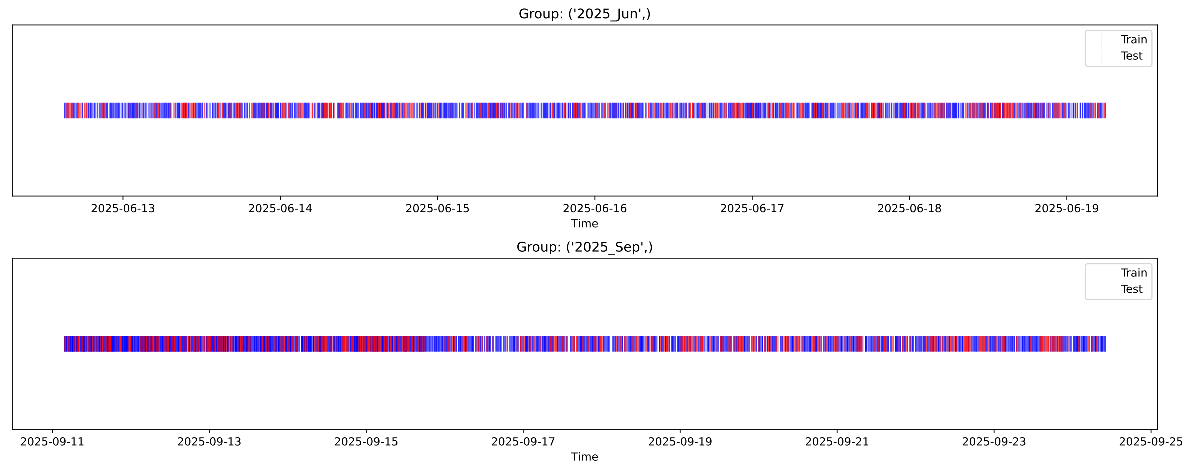
Fold 1/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



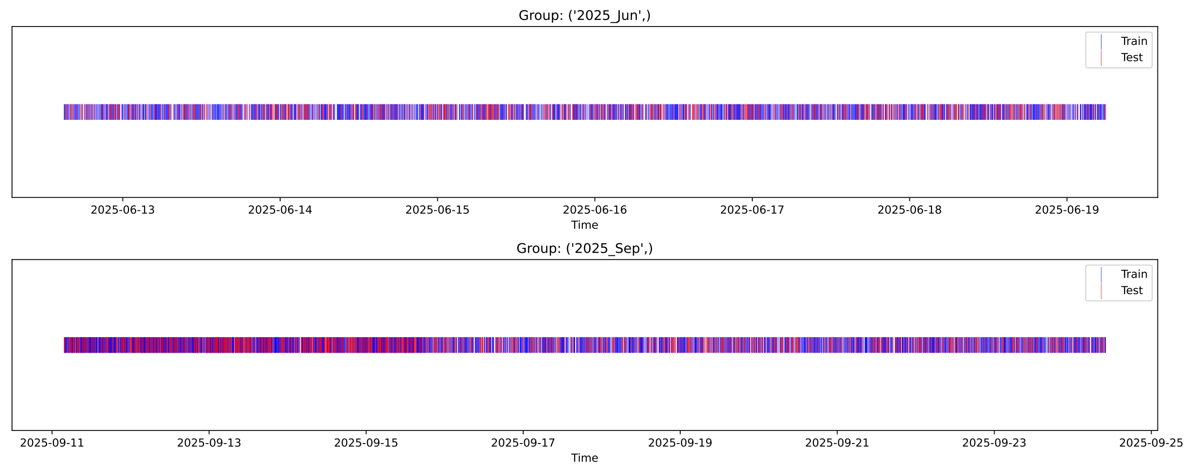
Fold 2/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



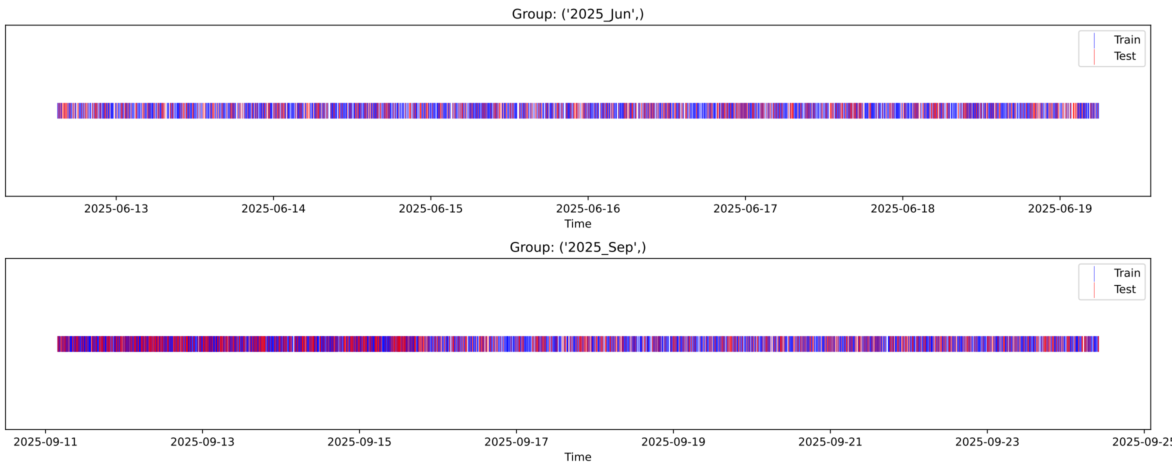
Fold 3/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



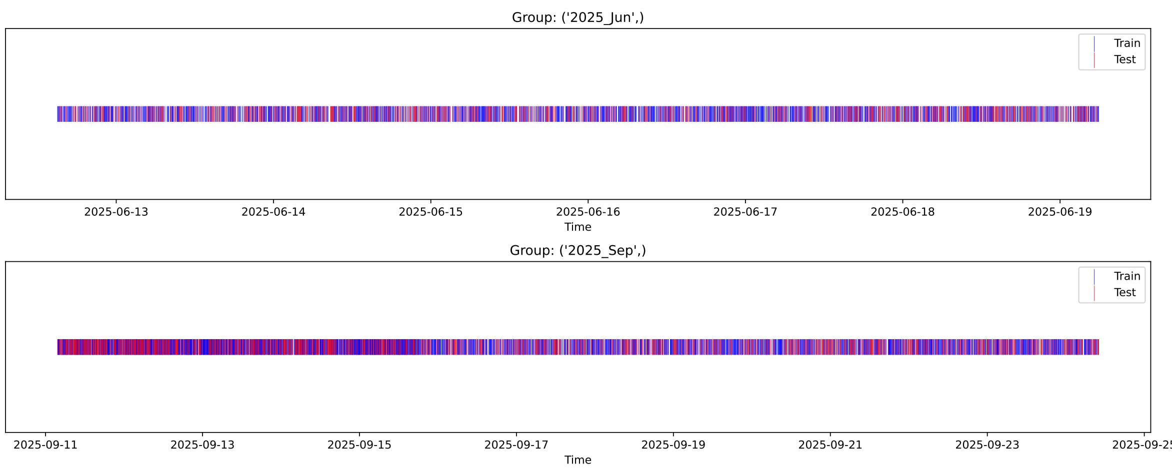
Fold 4/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.350, std=0.806, min=0.959, max=13.845 y_test: mean=2.355, std=0.778, min=1.011, max=7.822

Fold 0 — y_test: mean=2.34, std=0.79, y_train: mean=2.36, std=0.81

Fold 1 — y_test: mean=2.39, std=0.84, y_train: mean=2.33, std=0.79

Fold 2 — y_test: mean=2.35, std=0.79, y_train: mean=2.35, std=0.81

Fold 3 — y_test: mean=2.35, std=0.86, y_train: mean=2.35, std=0.78

Fold 4 — y_test: mean=2.35, std=0.81, y_train: mean=2.35, std=0.81

train scores CV: [0.53128224 0.53583201 0.5228531 0.55084446 0.5204756]

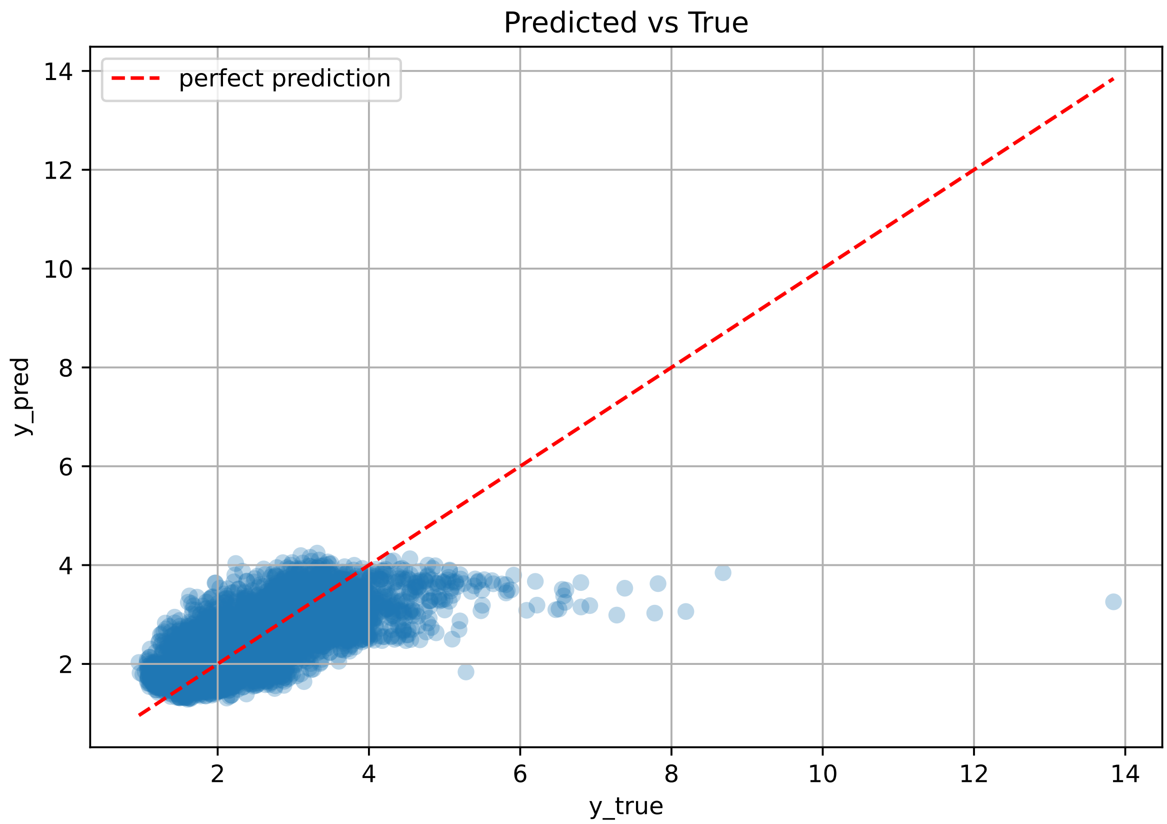
test scores CV: [0.5105229 0.50171851 0.53076711 0.47517321 0.53591097]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5323	0.5108	0.0218	0.5458	0.5459	0.0351	1.0688	1.042

Coefficients:

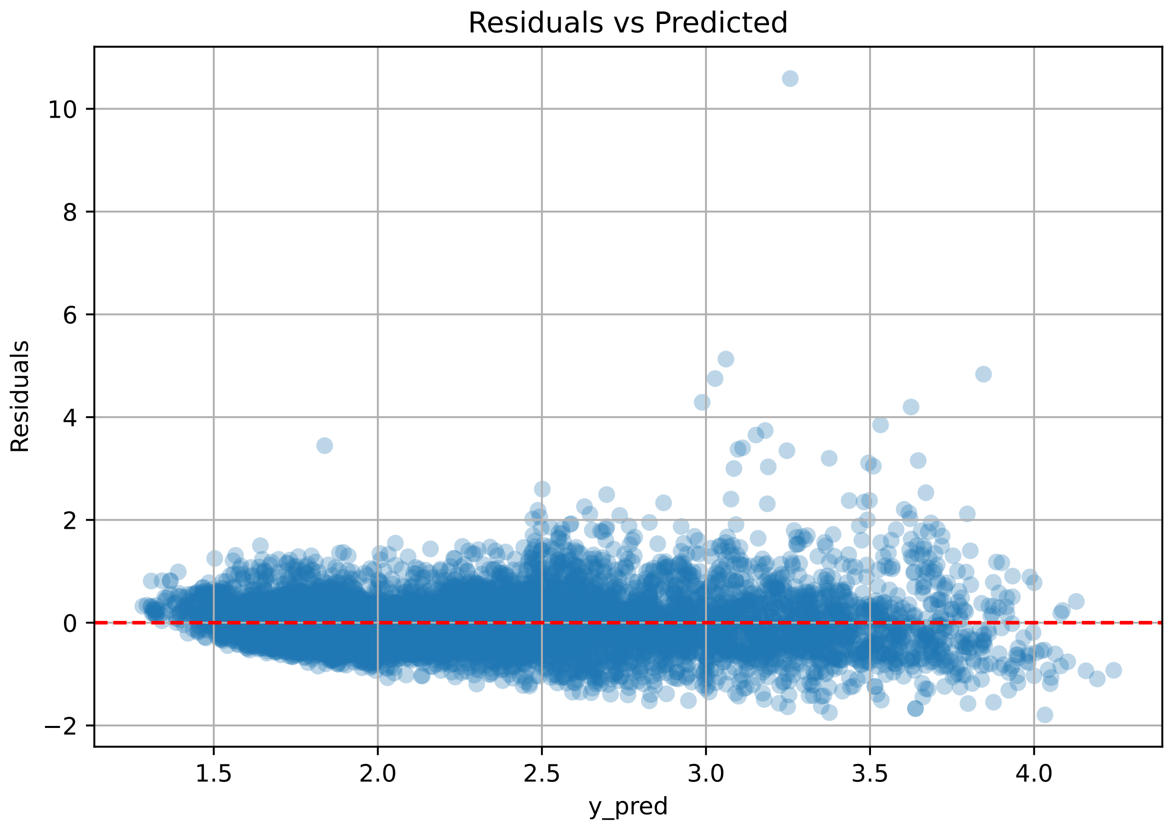
coef_r__CO2 = -0.0471, coef_r__NH3 = -0.3456, coef_r__NH3__st__4h__diff__last = 0.0342, coef_r__NH3__st__4h__max = 0.2959, coef_r__NH3__st__4h__slope = 0.0769, coef_r__NH3__st__4h__std = 0.0185, coef_r__THI = 0.8902, coef_r__temperature = -0.4262, coef_r__temperature__st__4h__diff__last = 0.0533, coef_r__temperature__st__4h__min = -0.6010, coef_r__temperature__st__4h__slope = -0.1594, coef_r__umidity = -0.5848, coef_r__umidity__st__4h__diff__last = -0.0818, coef_r__umidity__st__4h__std = -0.1037, intercept = 2.3503,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

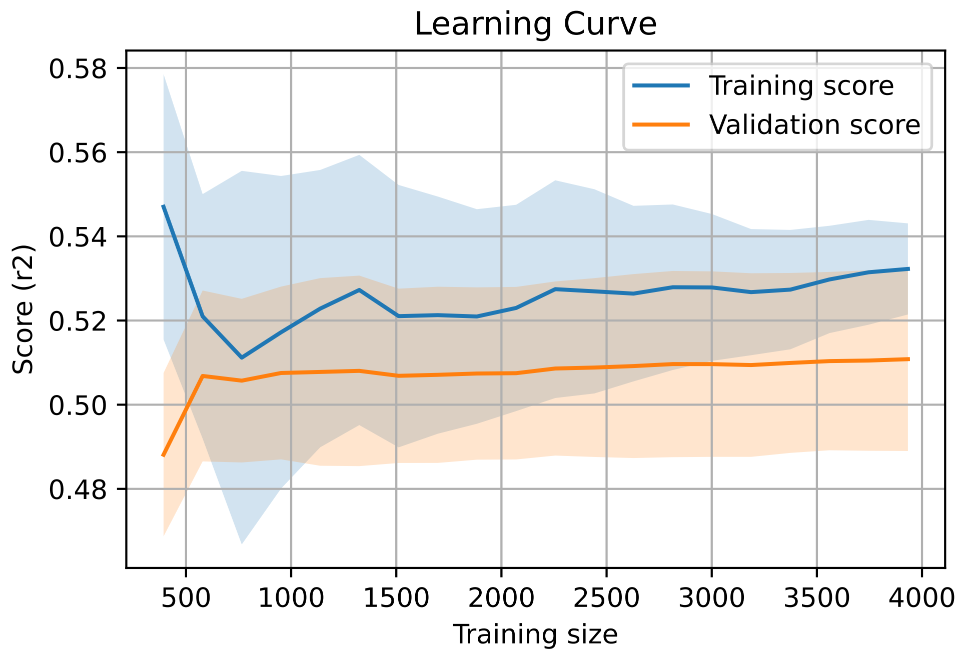
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 8028, **Len(training)** : 5620,

Len(training_real) : 3934, **Len(test_of_training)** : 1686, **Len(test)** : 2408

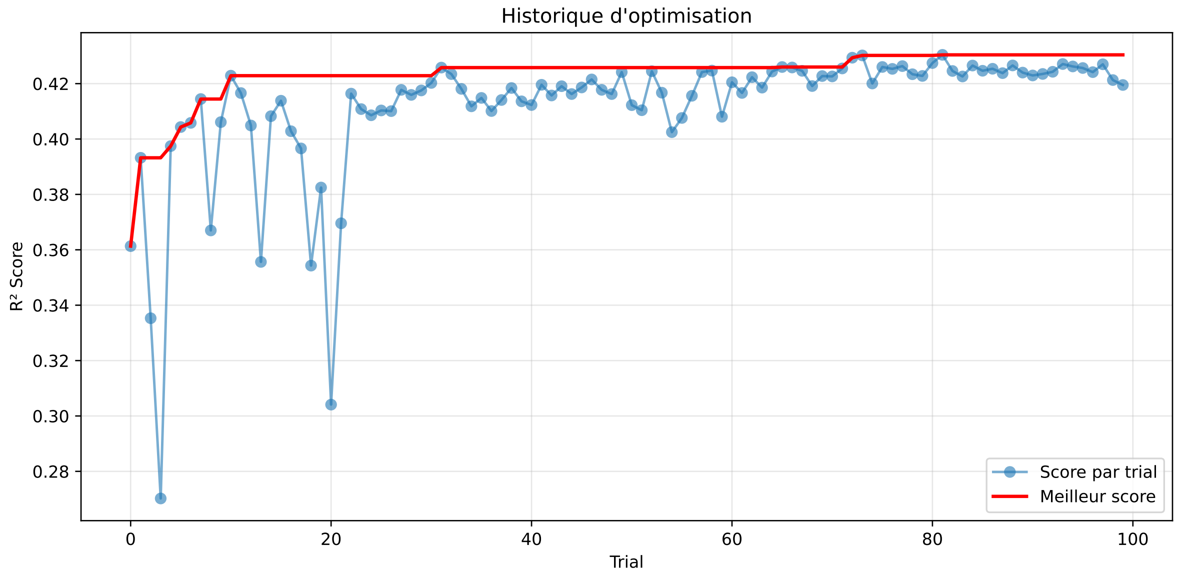


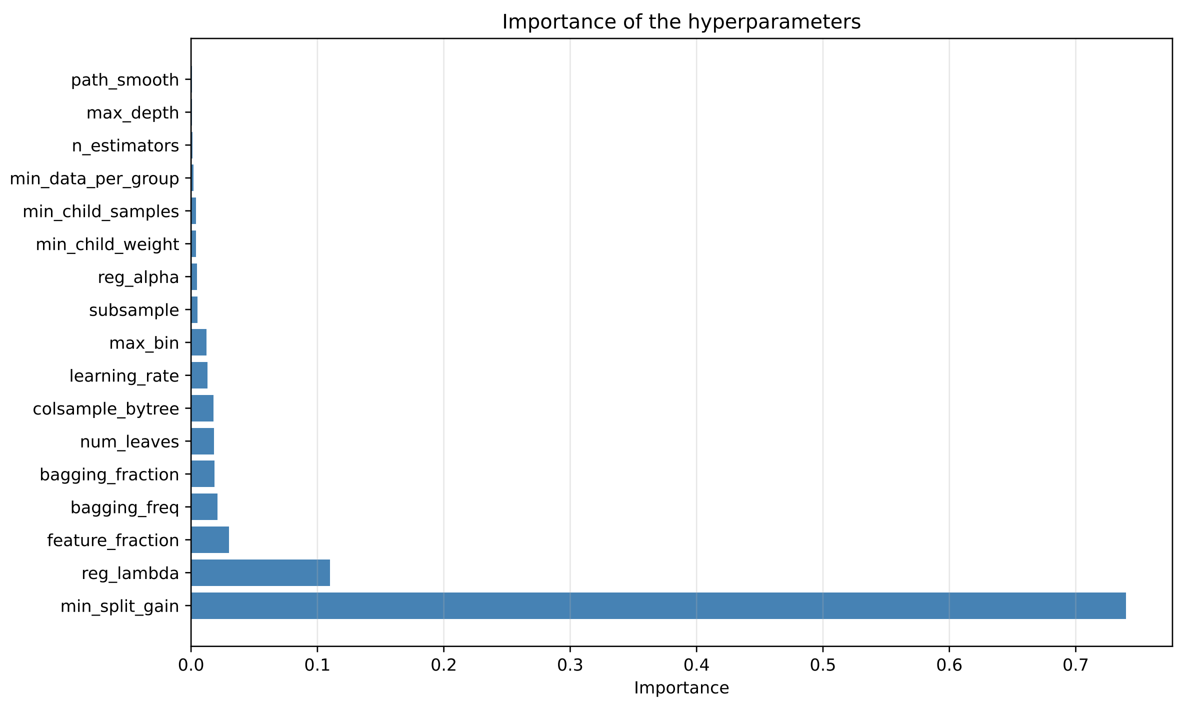
3.5 Model : LGBM

3.5.1 Gridsearch

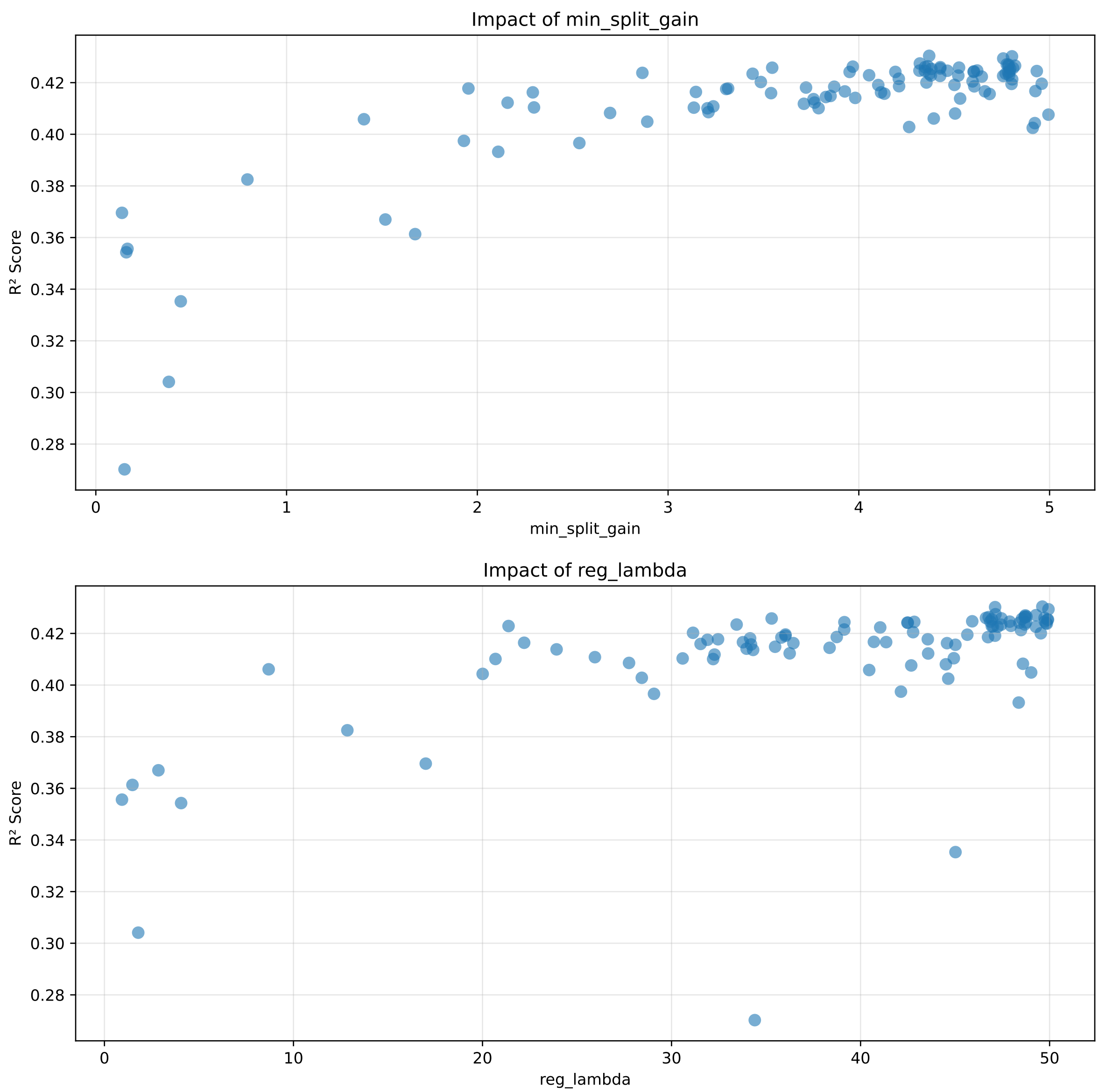
Distribution y_train vs y_test: y_train: mean=2.350, std=0.806, min=0.959, max=13.845 y_test: mean=2.355, std=0.778, min=1.011, max=7.822

===================== TOP Importance FEATURES ===================== feature importance r_CO2 93 r_temperature_st_4h_slope 88 r_umidity 81 r_NH3_st_4h_max 62 r_temperature_st_4h_diff_last 58 r_NH3_st_4h_std 47 r_NH3 42 r_umidity_st_4h_diff_last 38 r_temperature_st_4h_min 37 r_temperature 30 r_NH3_st_4h_diff_last 27 r_umidity_st_4h_std 15 r_THI 14 r_NH3_st_4h_slope 13



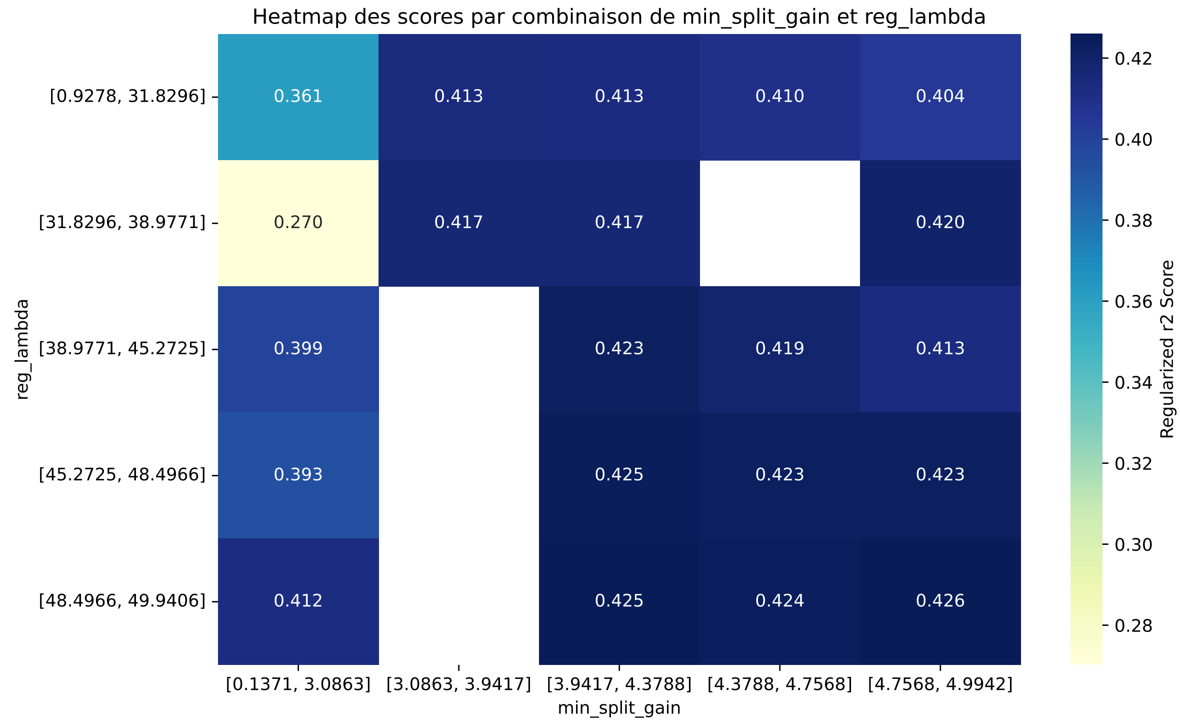


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_lambda']




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6154	0.5846	0.6141	0.0259	0.4304	0.6288	0.0443	1.0757	1.0528
2	0.6078	0.5782	0.6066	0.028	0.4302	0.6193	0.0411	1.0712	1.0512
3	0.6063	0.5768	0.6048	0.0312	0.4294	0.6258	0.049	1.0849	1.0511
4	0.6144	0.5832	0.613	0.0261	0.4274	0.6344	0.0511	1.0877	1.0534
5	0.6122	0.5813	0.6098	0.0255	0.4271	0.6274	0.0461	1.0792	1.0531
6	0.6104	0.5798	0.6081	0.028	0.427	0.6288	0.0489	1.0844	1.0527
7	0.6096	0.5791	0.6079	0.0269	0.4266	0.6219	0.0428	1.0739	1.0527
8	0.6135	0.5823	0.6112	0.0273	0.4265	0.6347	0.0524	1.0899	1.0535
9	0.617	0.5852	0.6147	0.0257	0.4263	0.6386	0.0534	1.0913	1.0543
10	0.6212	0.5887	0.6163	0.0262	0.4262	0.6419	0.0533	1.0905	1.0552
---	---	---	---	---	---	---	---	---	---
100	0.7917	0.7048	0.7404	0.0331	0.2702	0.7562	0.0515	1.073	1.1233

Hyperparameters:

Rank 1: {'reg___n_estimators': 800, 'reg___max_depth': 7, 'reg___learning_rate': 0.03371989354715019, 'reg___num_leaves': 46, 'reg___subsample': 0.5151441319494701, 'reg___colsample_bytree': 0.8147140757919262, 'reg___min_child_samples': 48, 'reg___reg_alpha': 1.707852515415438, 'reg___reg_lambda': 49.61660813570568, 'reg___min_split_gain': 4.36893210444576, 'reg___path_smooth': 1.6952576051010761, 'reg___min_child_weight':

1.4627740619611271, 'reg__feature_fraction': 0.5578248531631538, 'reg__bagging_fraction': 0.605529160144536, 'reg__bagging_freq': 6, 'reg__max_bin': 84, 'reg__min_data_per_group': 189}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.03564264154446846, 'reg__num_leaves': 47, 'reg__subsample': 0.5060206767274266, 'reg__colsample_bytree': 0.7287293324692764, 'reg__min_child_samples': 47, 'reg__reg_alpha': 1.5351013213580202, 'reg__reg_lambda': 47.11698160477095, 'reg__min_split_gain': 4.803033299139342, 'reg__path_smooth': 1.8345389952571716, 'reg__min_child_weight': 0.3174299246722298, 'reg__feature_fraction': 0.5157112524489977, 'reg__bagging_fraction': 0.5843844972542885, 'reg__bagging_freq': 4, 'reg__max_bin': 84, 'reg__min_data_per_group': 127}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.035374901059682023, 'reg__num_leaves': 38, 'reg__subsample': 0.5114275362023952, 'reg__colsample_bytree': 0.6103479401224042, 'reg__min_child_samples': 48, 'reg__reg_alpha': 1.731016800950342, 'reg__reg_lambda': 49.94061791188285, 'reg__min_split_gain': 4.757144926627451, 'reg__path_smooth': 2.0797324685336616, 'reg__min_child_weight': 0.2882024828190974, 'reg__feature_fraction': 0.5173934317560623, 'reg__bagging_fraction': 0.5818464227652316, 'reg__bagging_freq': 4, 'reg__max_bin': 84, 'reg__min_data_per_group': 173}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.03206028198588131, 'reg__num_leaves': 46, 'reg__subsample': 0.5151690307773368, 'reg__colsample_bytree': 0.7274585242701381, 'reg__min_child_samples': 47, 'reg__reg_alpha': 1.687373605381884, 'reg__reg_lambda': 47.136733287394755, 'reg__min_split_gain': 4.31993092653033, 'reg__path_smooth': 1.8375344134191265, 'reg__min_child_weight': 1.5668227866046909, 'reg__feature_fraction': 0.5650298598550056, 'reg__bagging_fraction': 0.5994157683456586, 'reg__bagging_freq': 6, 'reg__max_bin': 85, 'reg__min_data_per_group': 188}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.05133800920970828, 'reg__num_leaves': 50, 'reg__subsample': 0.5180888387132013, 'reg__colsample_bytree': 0.6695267222804522, 'reg__min_child_samples': 45, 'reg__reg_alpha': 0.9510914728713341, 'reg__reg_lambda': 49.27968389256621, 'reg__min_split_gain': 4.77561825497883, 'reg__path_smooth': 2.2946573724397825, 'reg__min_child_weight': 3.2918621384219278, 'reg__feature_fraction': 0.5791053832729324, 'reg__bagging_fraction': 0.6113050365750529, 'reg__bagging_freq': 4, 'reg__max_bin': 67, 'reg__min_data_per_group': 124}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.04014447588449473, 'reg__num_leaves': 50, 'reg__subsample': 0.5159232743607138, 'reg__colsample_bytree': 0.6524085000991953, 'reg__min_child_samples': 45, 'reg__reg_alpha': 1.1071403584179982, 'reg__reg_lambda': 48.716439450092096, 'reg__min_split_gain': 4.78437222423153, 'reg__path_smooth': 2.3311468470456163, 'reg__min_child_weight':

2.583839640602076, 'reg___feature_fraction': 0.5765414933308701, 'reg___bagging_fraction': 0.6102452183126692, 'reg___bagging_freq': 6, 'reg___max_bin': 66, 'reg___min_data_per_group': 200}

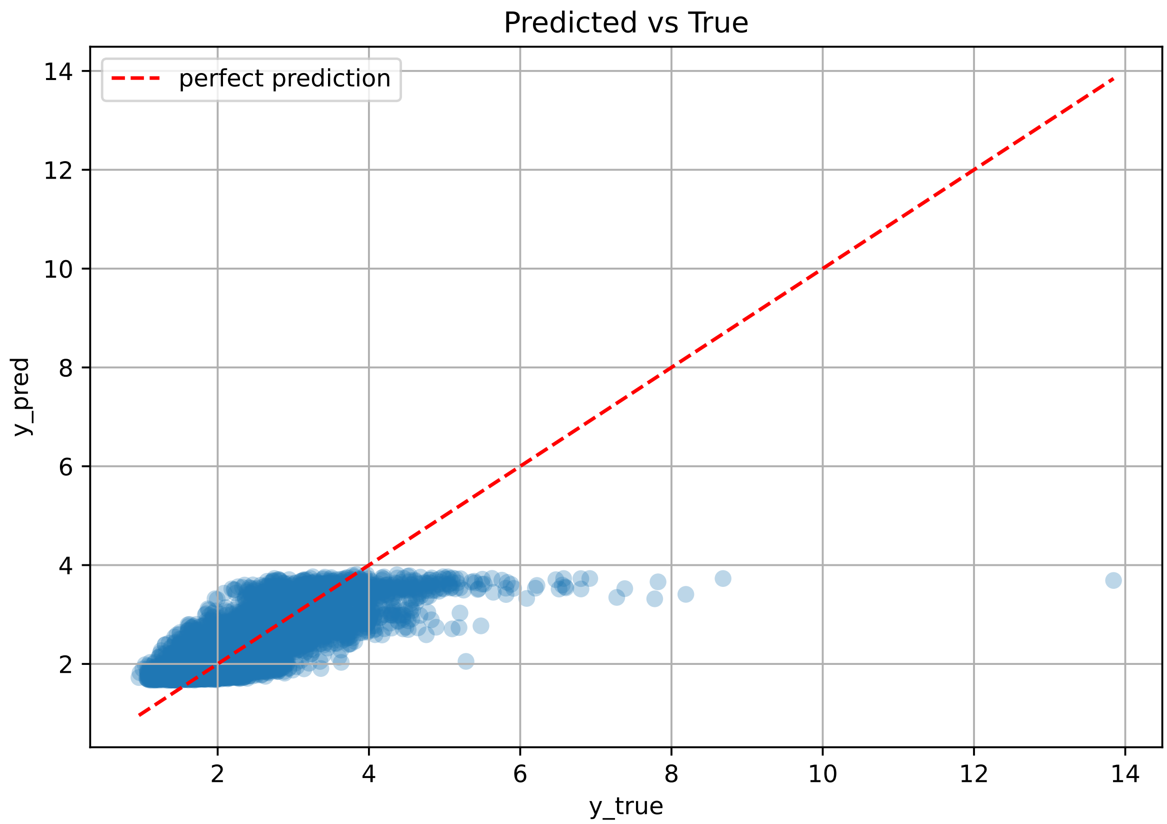
Rank 7: {'reg___n_estimators': 800, 'reg___max_depth': 7, 'reg___learning_rate': 0.05691570050418746, 'reg___num_leaves': 50, 'reg___subsample': 0.5208983098762988, 'reg___colsample_bytree': 0.8060876405319325, 'reg___min_child_samples': 47, 'reg___reg_alpha': 2.0391066911845224, 'reg___reg_lambda': 48.77652162902952, 'reg___min_split_gain': 4.819140042861619, 'reg___path_smooth': 1.5581944118753046, 'reg___min_child_weight': 3.0602308537119125, 'reg___feature_fraction': 0.554841947345079, 'reg___bagging_fraction': 0.5940606562383265, 'reg___bagging_freq': 4, 'reg___max_bin': 84, 'reg___min_data_per_group': 192}

Rank 8: {'reg___n_estimators': 800, 'reg___max_depth': 7, 'reg___learning_rate': 0.03243762653287496, 'reg___num_leaves': 48, 'reg___subsample': 0.5189156520743391, 'reg___colsample_bytree': 0.663993664505303, 'reg___min_child_samples': 48, 'reg___reg_alpha': 0.7475368542207192, 'reg___reg_lambda': 48.67892425310367, 'reg___min_split_gain': 4.784695305933188, 'reg___path_smooth': 1.5611448851873795, 'reg___min_child_weight': 2.848768924624128, 'reg___feature_fraction': 0.5644636849353116, 'reg___bagging_fraction': 0.6107786620355289, 'reg___bagging_freq': 4, 'reg___max_bin': 84, 'reg___min_data_per_group': 188}

Rank 9: {'reg___n_estimators': 700, 'reg___max_depth': 7, 'reg___learning_rate': 0.03599087515103025, 'reg___num_leaves': 47, 'reg___subsample': 0.5113012716248403, 'reg___colsample_bytree': 0.8097065306323653, 'reg___min_child_samples': 47, 'reg___reg_alpha': 1.2371819066730643, 'reg___reg_lambda': 46.776959128966546, 'reg___min_split_gain': 4.361891532750766, 'reg___path_smooth': 1.991932325924345, 'reg___min_child_weight': 1.7005372823073222, 'reg___feature_fraction': 0.5588962965516967, 'reg___bagging_fraction': 0.604299074006157, 'reg___bagging_freq': 6, 'reg___max_bin': 86, 'reg___min_data_per_group': 169}

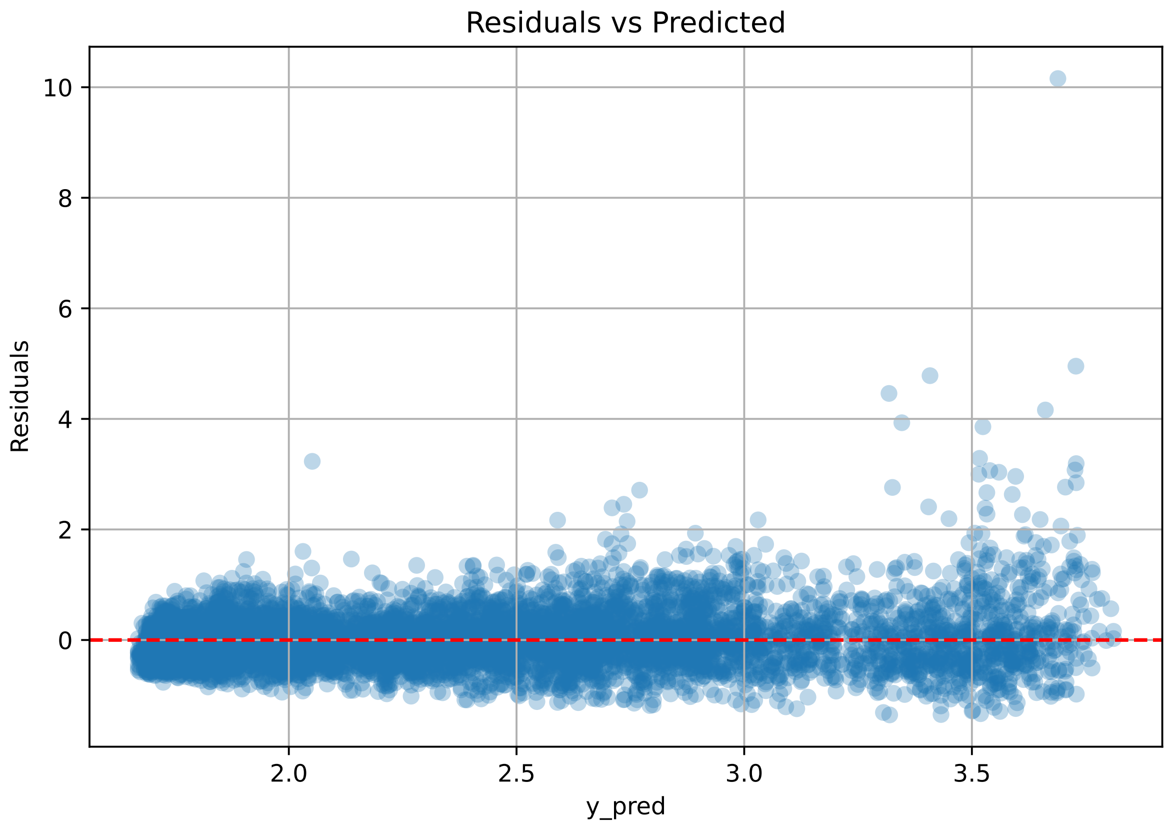
Rank 10: {'reg___n_estimators': 800, 'reg___max_depth': 8, 'reg___learning_rate': 0.047506615277442266, 'reg___num_leaves': 50, 'reg___subsample': 0.5179546757534053, 'reg___colsample_bytree': 0.6611566739416674, 'reg___min_child_samples': 45, 'reg___reg_alpha': 0.9567049831415435, 'reg___reg_lambda': 48.69629700382907, 'reg___min_split_gain': 3.96869019135928, 'reg___path_smooth': 2.367273712225975, 'reg___min_child_weight': 3.1552315406916236, 'reg___feature_fraction': 0.5784557610392931, 'reg___bagging_fraction': 0.6067262693108807, 'reg___bagging_freq': 7, 'reg___max_bin': 65, 'reg___min_data_per_group': 200}

3.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

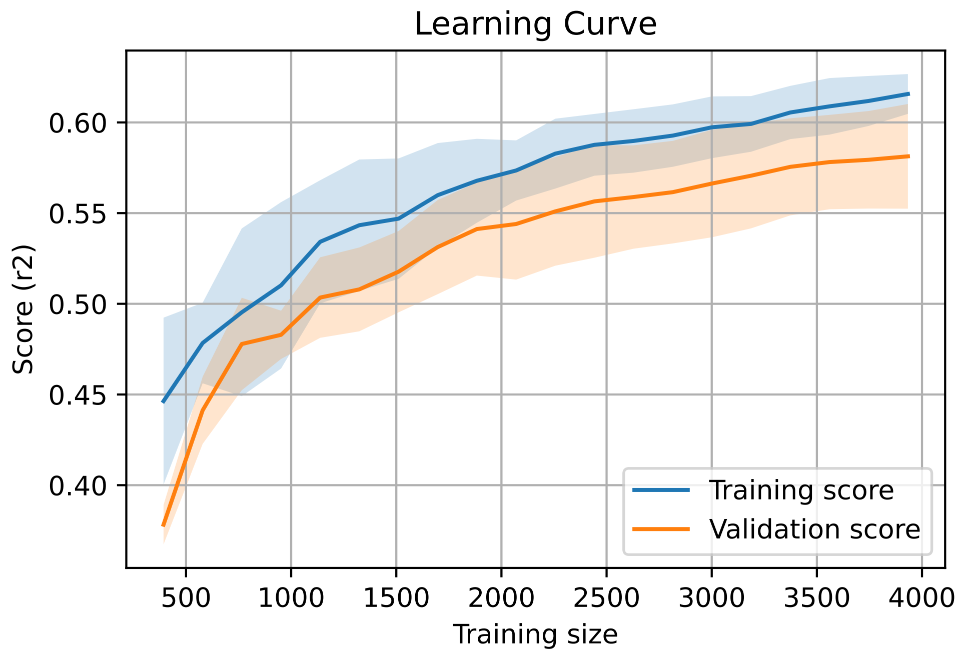
3.5.3 Scatter residuals



3.5.4 Learning curve

$\mathbf{Len(group)}$: 8028, $\mathbf{Len(training)}$: 5620,

$\mathbf{Len(training\_real)}$: 3934, $\mathbf{Len(test\_of\_training)}$: 1686, $\mathbf{Len(test)}$: 2408

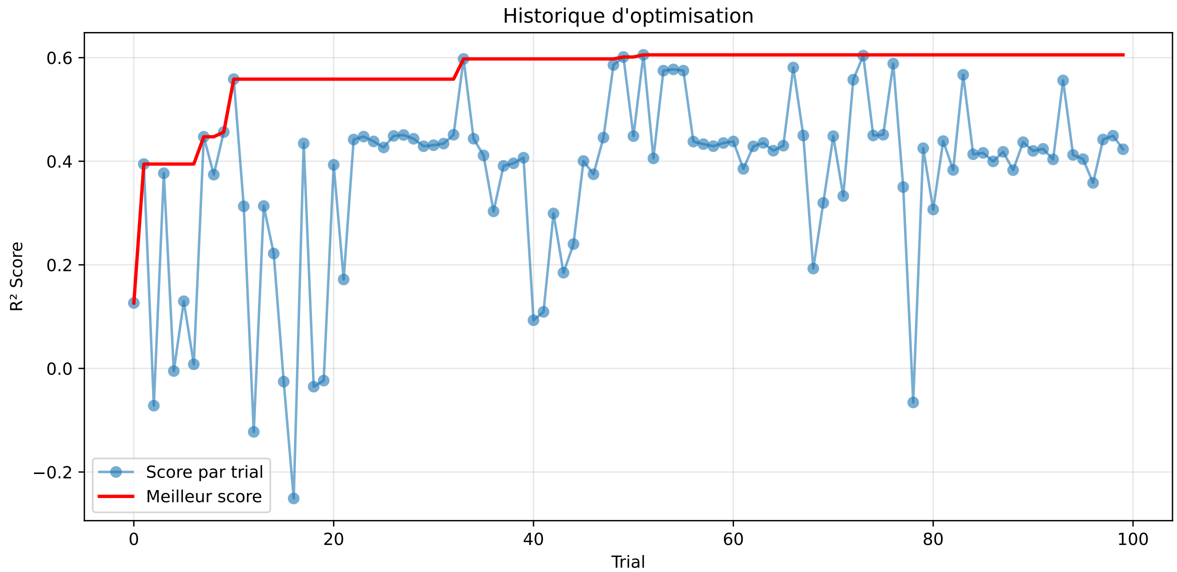


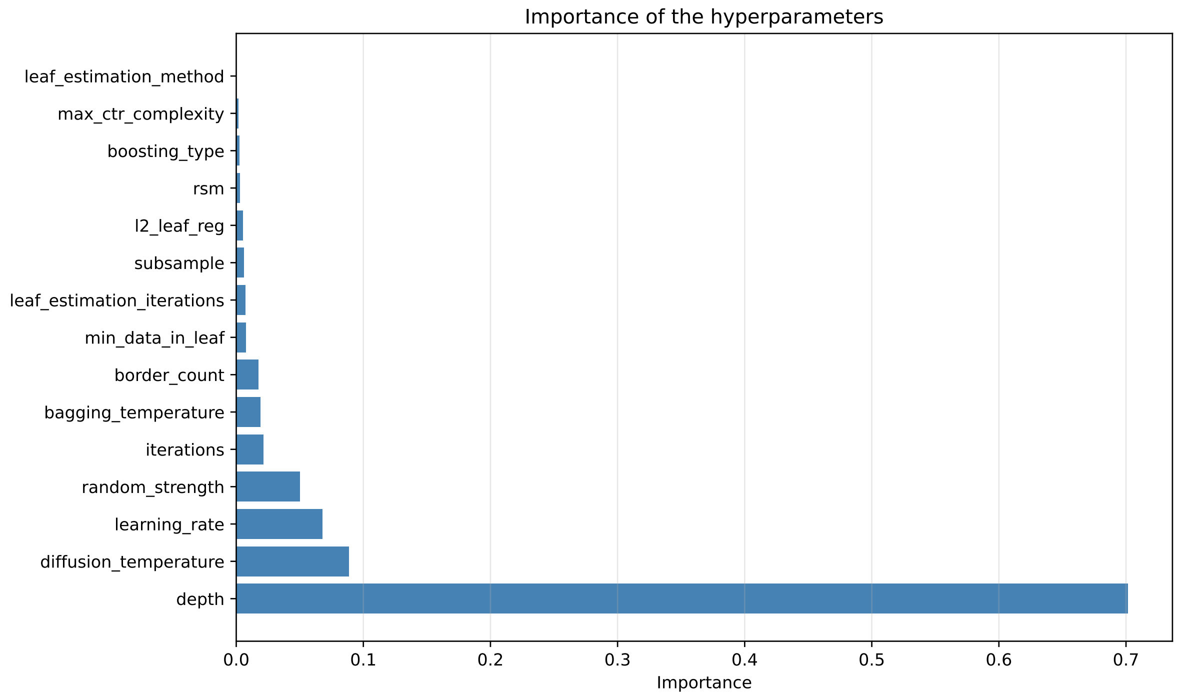
3.6 Model : CatBoost

3.6.1 Gridsearch

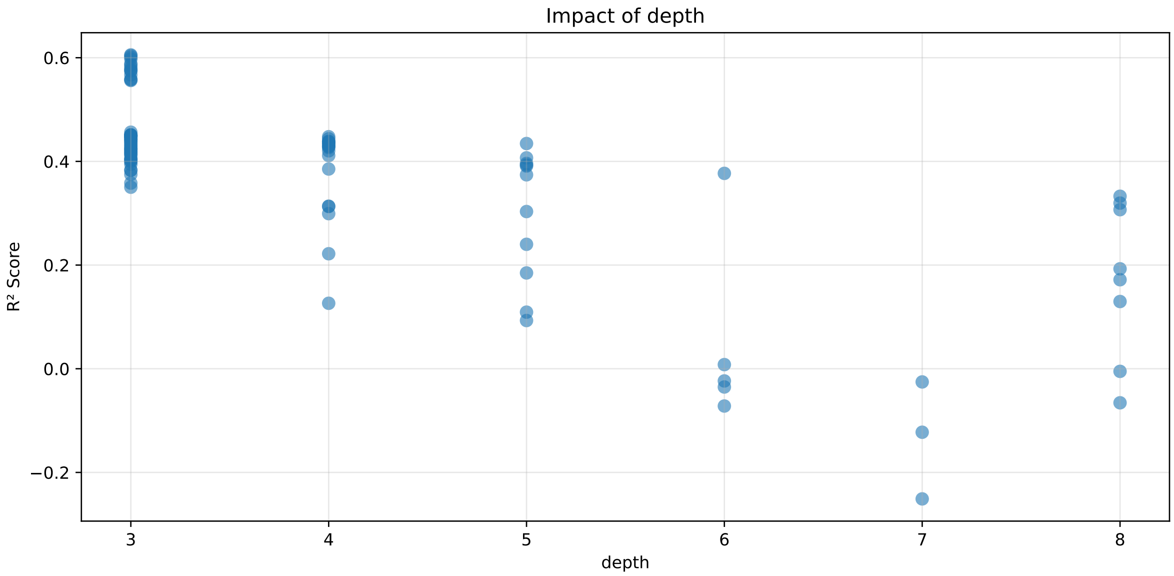
Distribution y_train vs y_test: y_train: mean=2.350, std=0.806, min=0.959, max=13.845 y_test: mean=2.355, std=0.778, min=1.011, max=7.822

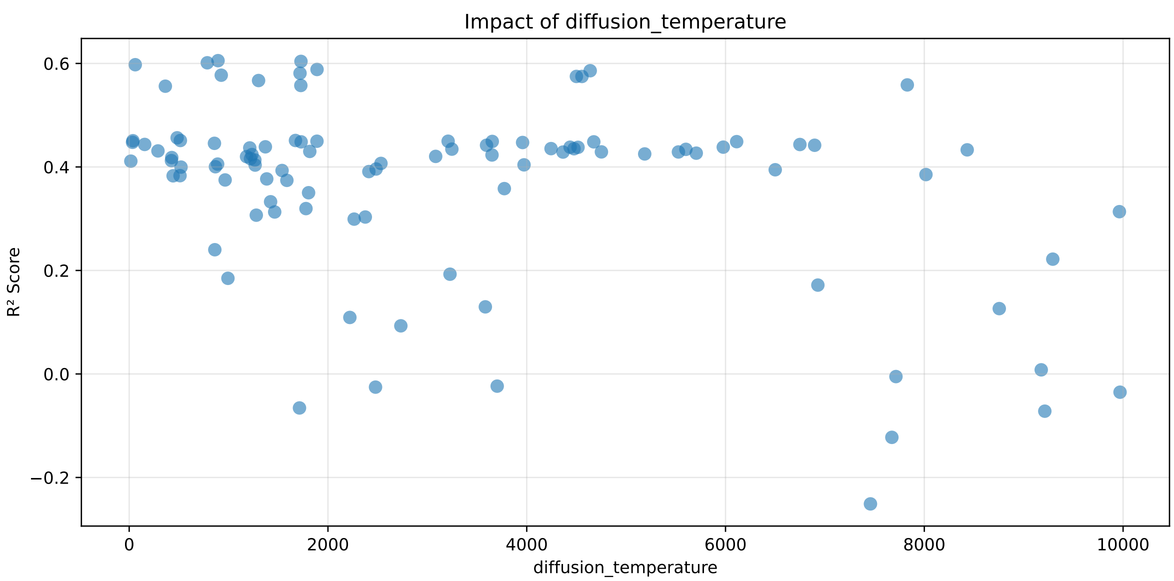
===================== TOP Importance FEATURES ===================== feature importance r_umidity 25.143023 r_temperature_st_4h_slope 20.726835 r_temperature_st_4h_diff_last 12.010734 r_temperature 6.791058 r_temperature_st_4h_min 5.704101 r_CO2 5.154679 r_NH3_st_4h_std 4.863997 r_NH3_st_4h_max 4.050972 r_umidity_st_4h_diff_last 3.882954 r_NH3 3.365054 r_umidity_st_4h_std 2.473005 r_THI 2.247746 r_NH3_st_4h_slope 2.156686 r_NH3_st_4h_diff_last 1.429155

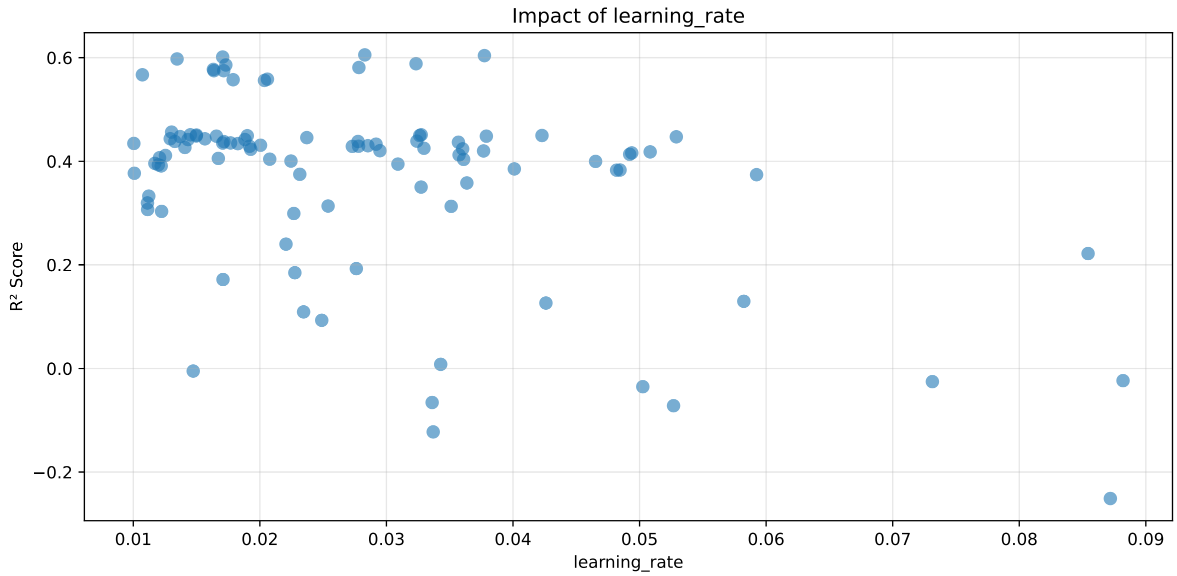


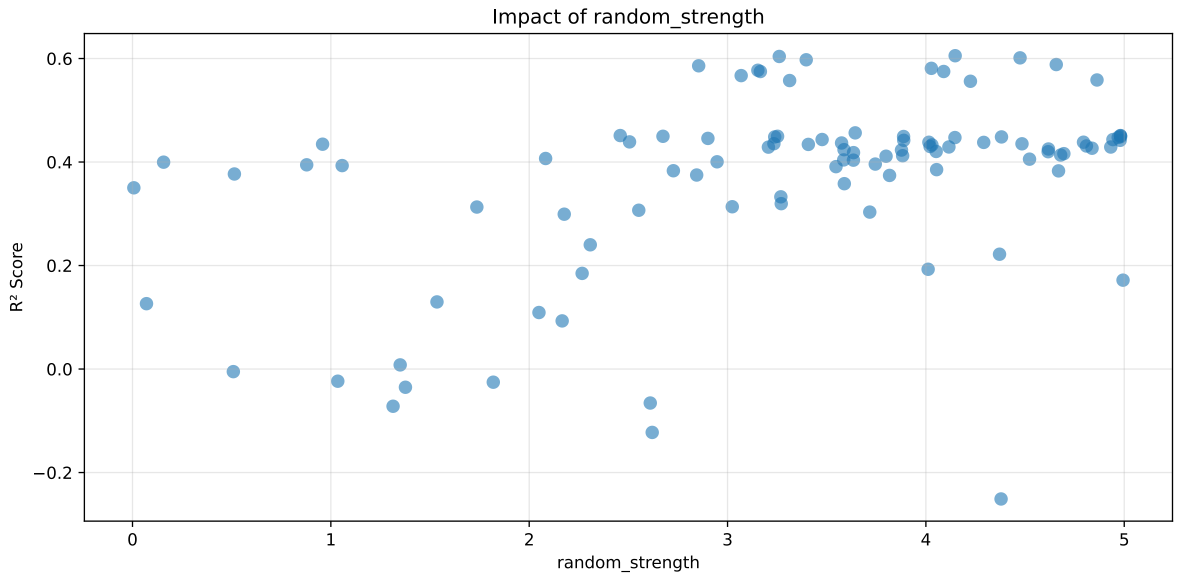


Paramètres avec >5% d'importance : ['depth', 'diffusion_temperature', 'learning_rate', 'random_strength']


Impact of depth
R² Score
0.6
0.4
0.2
0.0
−0.2
3
4
5
6
7
8
depth

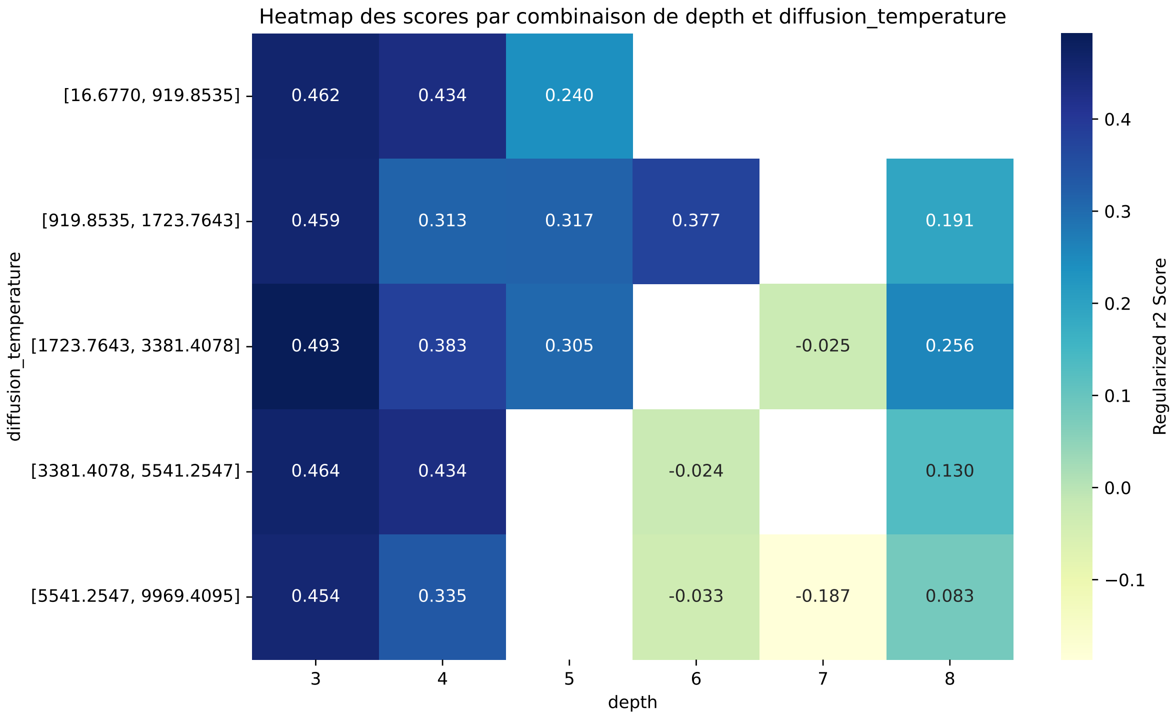

Impact of diffusion_temperature
R² Score
0.6
0.4
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000
diffusion_temperature





`C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main`

`The default value of observed=False is deprecated and will change to observed=True in a futu`



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6344	0.6054	0.6328	0.0284	0.6054	0.6362	0.0308	1.0508	1.0479
2	0.6338	0.604	0.6298	0.0303	0.604	0.6377	0.0337	1.0558	1.0494
3	0.6309	0.6013	0.6305	0.0278	0.6013	0.6317	0.0304	1.0506	1.0493
4	0.6247	0.5976	0.6237	0.0275	0.5976	0.6252	0.0276	1.0462	1.0454
5	0.6151	0.5884	0.6136	0.0276	0.5884	0.6162	0.0279	1.0474	1.0454
6	0.6146	0.5859	0.6151	0.027	0.5859	0.6174	0.0315	1.0537	1.049
7	0.6061	0.581	0.6049	0.0272	0.581	0.608	0.027	1.0464	1.0431
8	0.6048	0.5773	0.6043	0.0275	0.5773	0.6082	0.0308	1.0534	1.0476
9	0.6021	0.5749	0.6025	0.0277	0.5749	0.6072	0.0323	1.0561	1.0472
10	0.6012	0.5749	0.6021	0.027	0.5749	0.6054	0.0305	1.053	1.0457
---	---	---	---	---	---	---	---	---	---
100	0.9584	0.7569	0.7928	0.0337	-0.2511	0.8097	0.0528	1.0698	1.2663

Hyperparameters:

Rank 1: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.028290465178868622, 'reg___l2_leaf_reg': 23.263423338876873, 'reg___bagging_temperature': 3.06105150123587, 'reg___random_strength': 4.147631769775964, 'reg___subsample': 0.8285292778268712, 'reg___min_data_in_leaf': 37, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.5275934615855823, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 893.9827477180896}

Rank 2: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.03774340669427119, 'reg___l2_leaf_reg': 4.387862627485787, 'reg___bagging_temperature': 3.446066777588433, 'reg___random_strength': 3.2607720832173337, 'reg___subsample': 0.8064148161579154, 'reg___min_data_in_leaf': 44, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5010717060173846, 'reg___border_count': 38, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 1728.3775027139716}

Rank 3: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.017061014271951454, 'reg___l2_leaf_reg': 1.0790275667257383, 'reg___bagging_temperature': 3.679463938428468, 'reg___random_strength': 4.474806959146424, 'reg___subsample': 0.7698188506377994, 'reg___min_data_in_leaf': 37, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6717917162274344, 'reg___border_count': 34, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 785.8371787082764}

Rank 4: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.013459437555724731, 'reg___l2_leaf_reg': 13.607117163188548, 'reg___bagging_temperature': 4.102096766399642, 'reg___random_strength': 3.396697076744791, 'reg___subsample': 0.8535113425375807, 'reg___min_data_in_leaf': 17, 'reg___leaf_estimation_iterations': 5, 'reg___rsm': 0.6043253227617666, 'reg___border_count': 74, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 61.08585930234676}

Rank 5: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.03233855758590728, 'reg___l2_leaf_reg': 3.3958511250299934, 'reg___bagging_temperature': 3.5391725587037035, 'reg___random_strength': 4.657301596484027, 'reg___subsample': 0.7135000261990141, 'reg___min_data_in_leaf': 45, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5115731326255757, 'reg___border_count': 39, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 1890.0856291787672}

Rank 6: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.017310868903790747, 'reg___l2_leaf_reg': 8.613825515687466, 'reg___bagging_temperature': 3.7330440738725206, 'reg___random_strength': 2.854441081168527, 'reg___subsample': 0.7644704410294924, 'reg___min_data_in_leaf': 20, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6723848821870038, 'reg___border_count': 69, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 4638.588307302314}

Rank 7: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.027824136668052214, 'reg___l2_leaf_reg': 2.703944706838225, 'reg___bagging_temperature': 3.365684913754104, 'reg___random_strength': 4.027687146781712, 'reg___subsample': 0.7022035327651398,

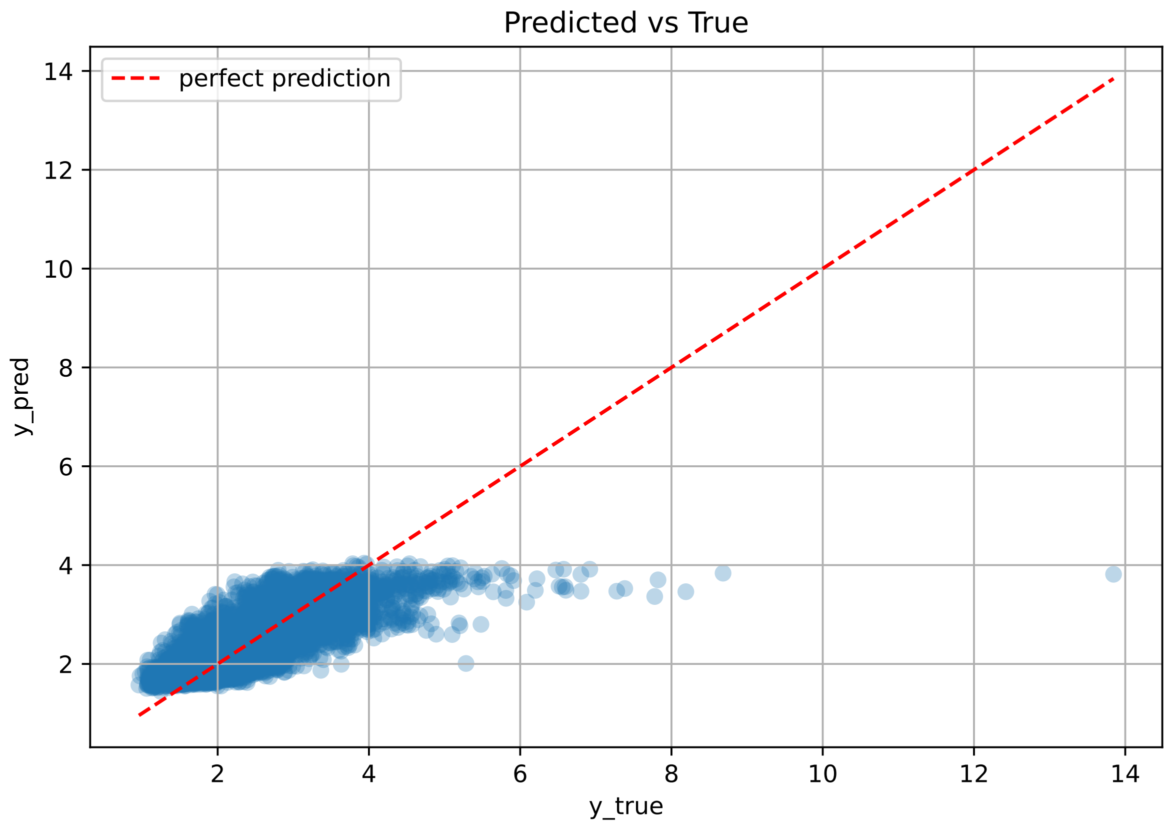
’reg__min_data_in_leaf’: 43, ’reg__leaf_estimation_iterations’: 10, ’reg__rsm’: 0.5024231259246964, ’reg__border_count’: 39, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 1718.8402689806587}

Rank 8: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.01632672603494883, ’reg__l2_leaf_reg’: 1.0961257524384855, ’reg__bagging_temperature’: 3.06716497888798, ’reg__random_strength’: 3.152847670655575, ’reg__subsample’: 0.8723178517483733, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 9, ’reg__rsm’: 0.7178946517702116, ’reg__border_count’: 88, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 926.3212332819215}

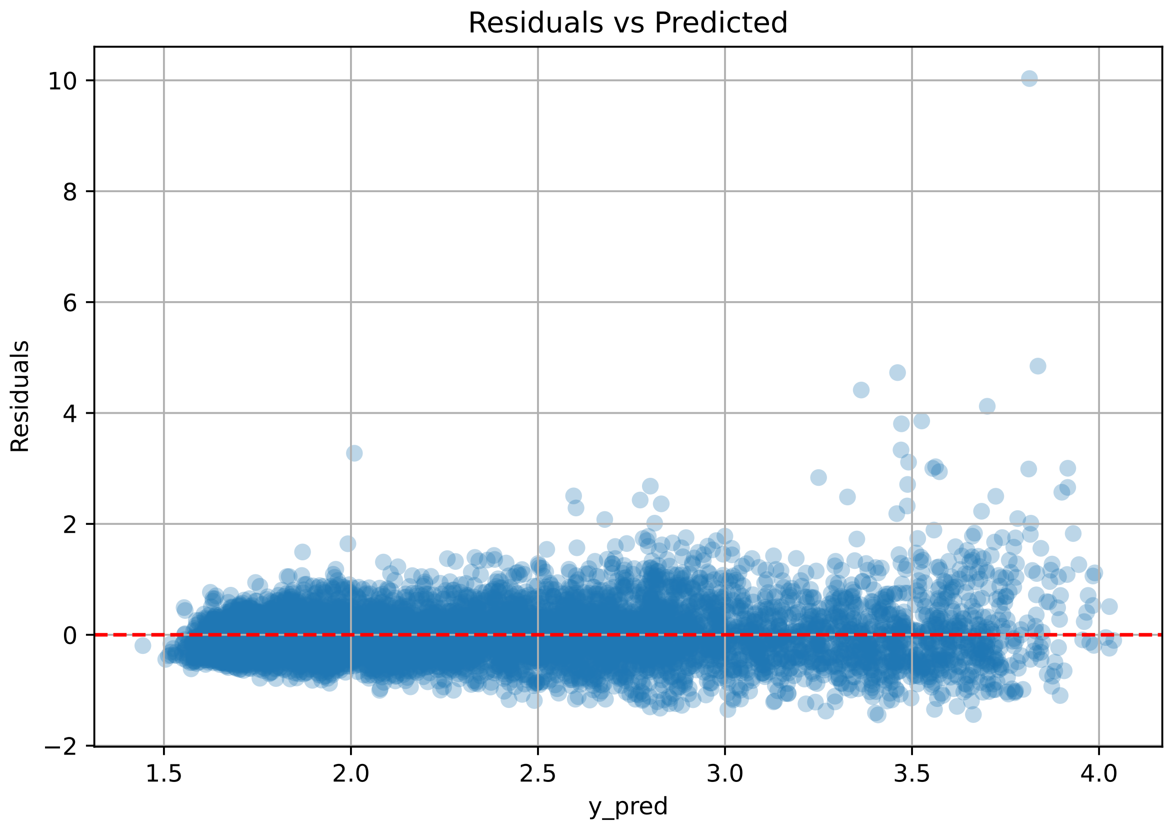
Rank 9: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.017140959950561682, ’reg__l2_leaf_reg’: 1.6521704497722798, ’reg__bagging_temperature’: 3.0675387308447144, ’reg__random_strength’: 4.089733991299584, ’reg__subsample’: 0.7814495972581076, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 9, ’reg__rsm’: 0.7230409784826719, ’reg__border_count’: 37, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 4499.57043671266}

Rank 10: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.016369667325609794, ’reg__l2_leaf_reg’: 19.72855181082271, ’reg__bagging_temperature’: 2.8316286535520723, ’reg__random_strength’: 3.1655451160664594, ’reg__subsample’: 0.8264288177206579, ’reg__min_data_in_leaf’: 13, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.6586424189060903, ’reg__border_count’: 33, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 4555.440362228545}

3.6.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

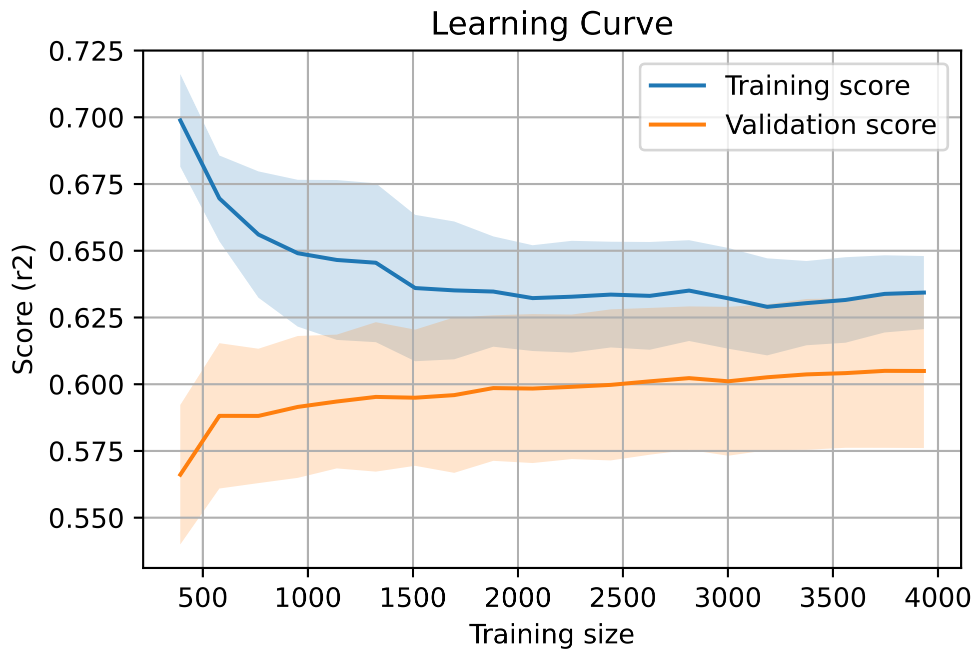
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 8028, **Len(training)** : 5620,

Len(training_real) : 3934, **Len(test_of_training)** : 1686, **Len(test)** : 2408



3.7 Sumup-table

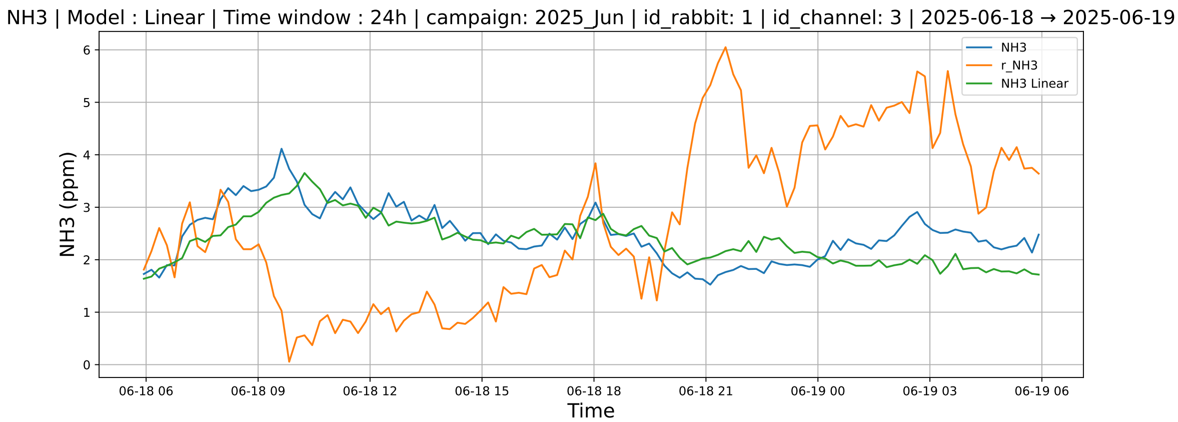
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.532	0.511	0.022	0.546	0.546	0.035	1.069	1.042	nan
LGBM	0.615	0.585	0.026	0.614	0.629	0.044	1.076	1.053	0.4304
CatBoost	0.634	0.605	0.028	0.633	0.636	0.031	1.051	1.048	0.6054

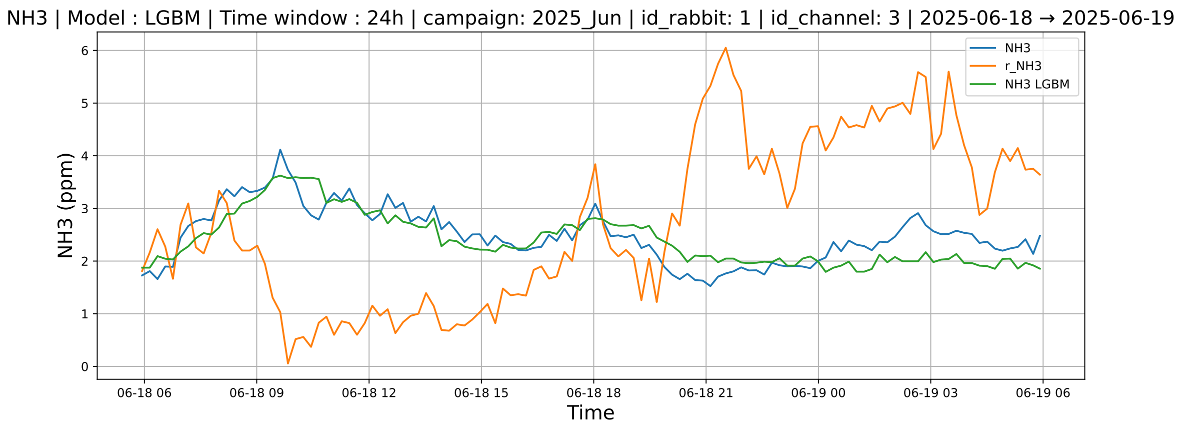
4 Plot

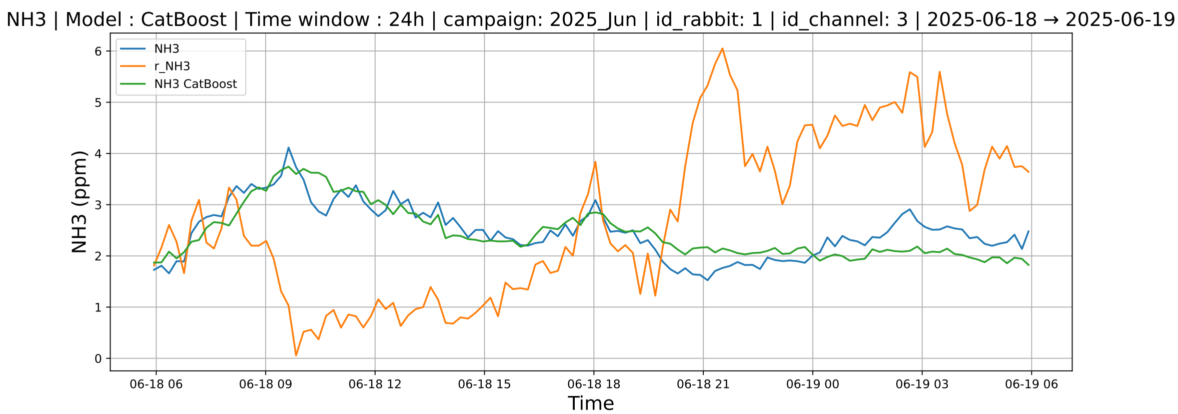
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

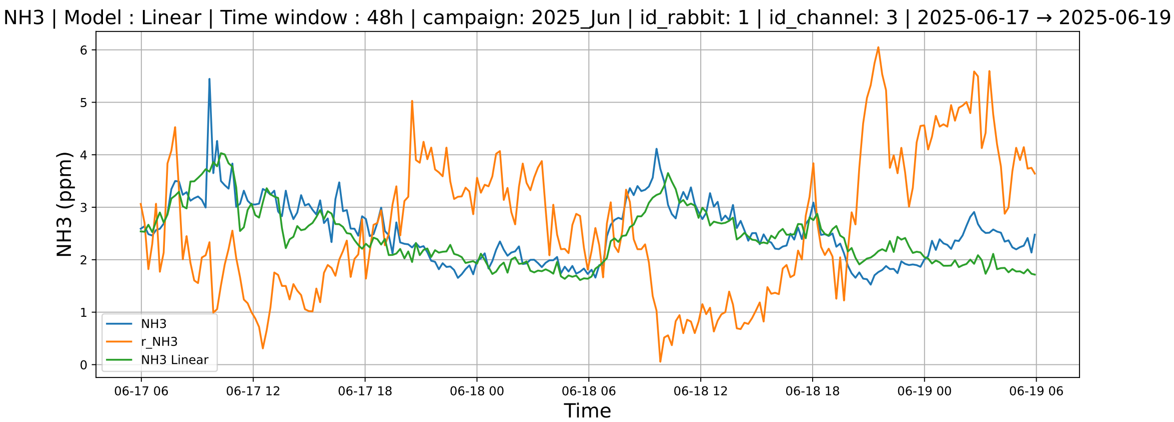
4.1.1.1 Time window : 24

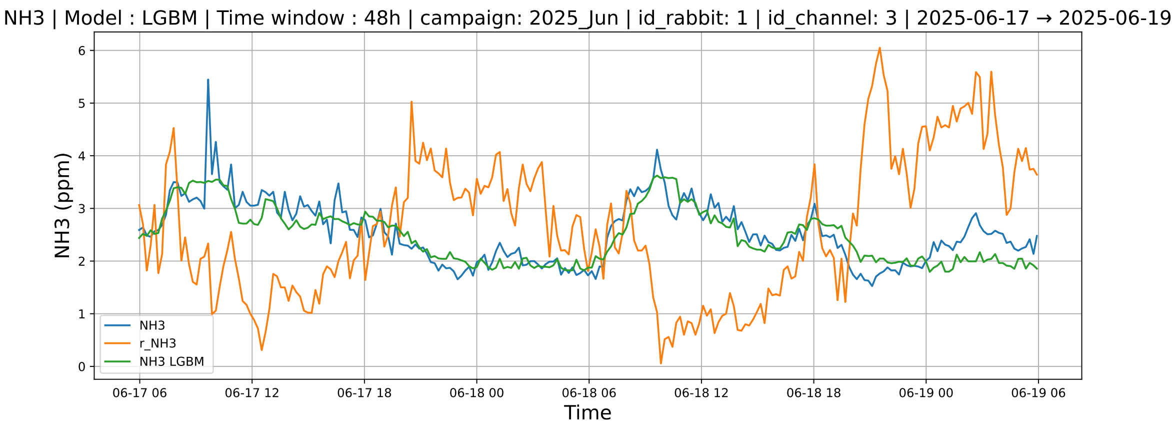


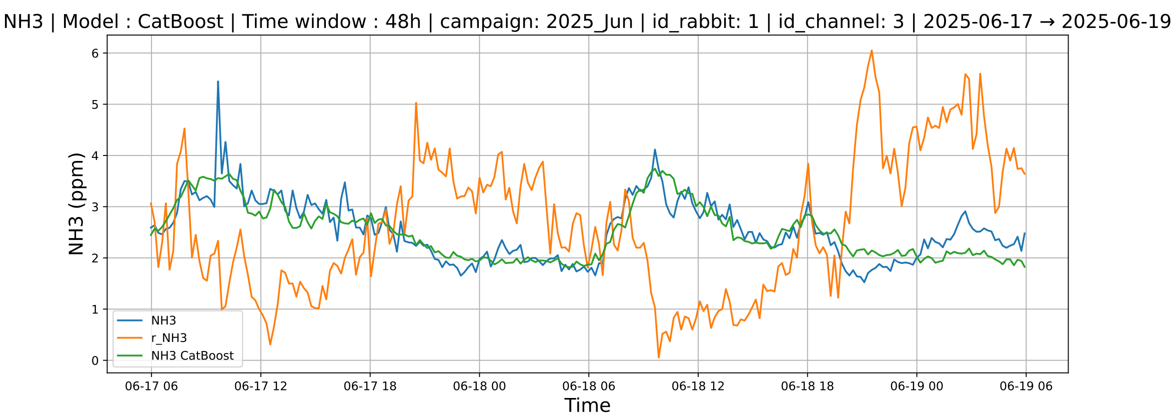




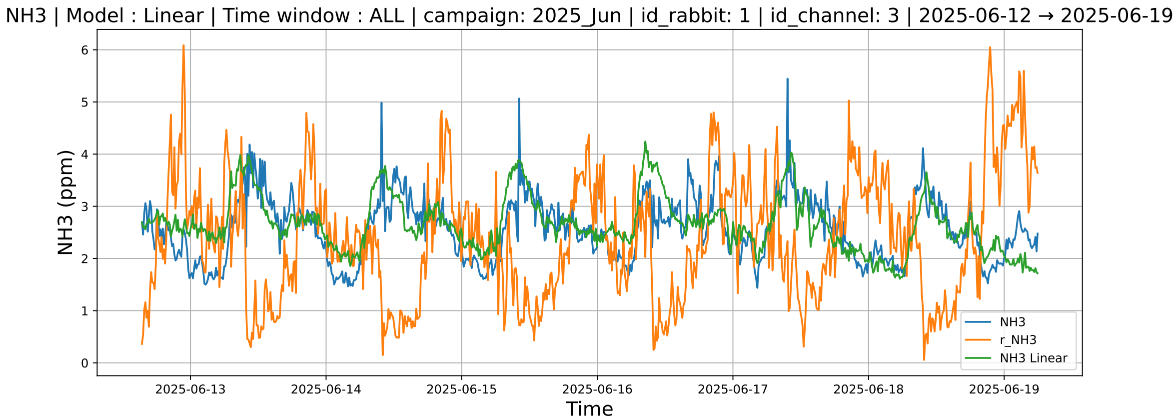
4.1.1.2 Time window : 48

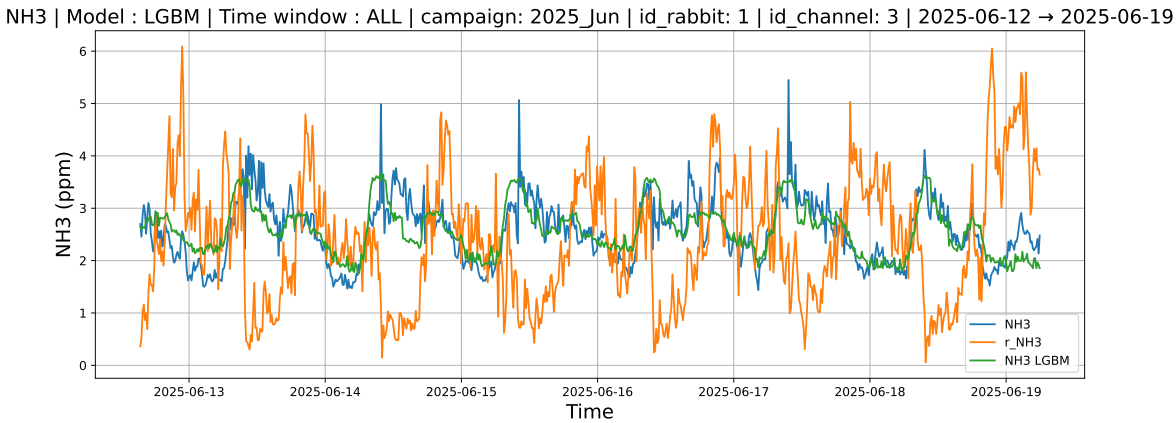


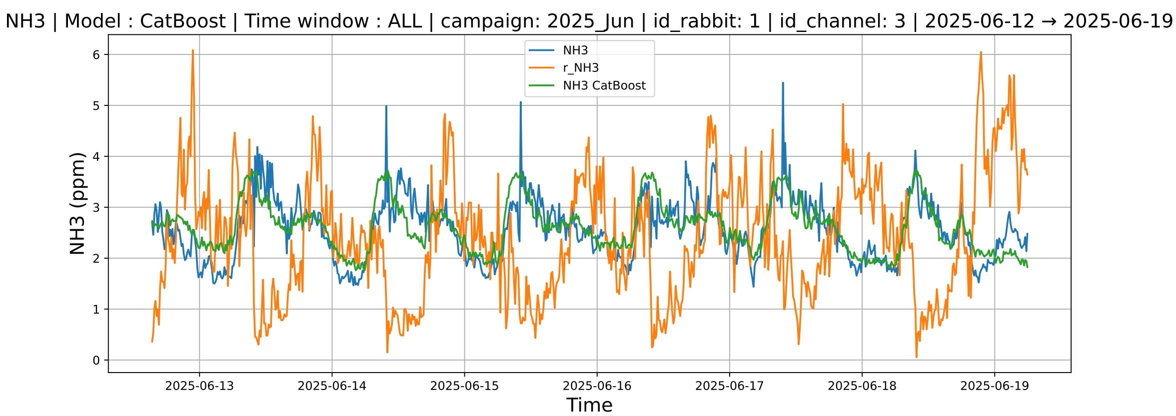




4.1.1.3 Time window : ALL



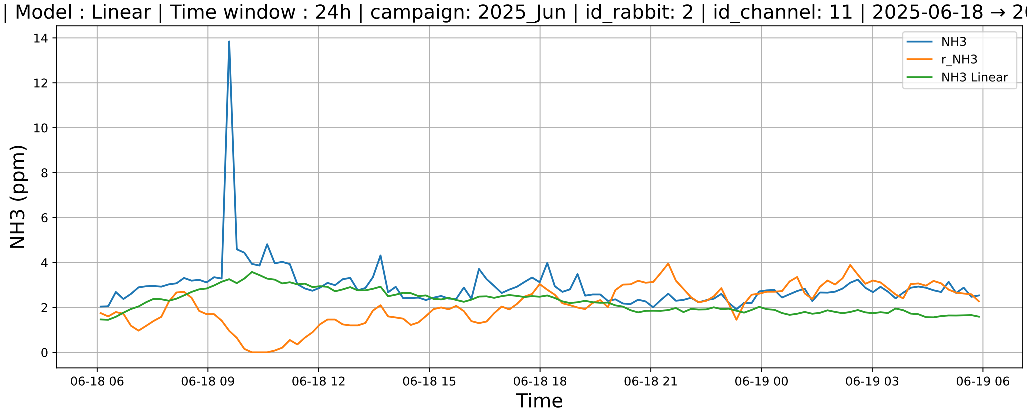




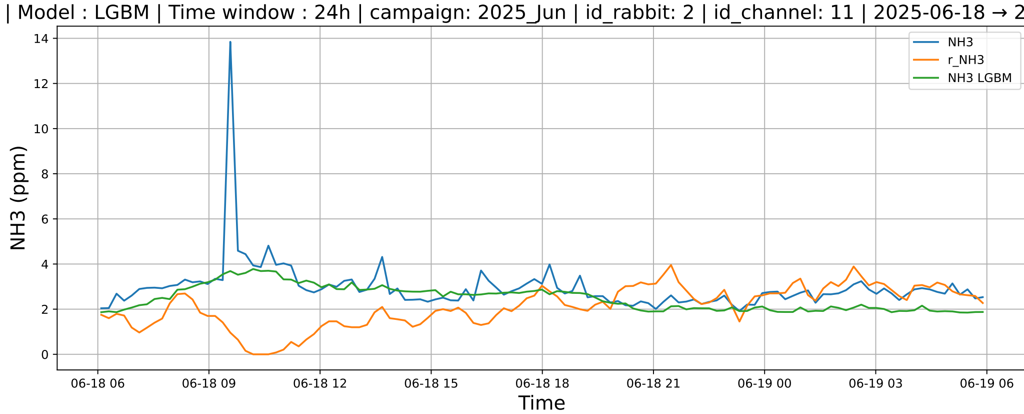
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

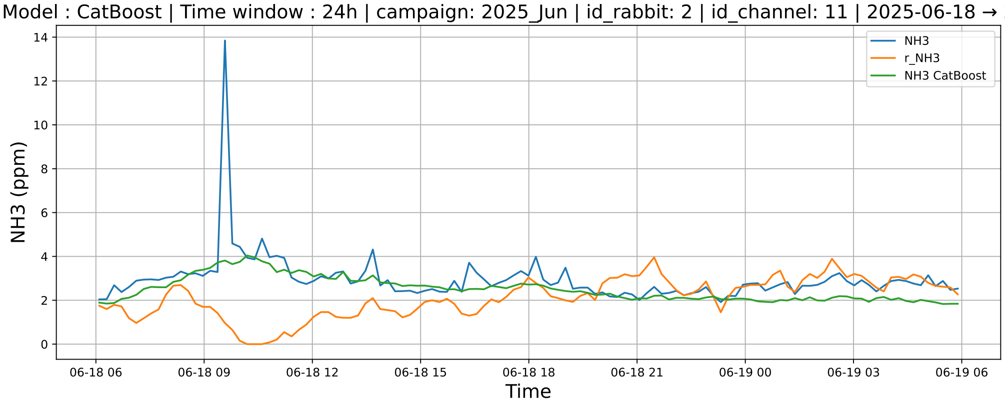
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

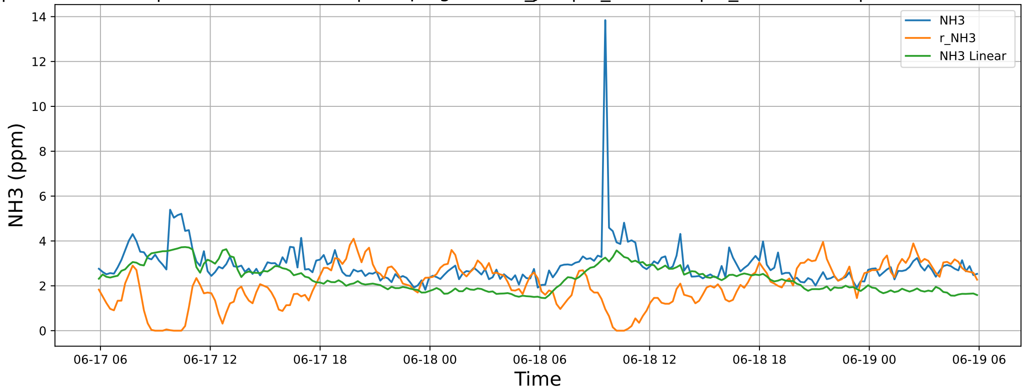


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

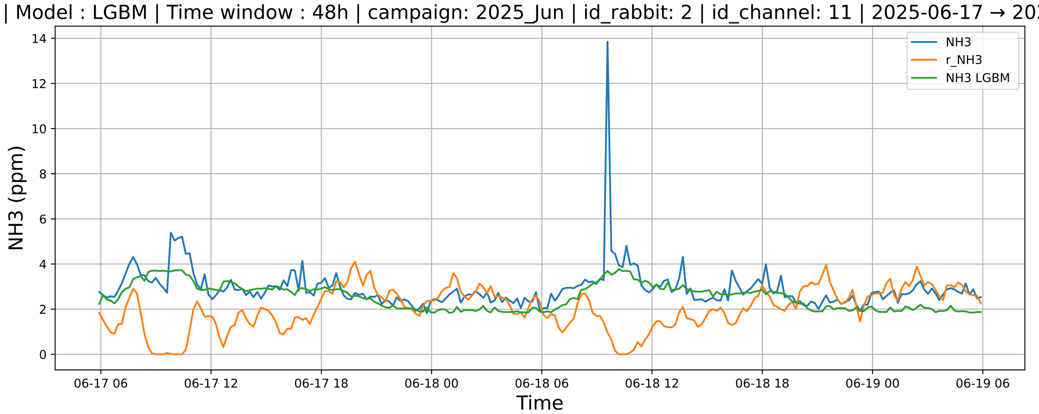


4.1.2.2 Time window : 48

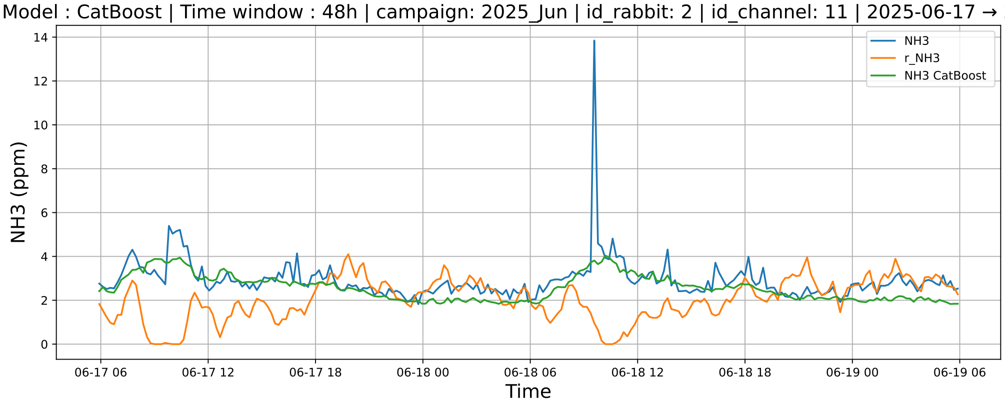
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

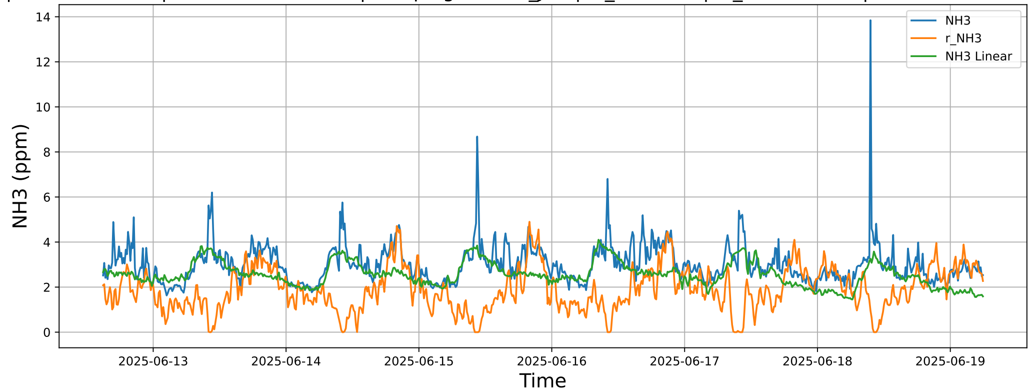


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

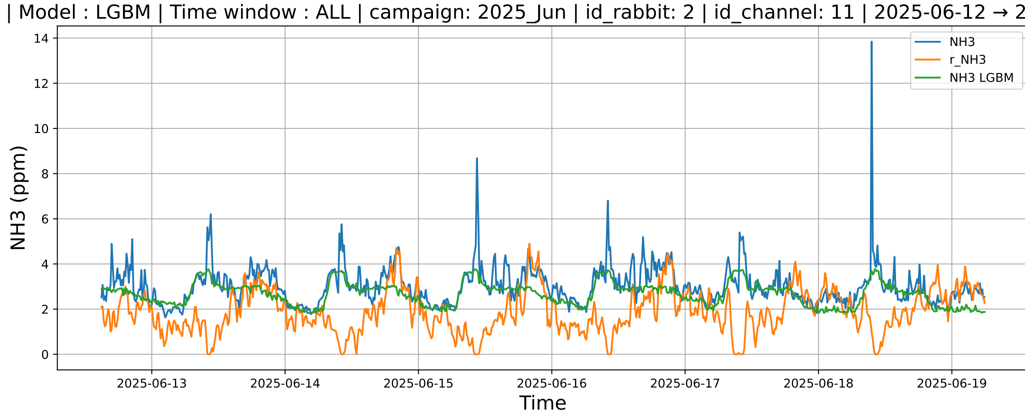


4.1.2.3 Time window : ALL

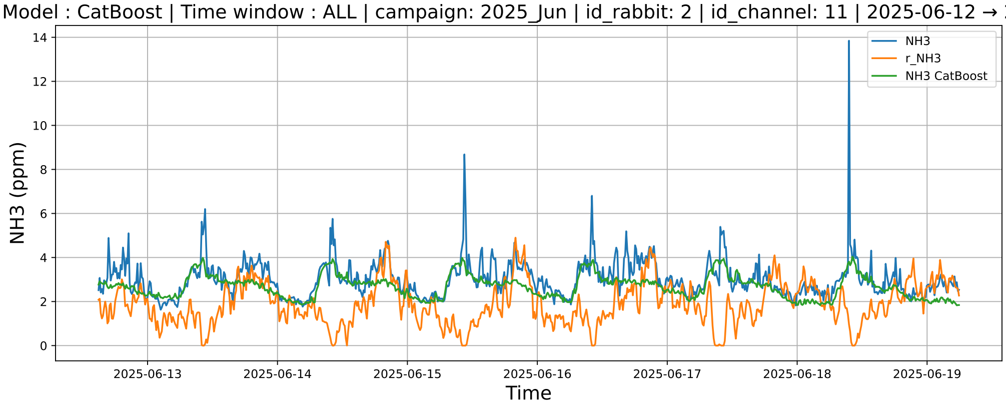
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



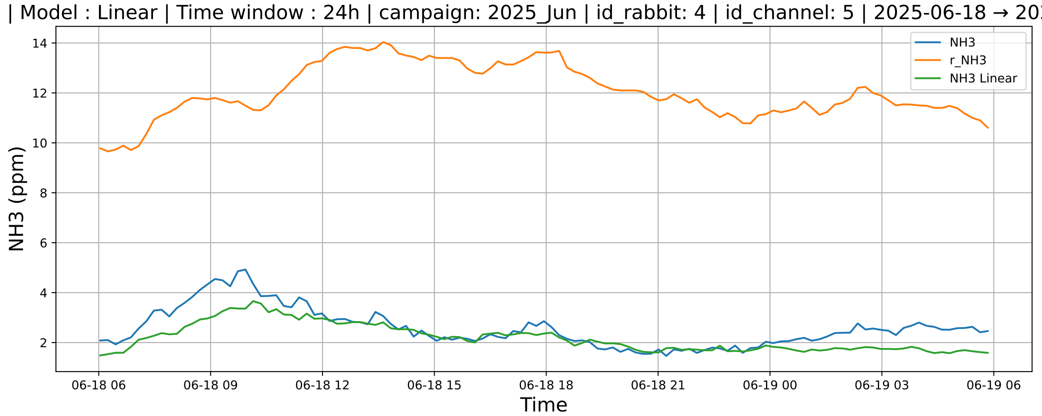
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



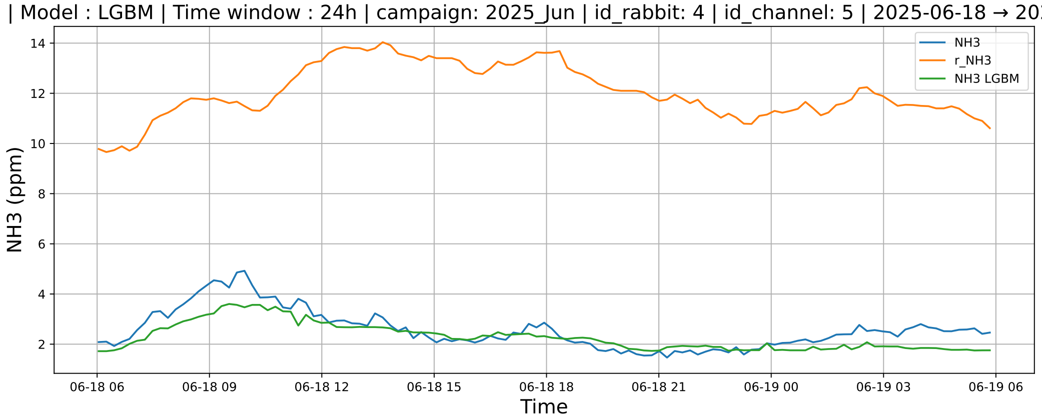
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

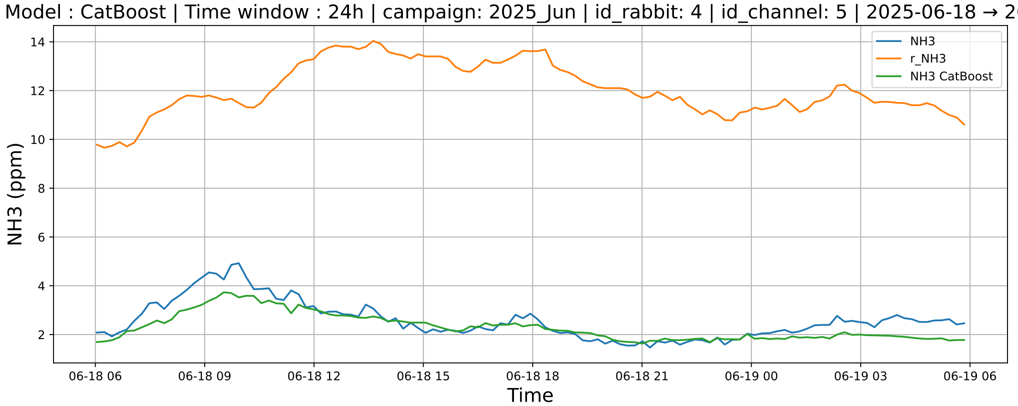
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

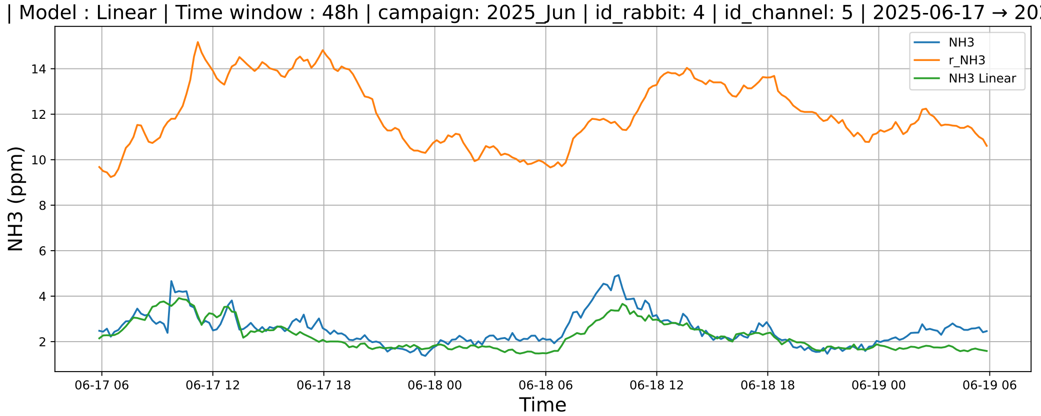


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

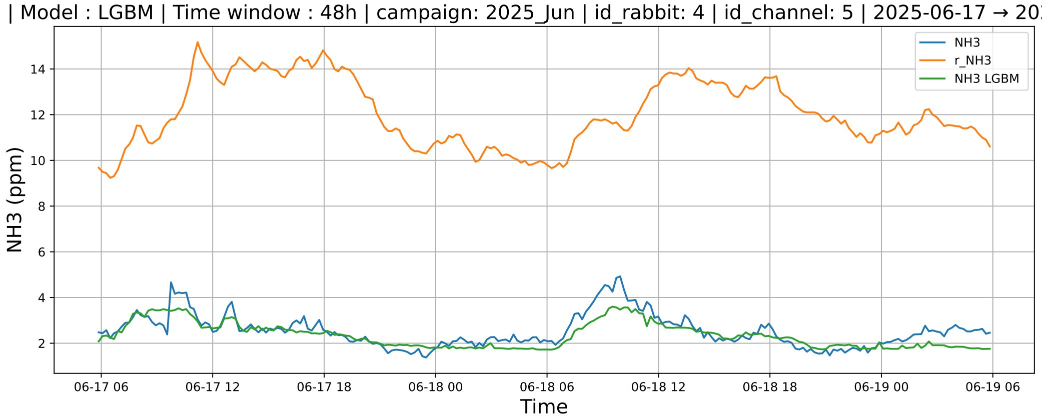


4.1.3.2 Time window : 48

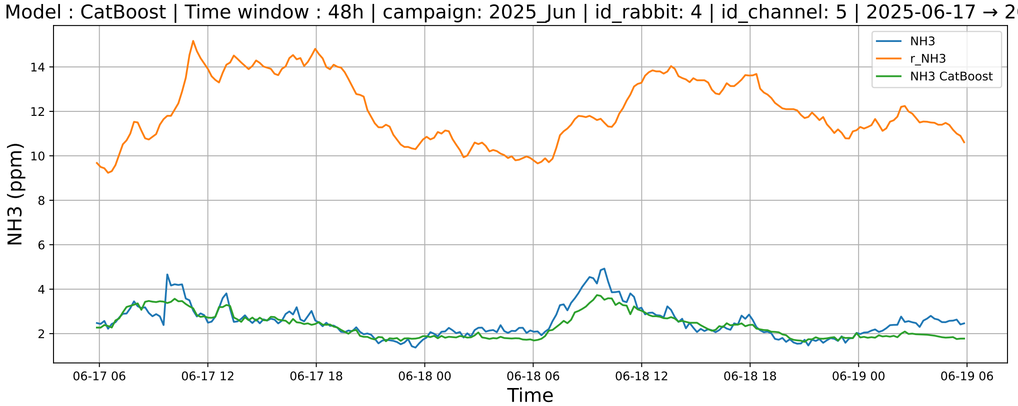
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



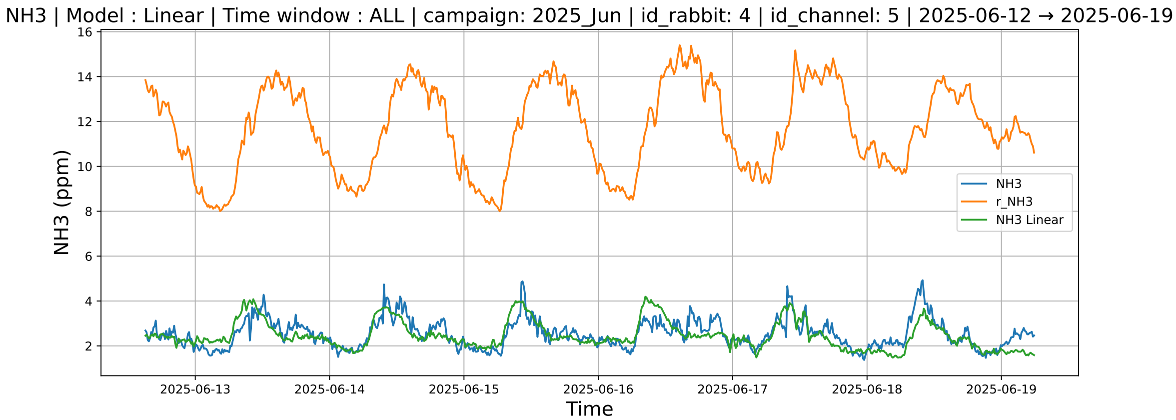
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

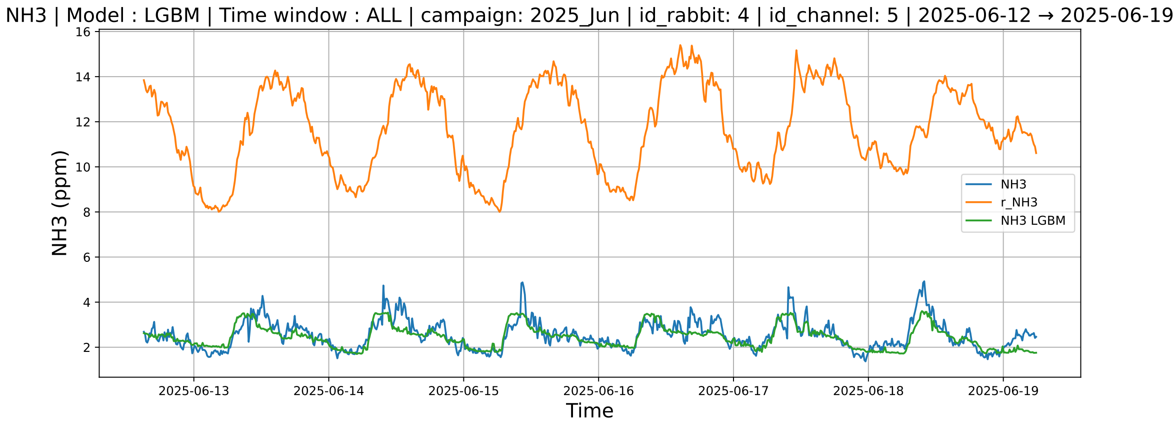


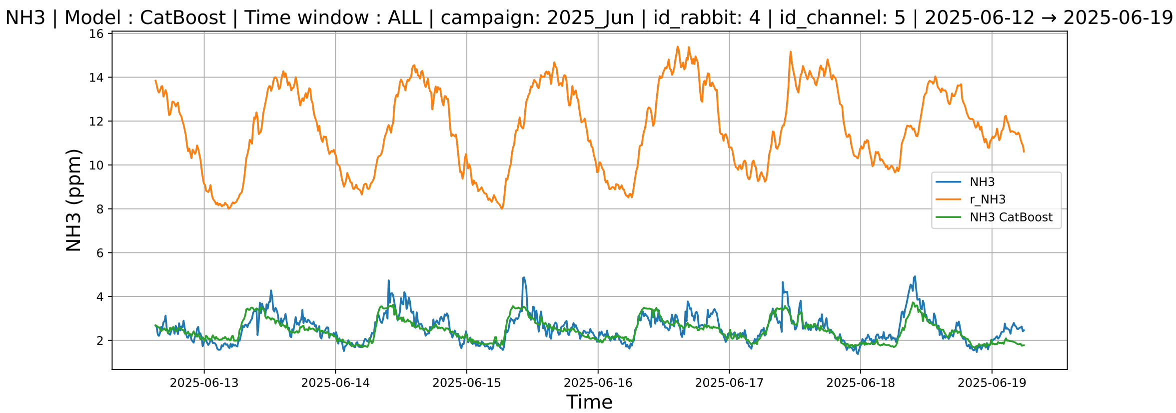
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.1.3.3 Time window : ALL



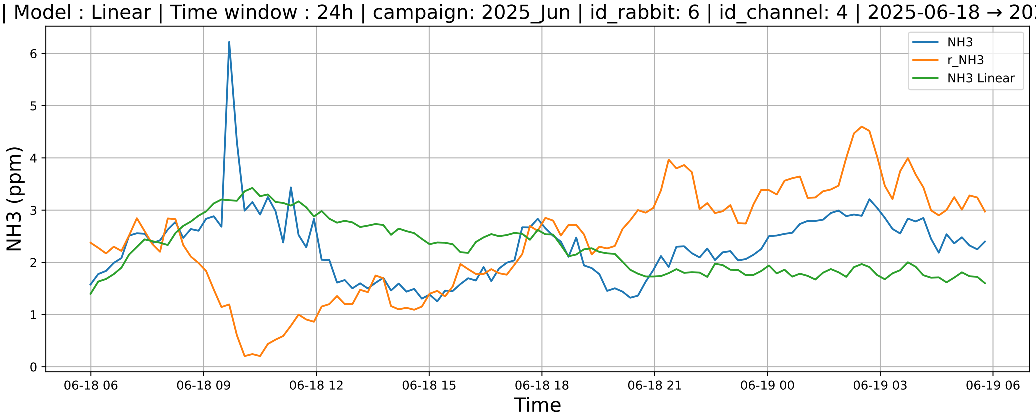




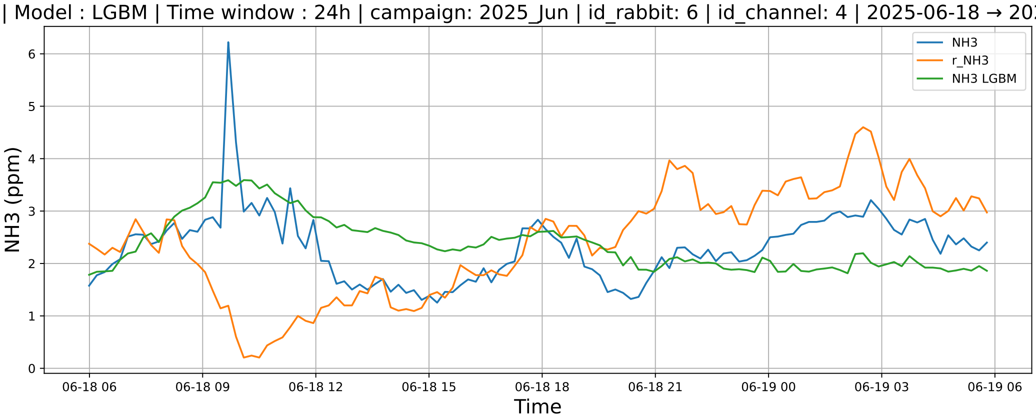
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

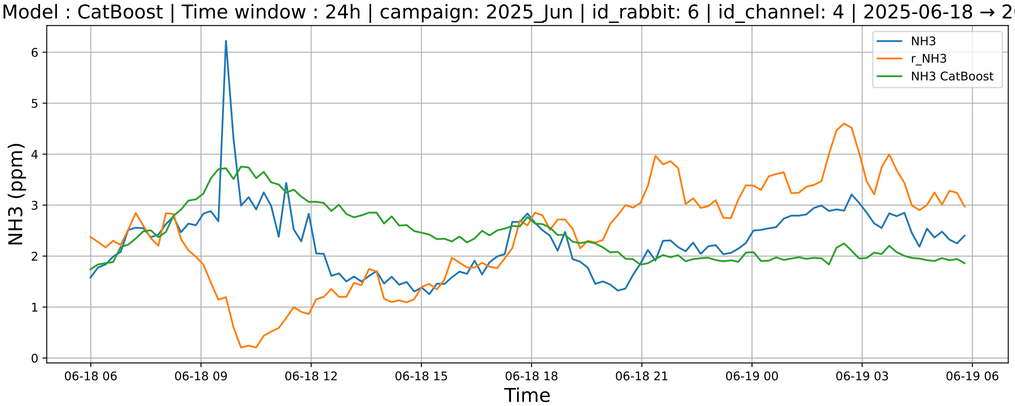
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

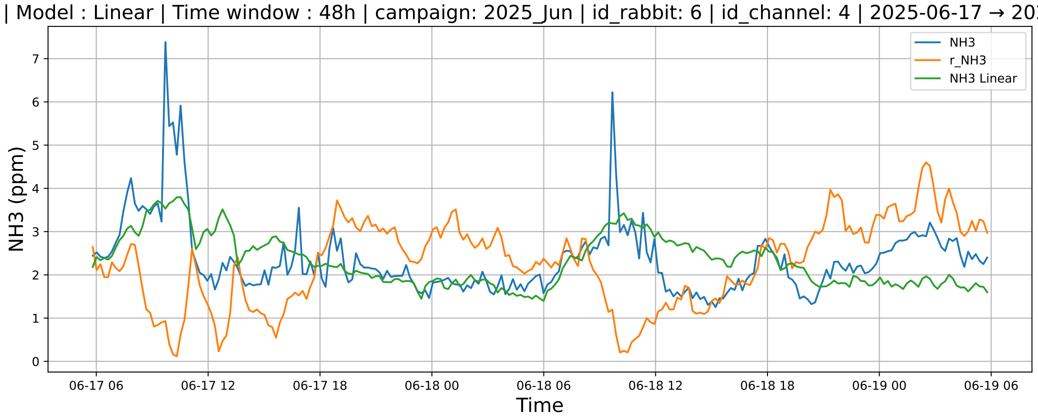


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

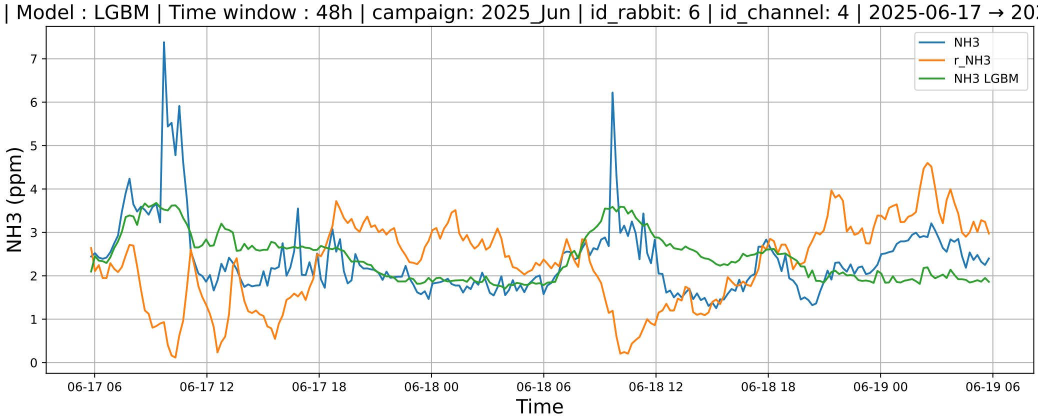


4.1.4.2 Time window : 48

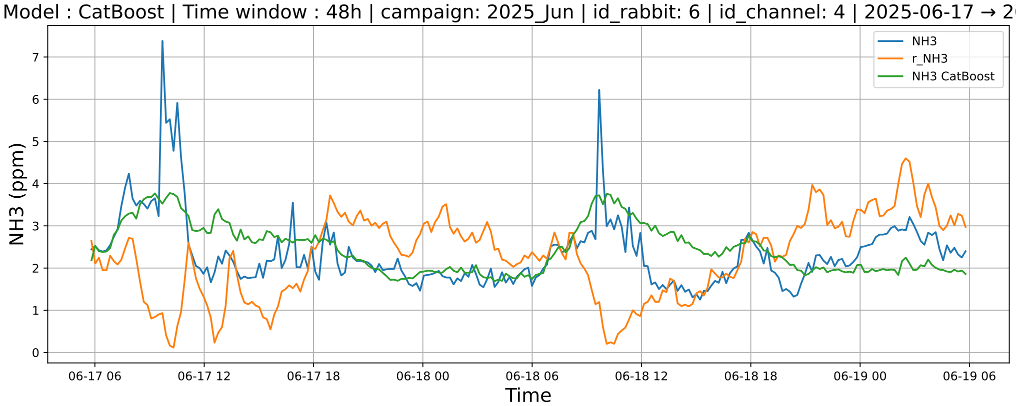
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



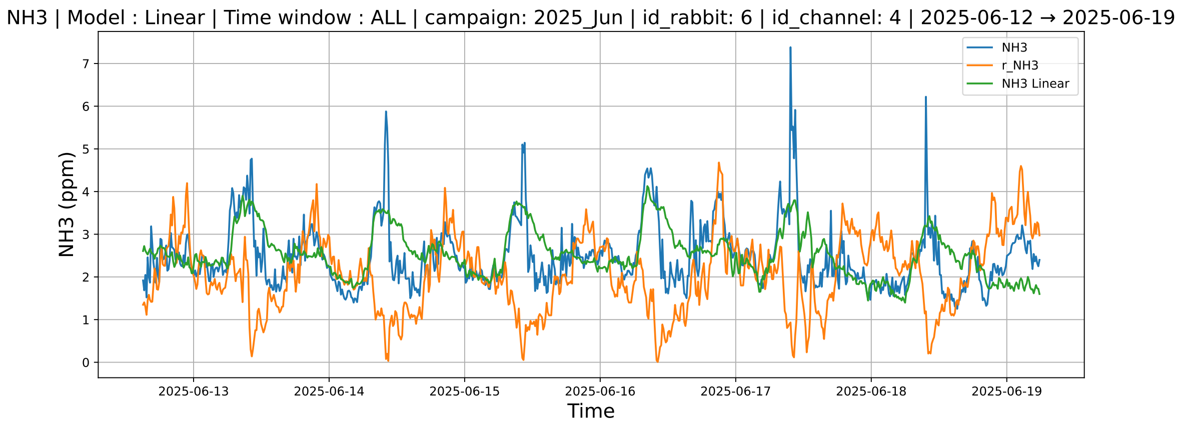
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

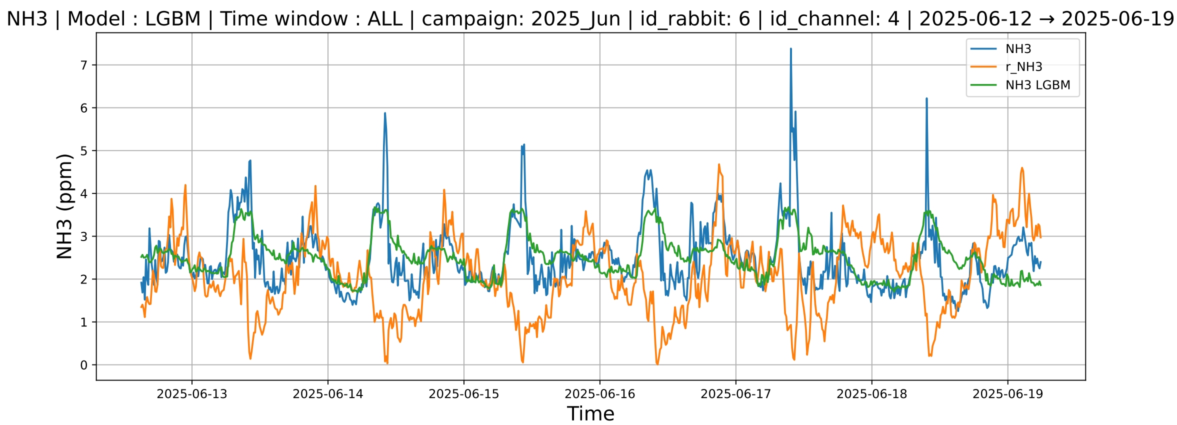


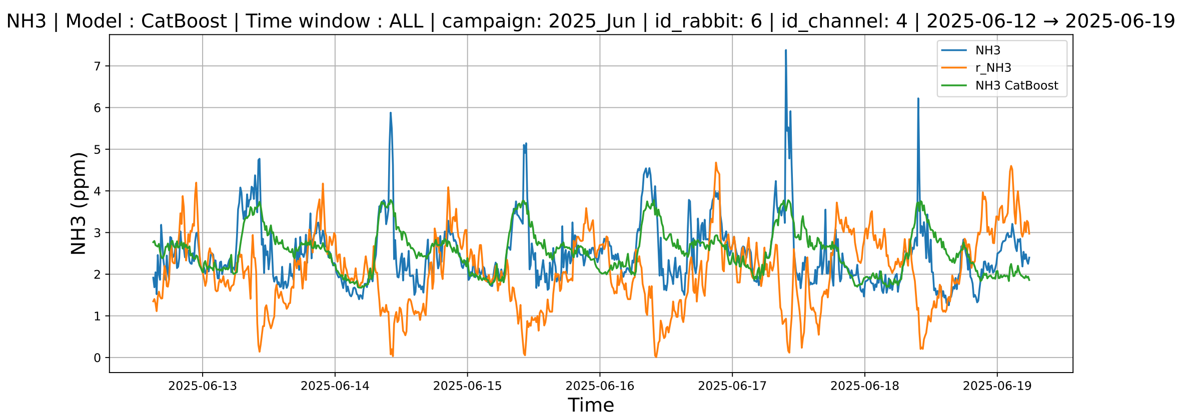
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.1.4.3 Time window : ALL

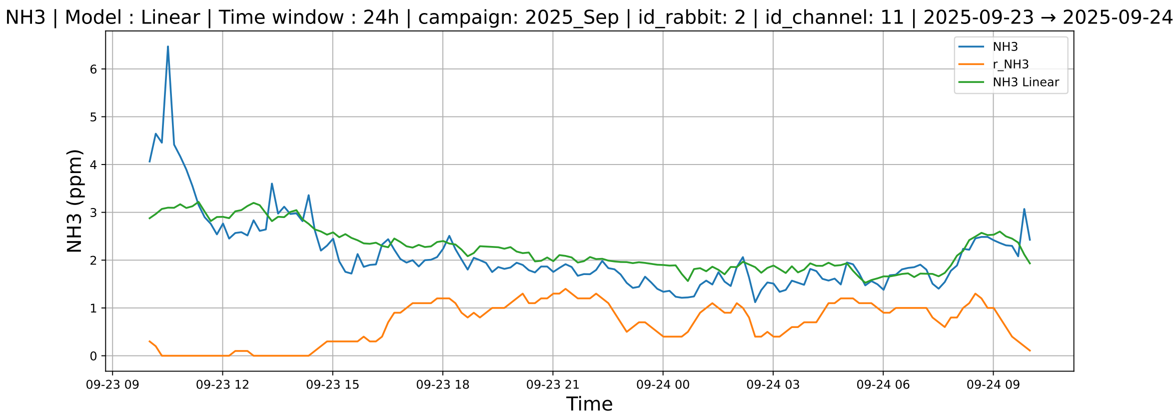


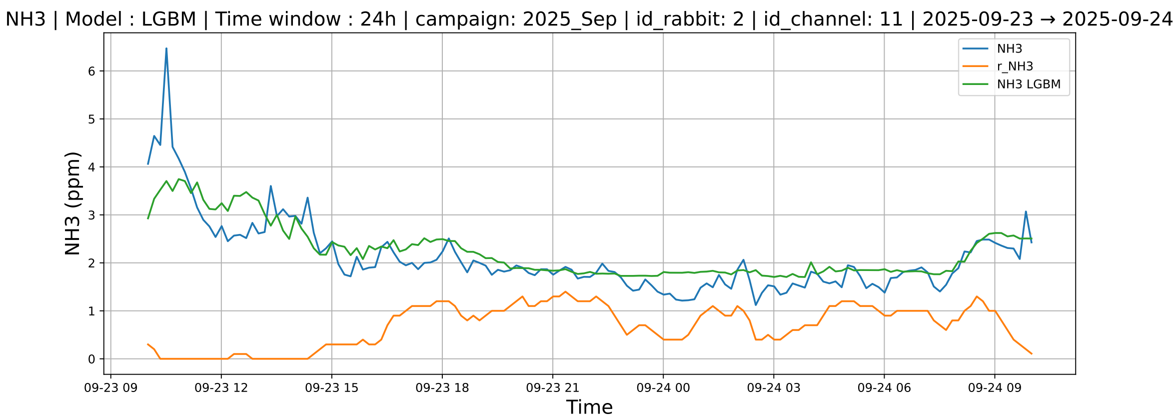


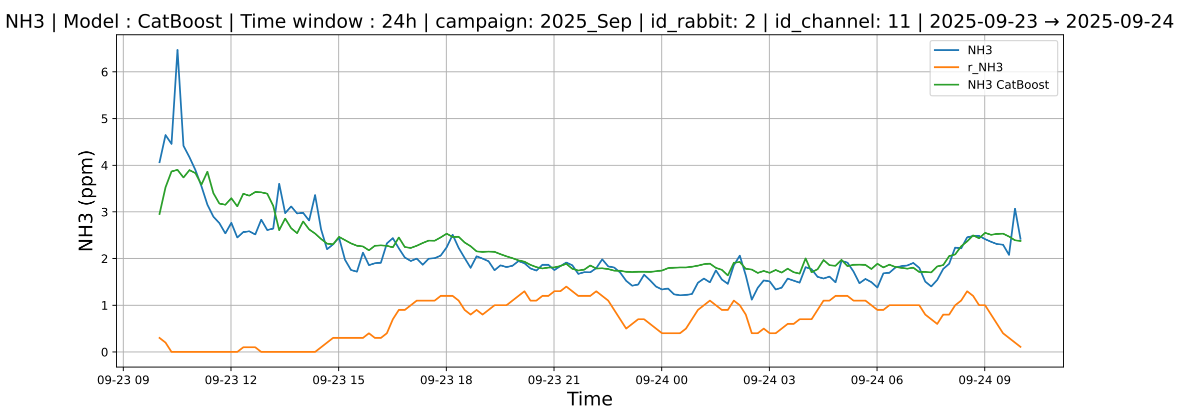


4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

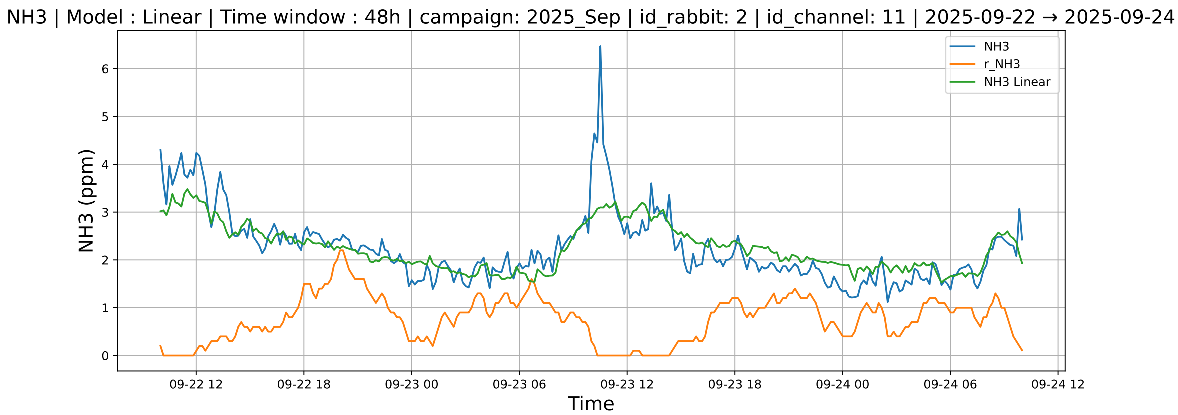
4.1.5.1 Time window : 24

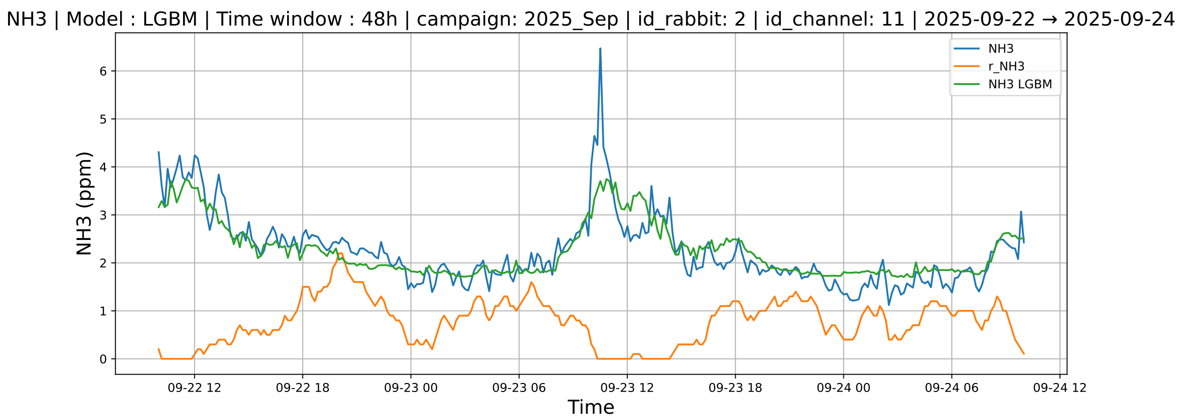


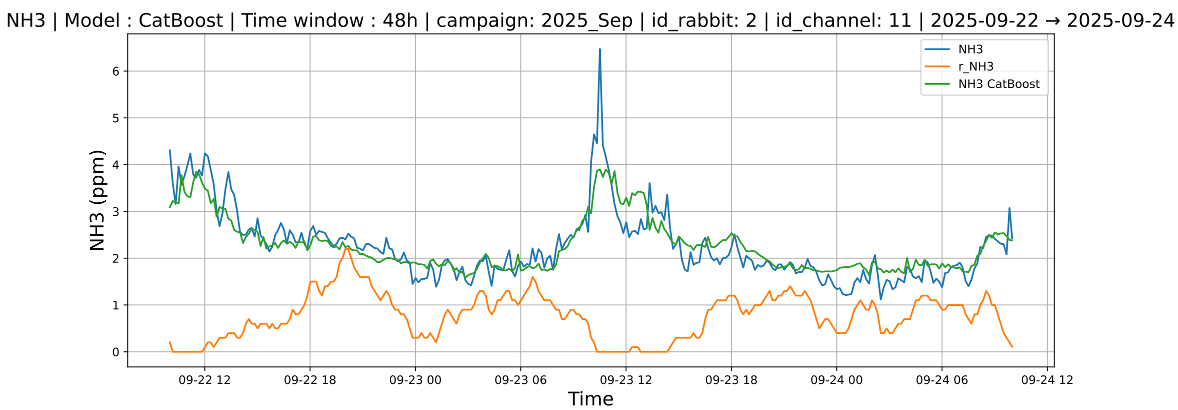




4.1.5.2 Time window : 48

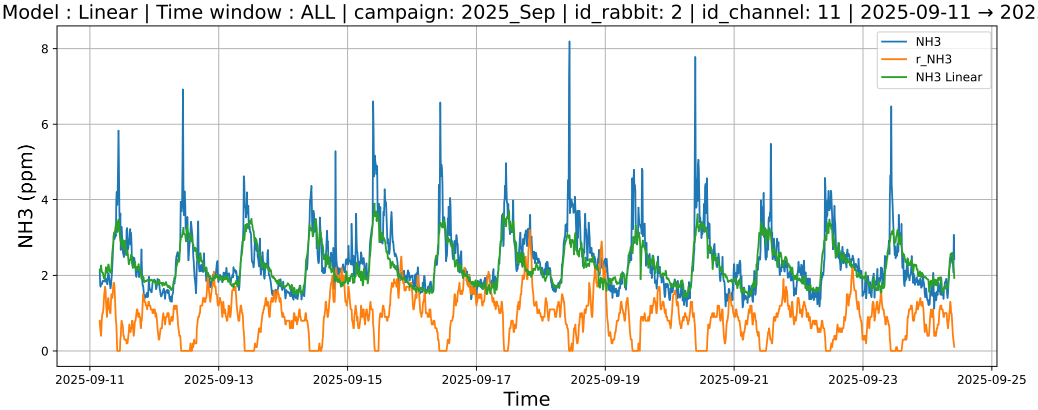




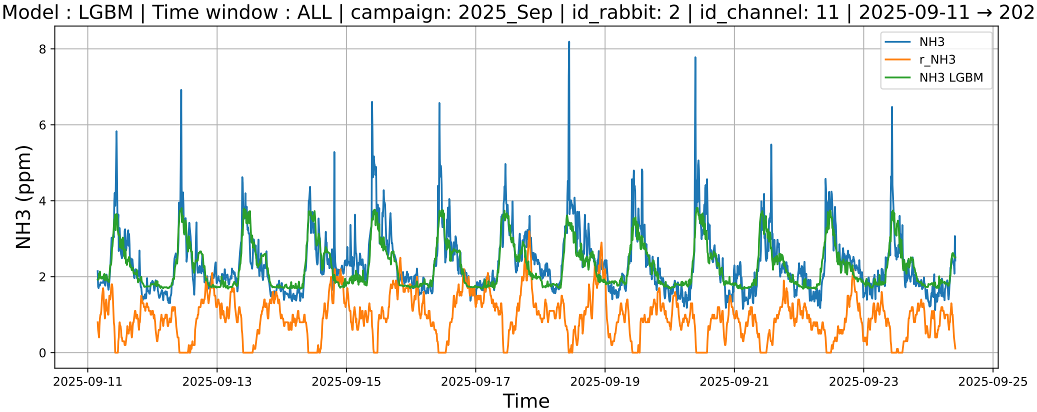


4.1.5.3 Time window : ALL

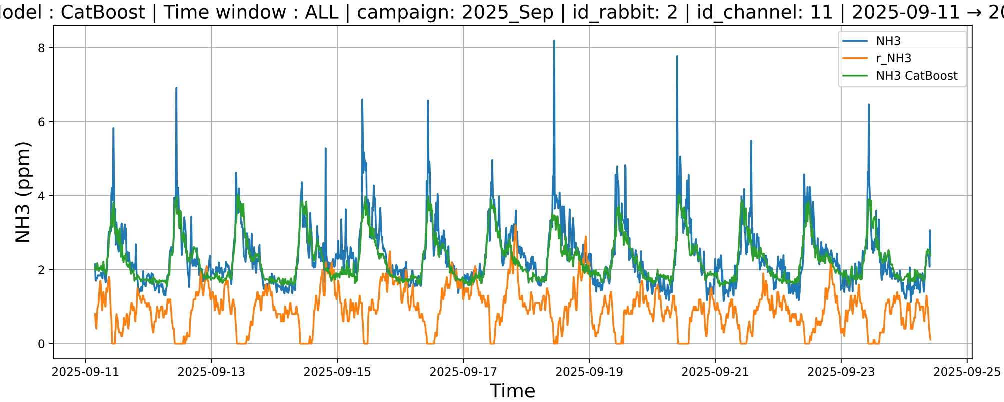
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



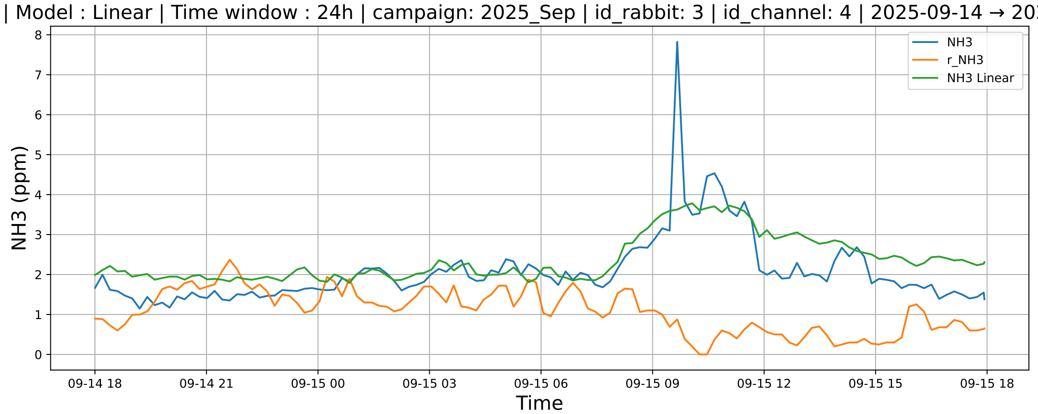
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



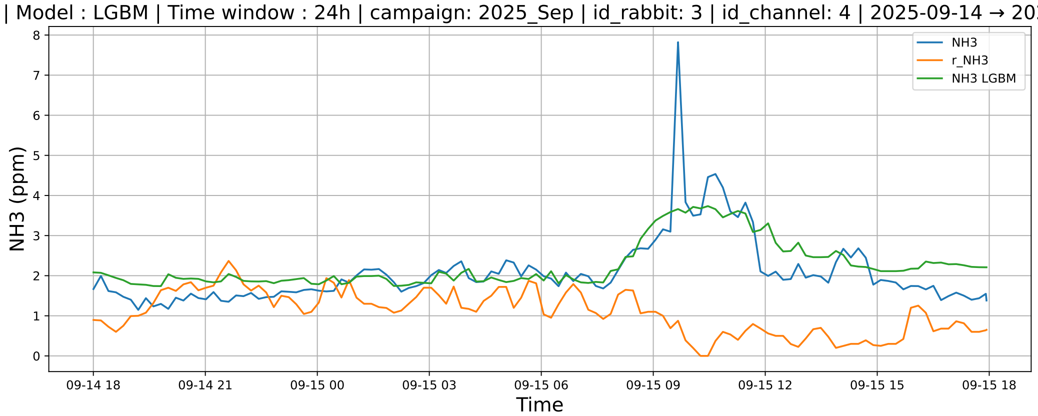
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

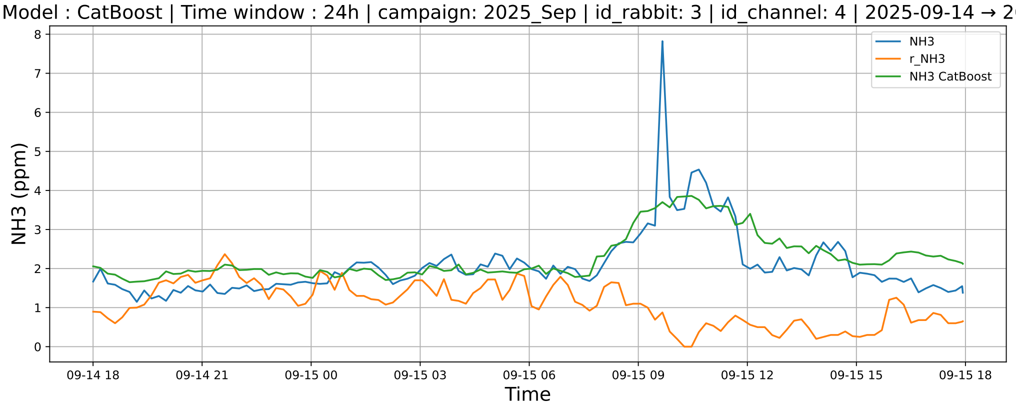
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

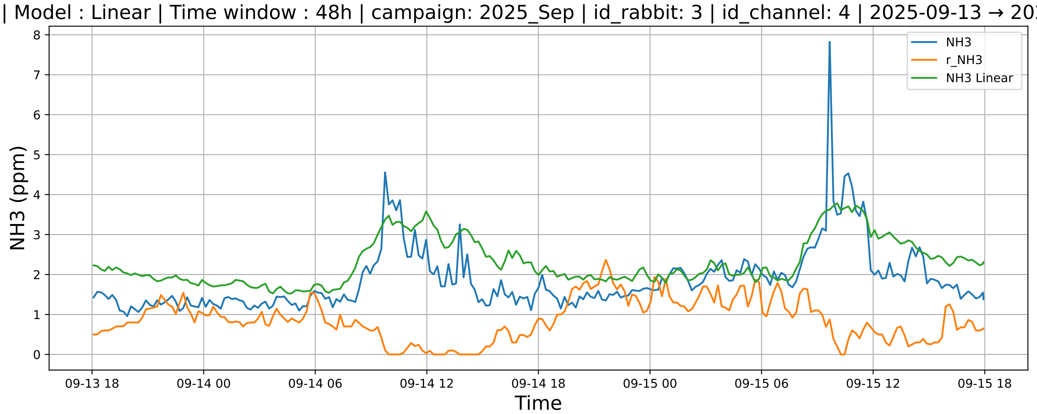


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

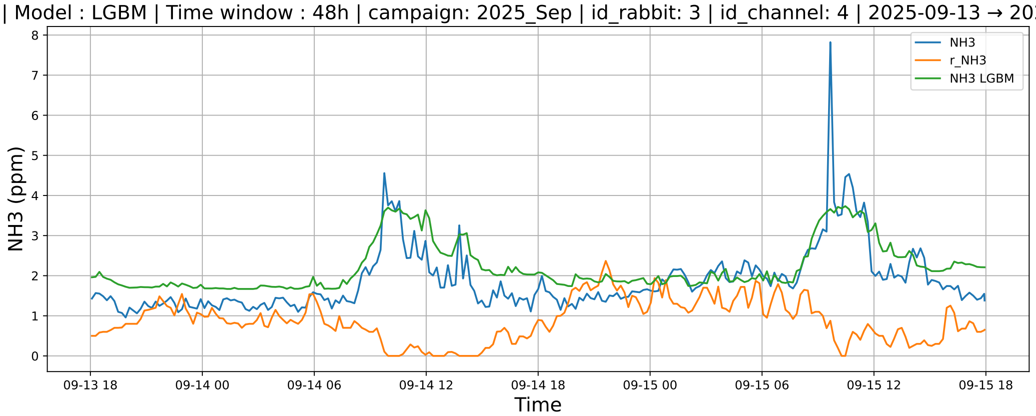


4.1.6.2 Time window : 48

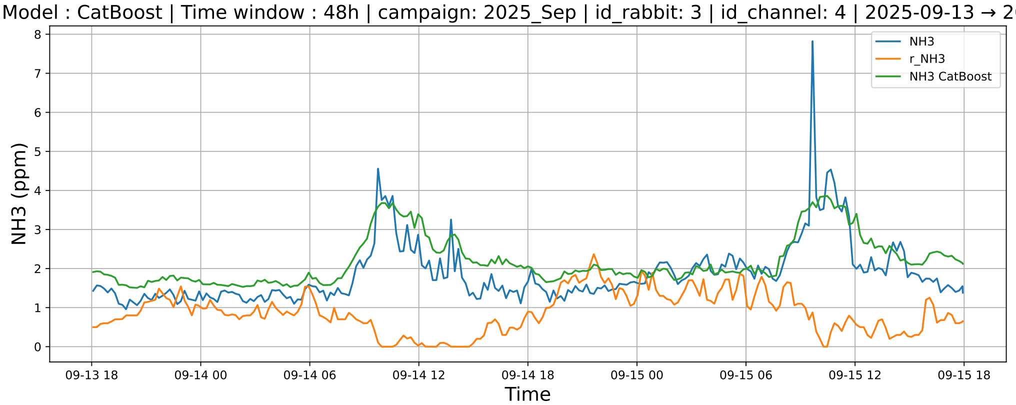
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

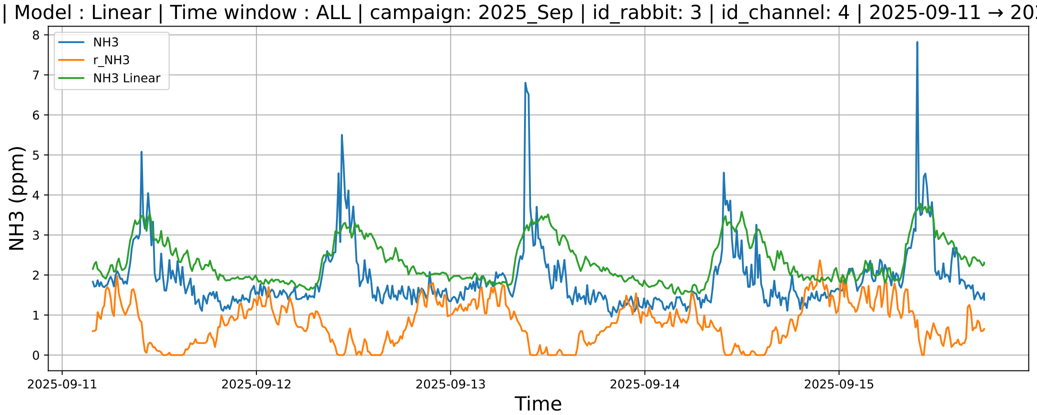


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

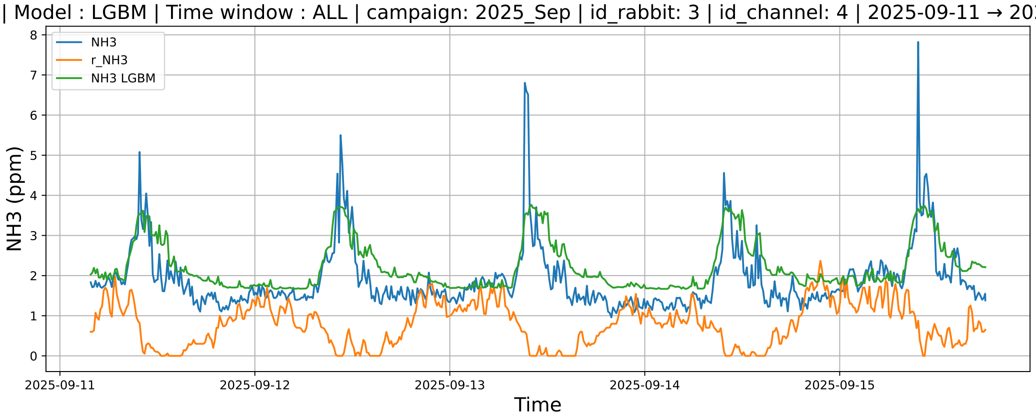


4.1.6.3 Time window : ALL

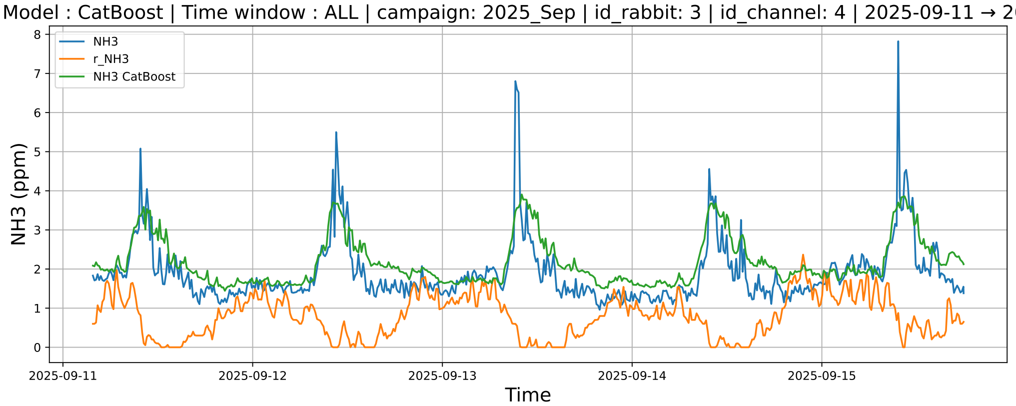
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

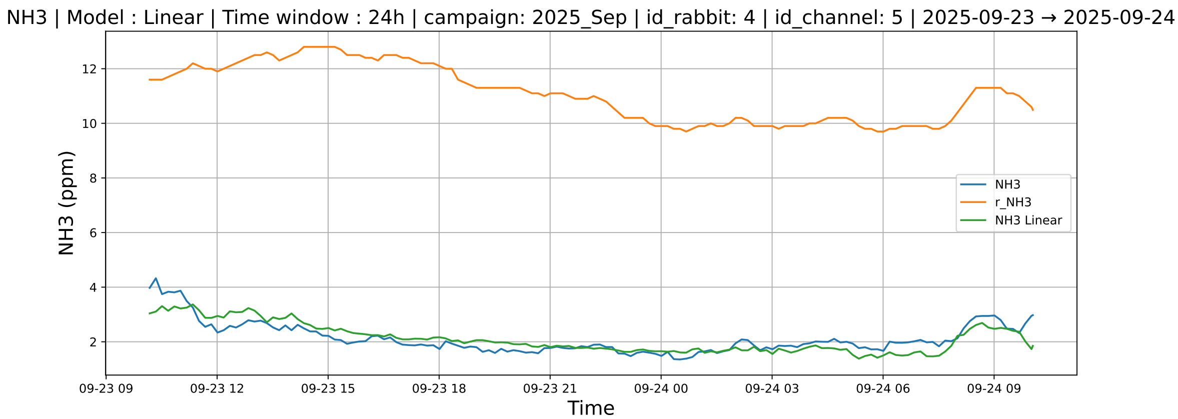


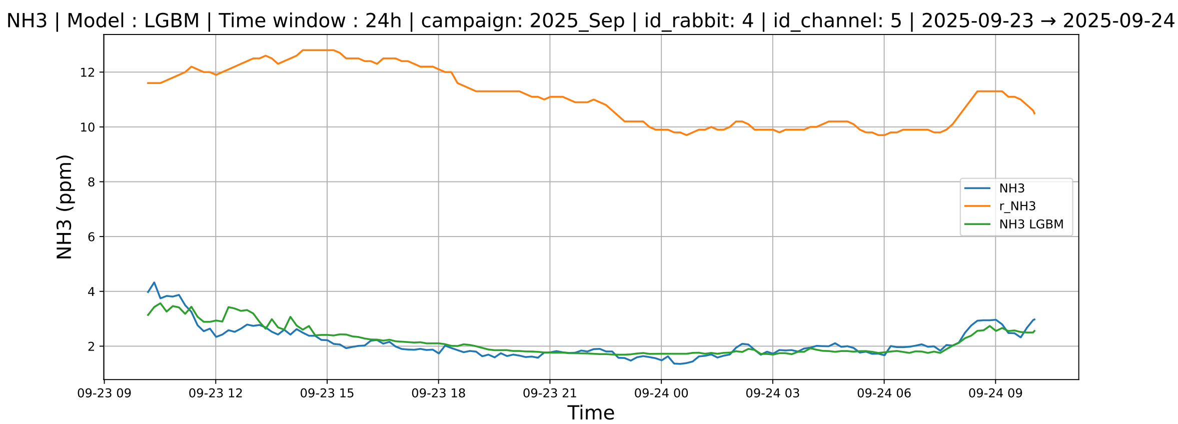
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

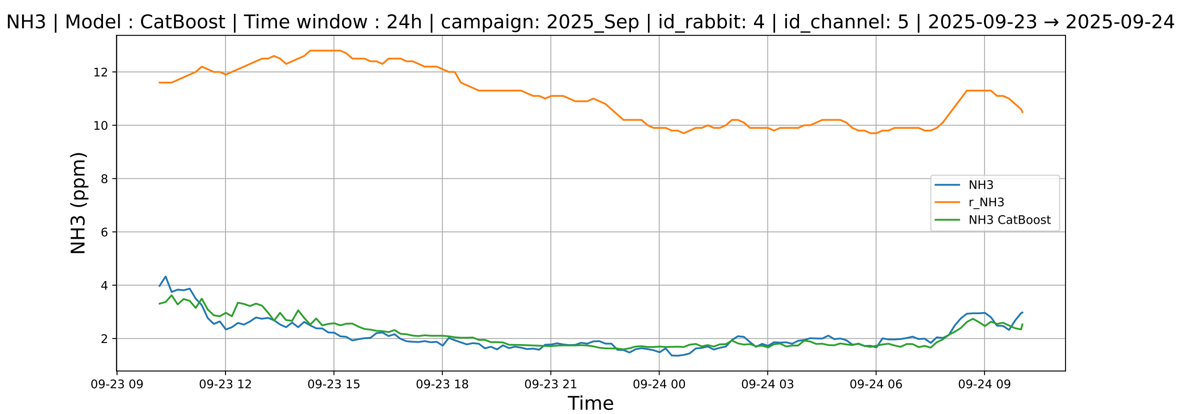


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

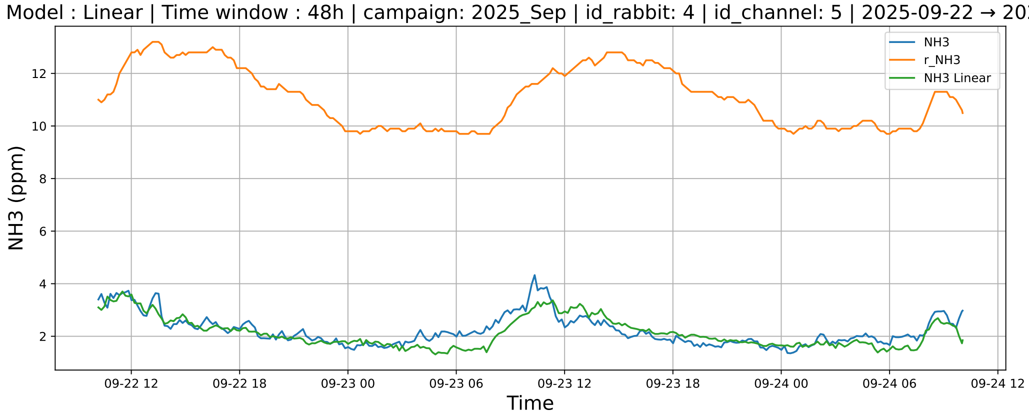




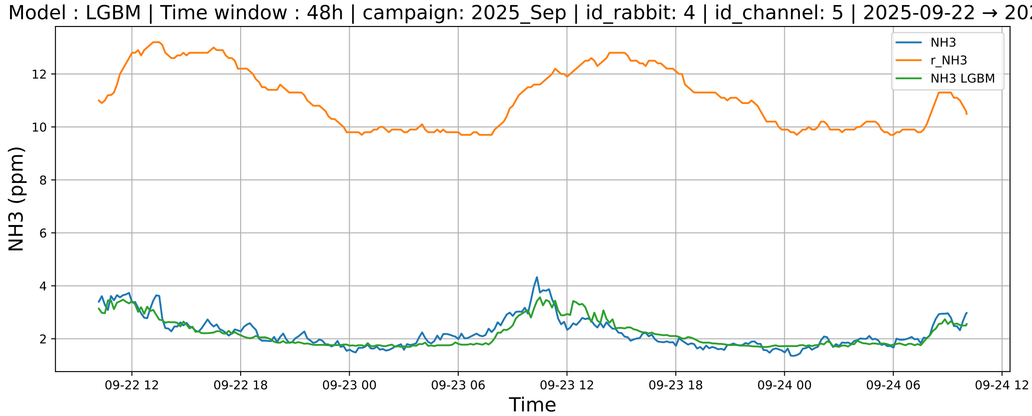


4.1.7.2 Time window : 48

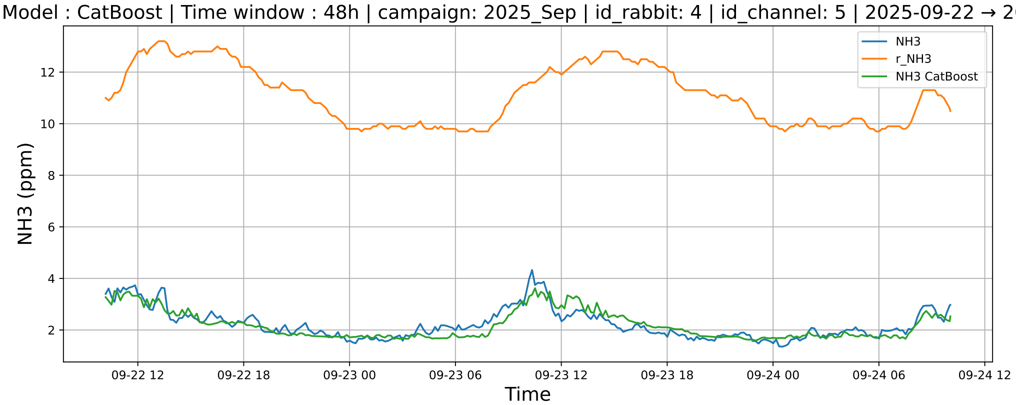
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

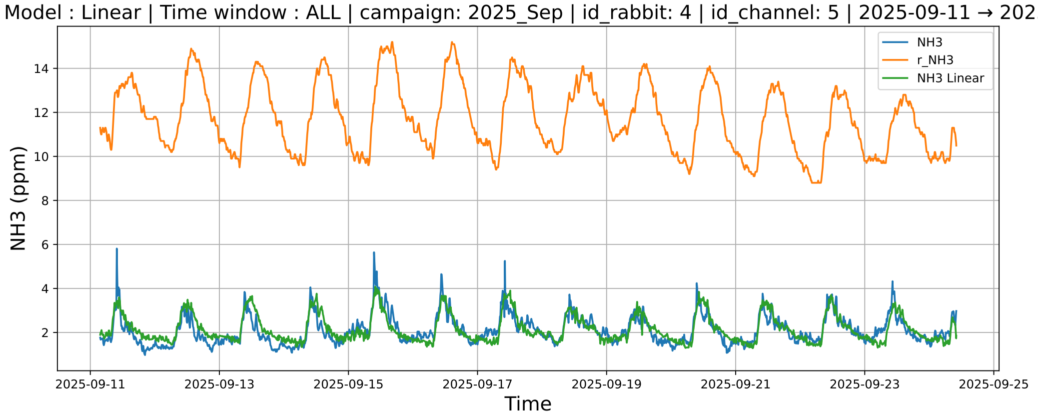


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

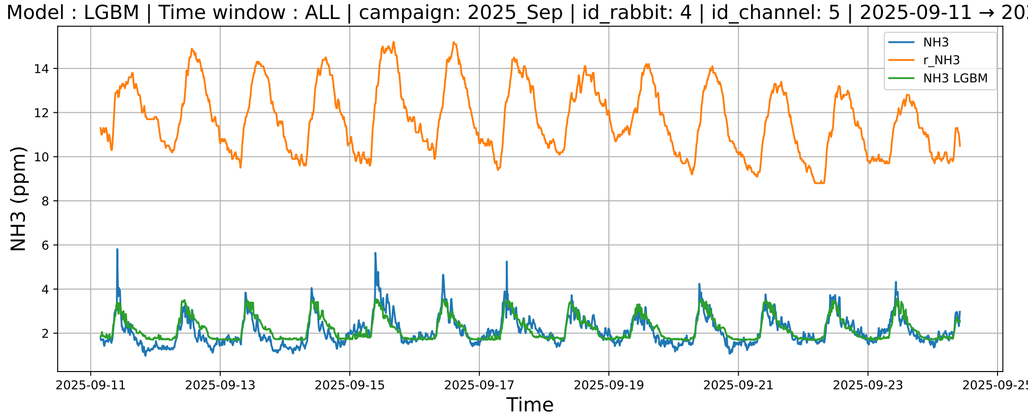


4.1.7.3 Time window : ALL

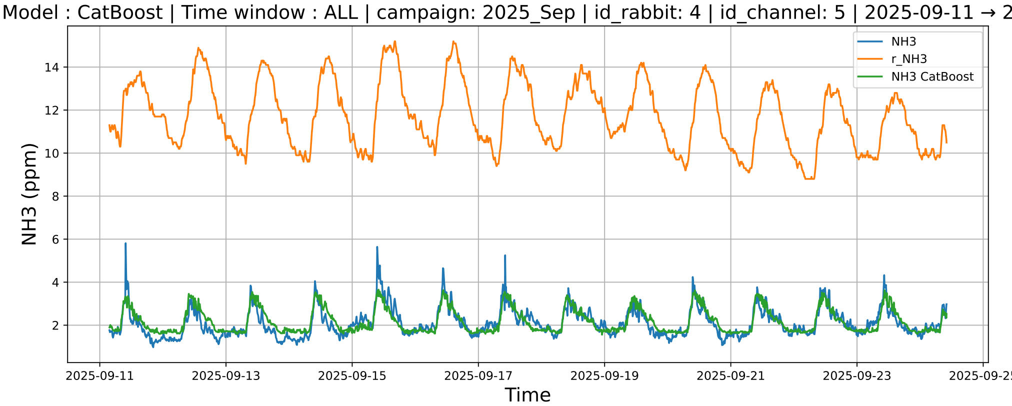
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

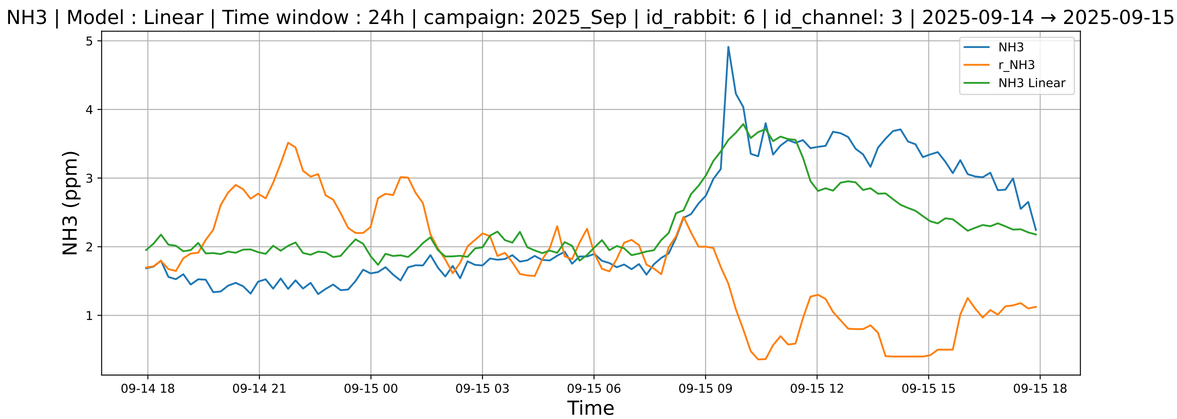


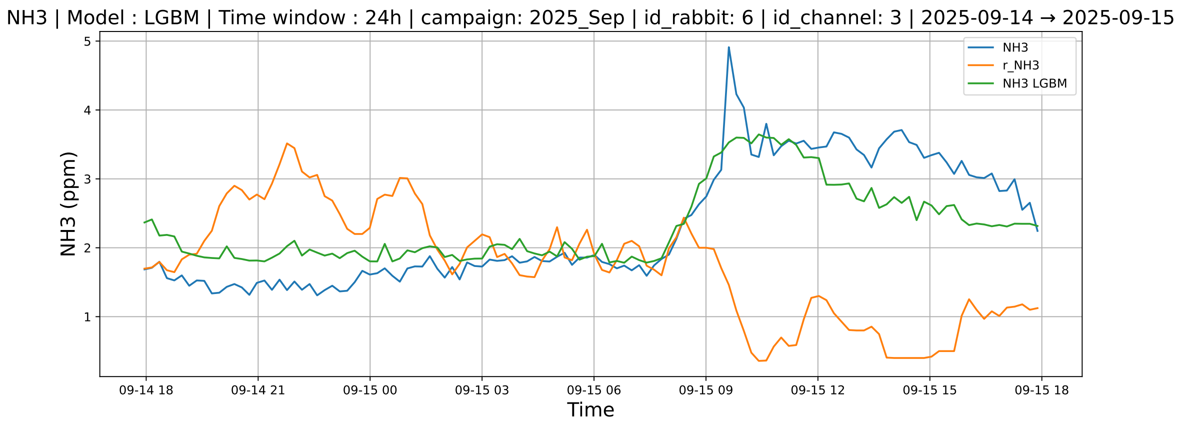
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

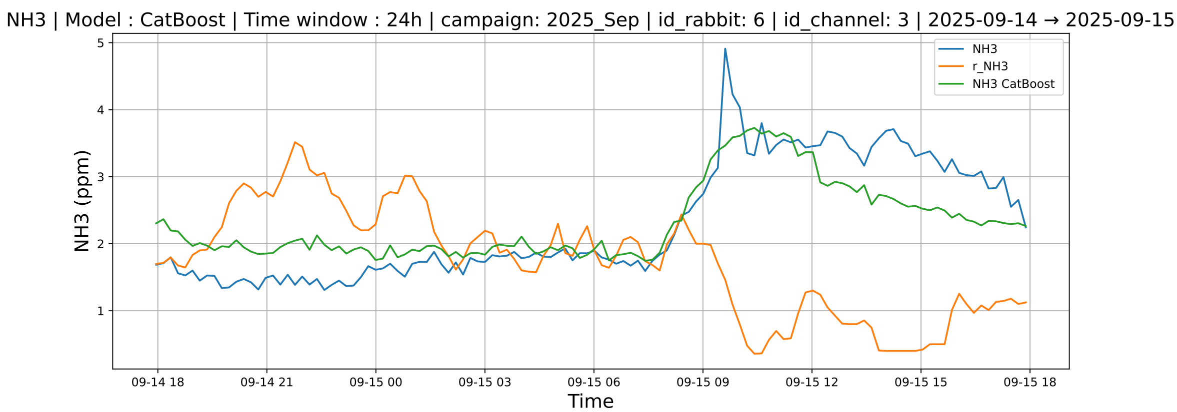


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

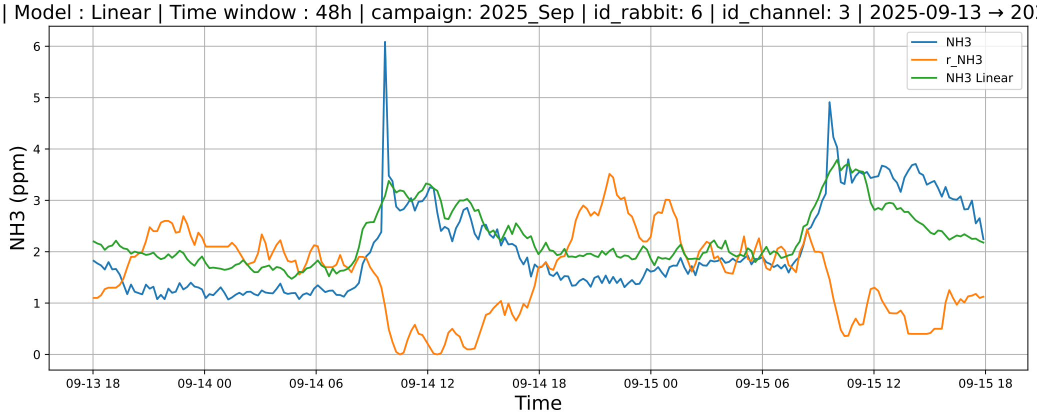




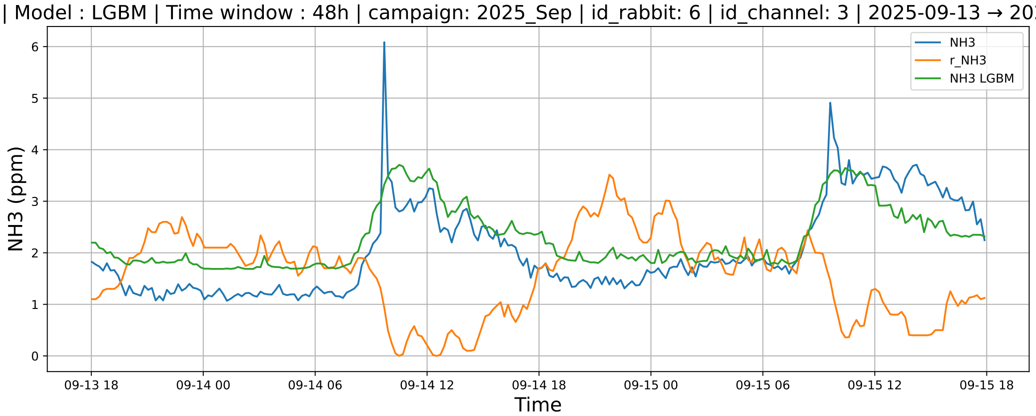


4.1.8.2 Time window : 48

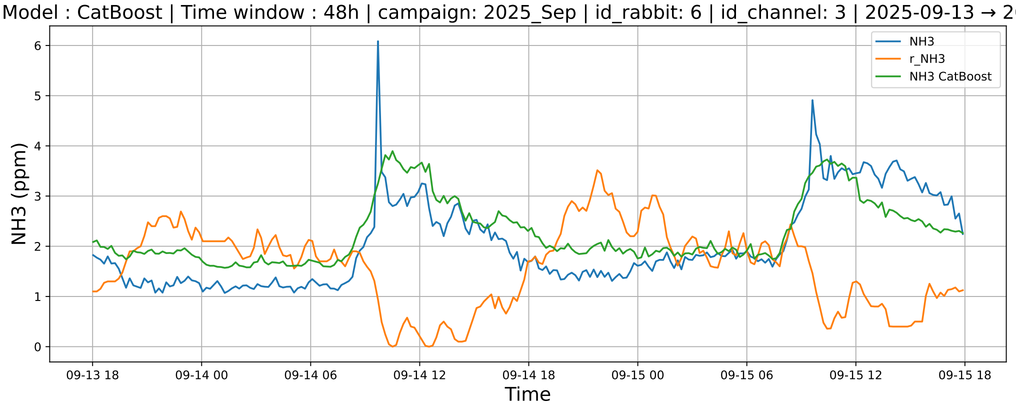
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

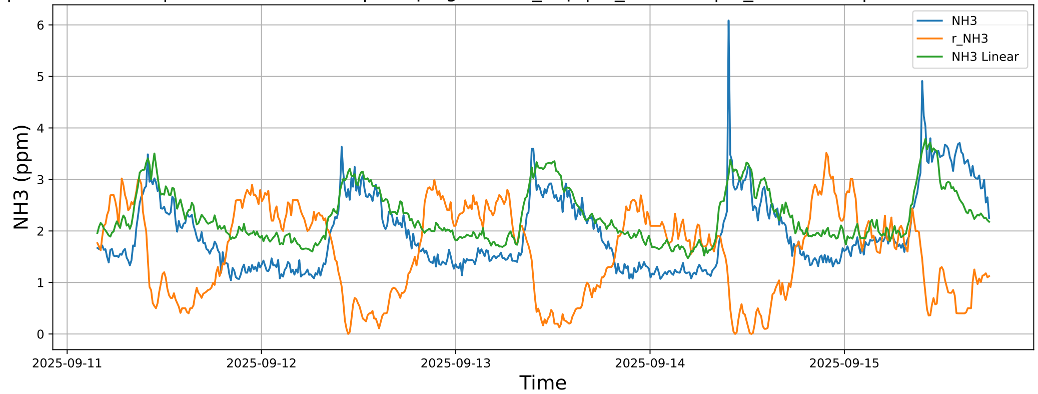


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

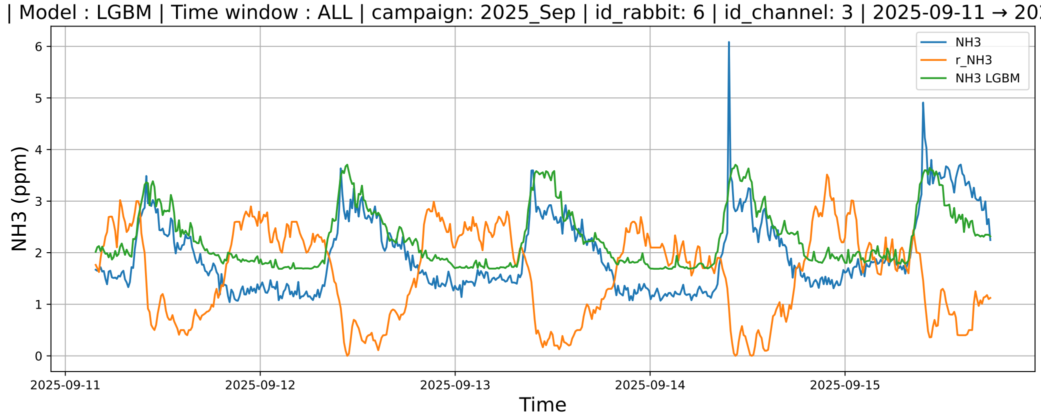


4.1.8.3 Time window : ALL

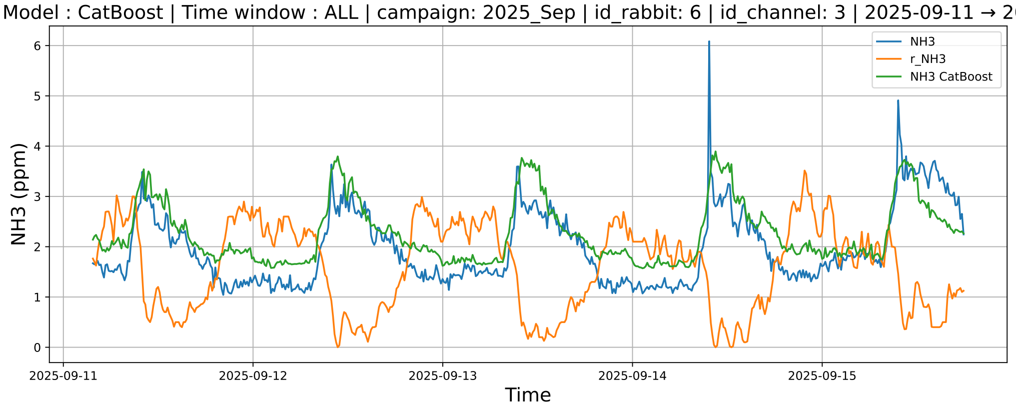
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



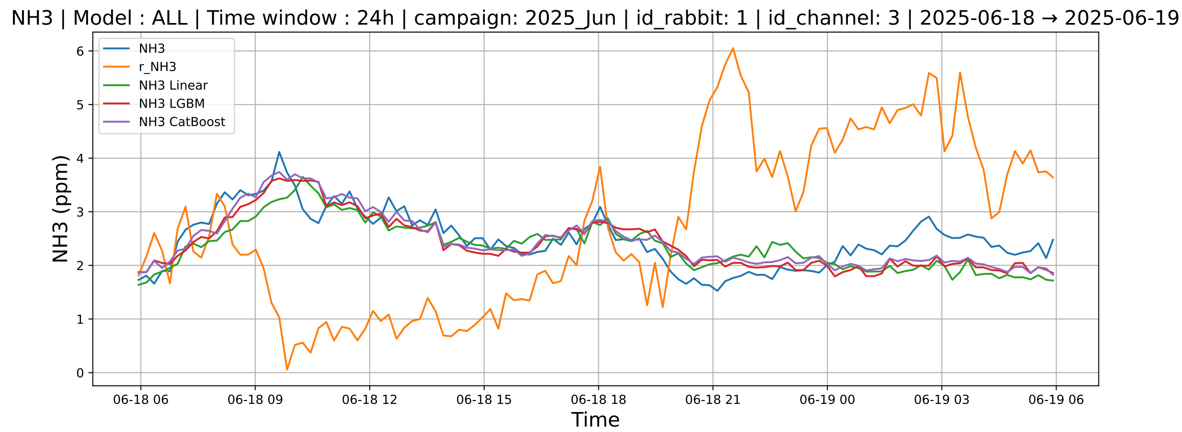
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



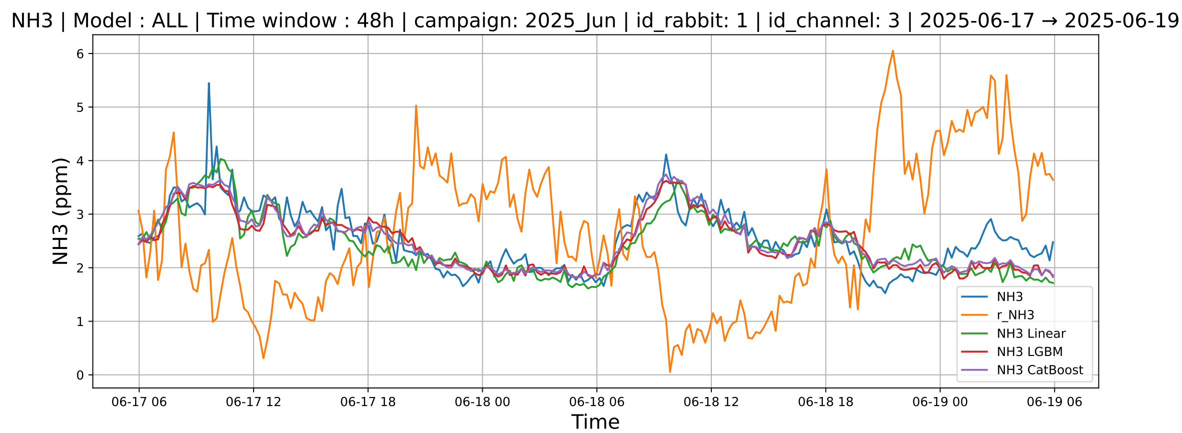
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

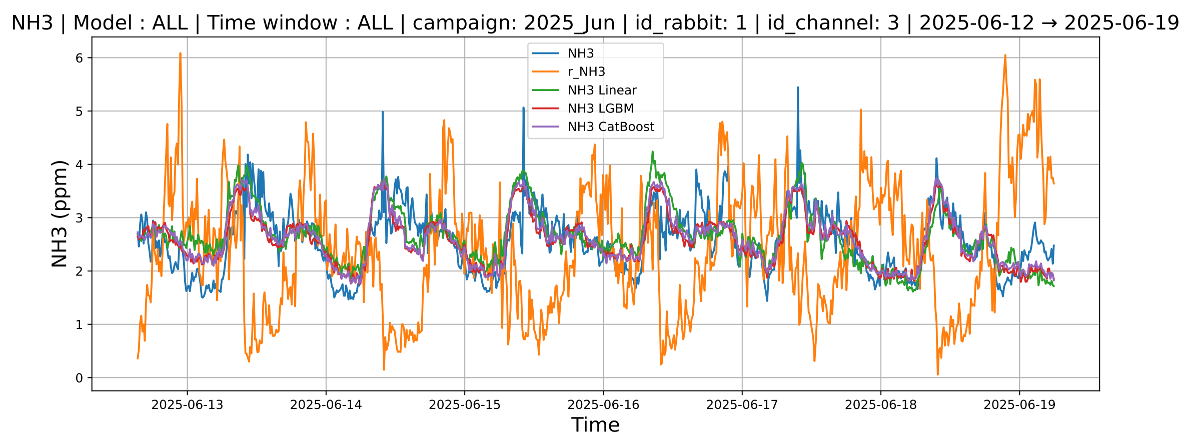
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48



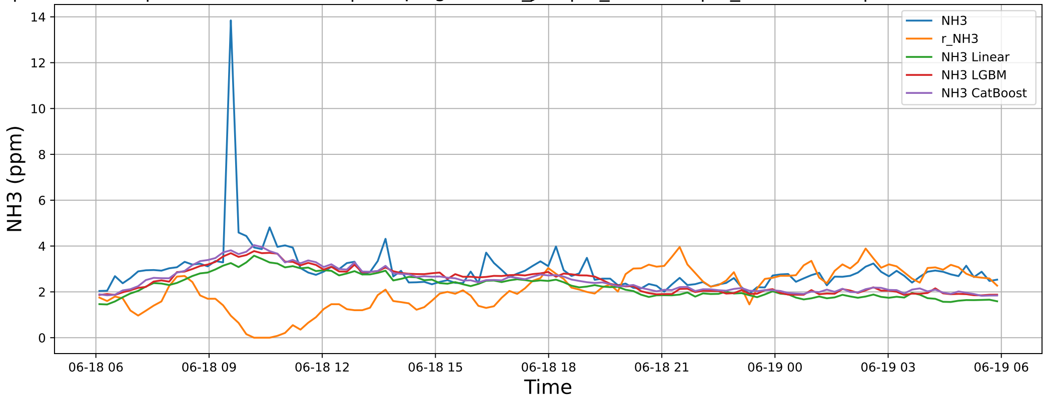
4.2.1.3 Time window : ALL



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

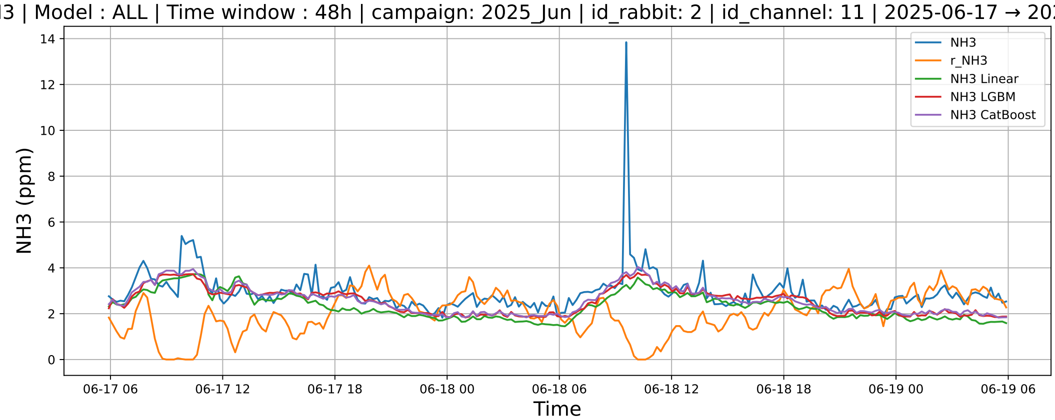
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



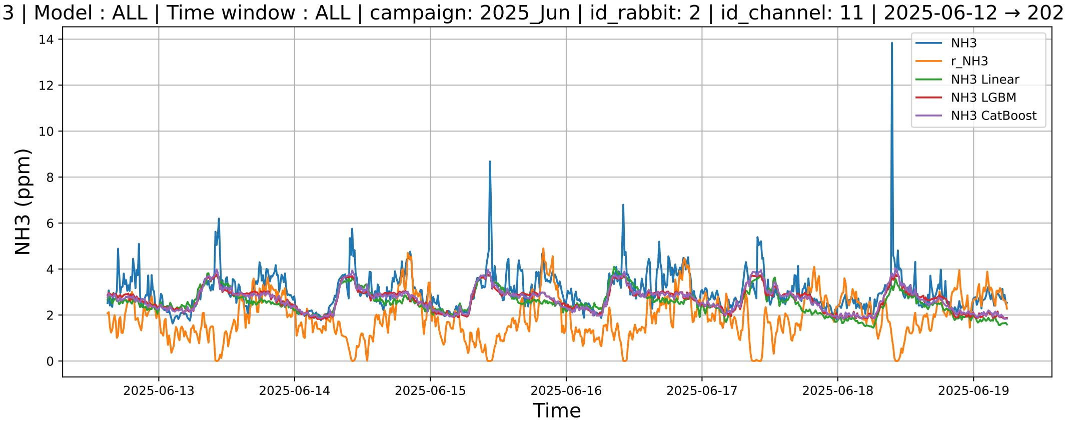
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

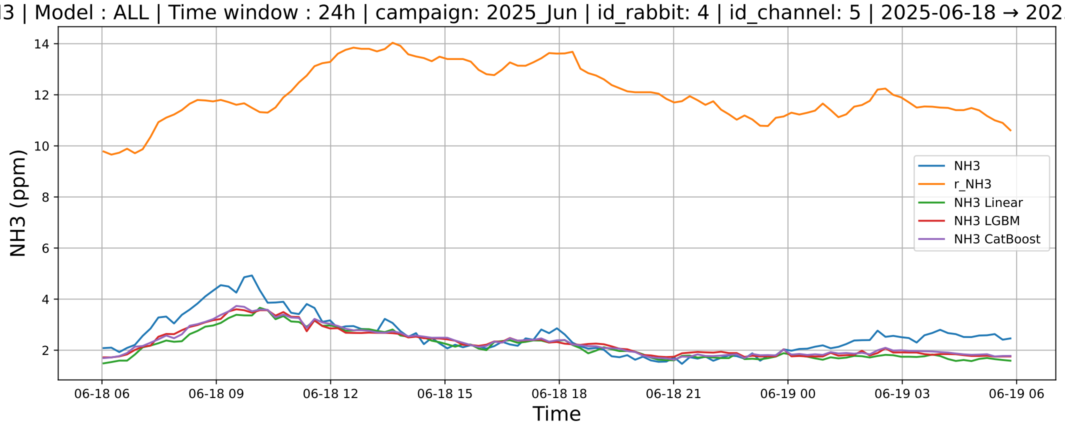
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

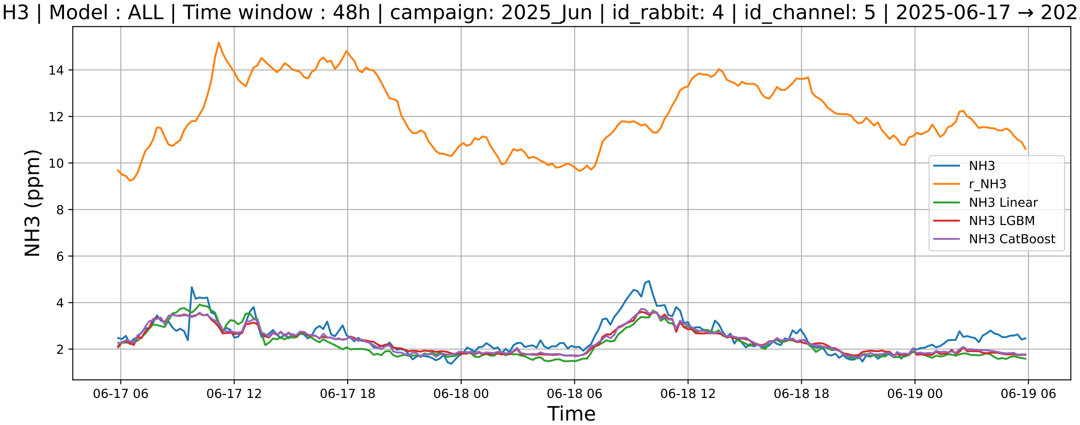
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



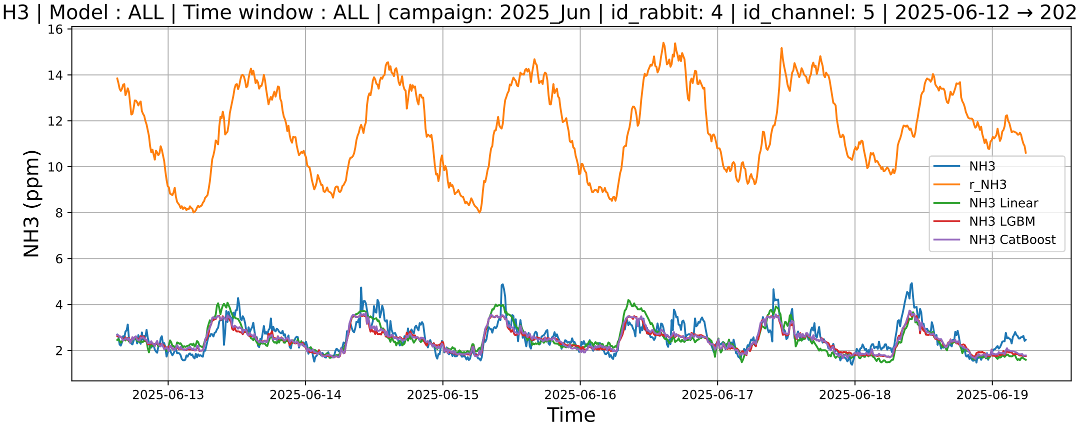
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



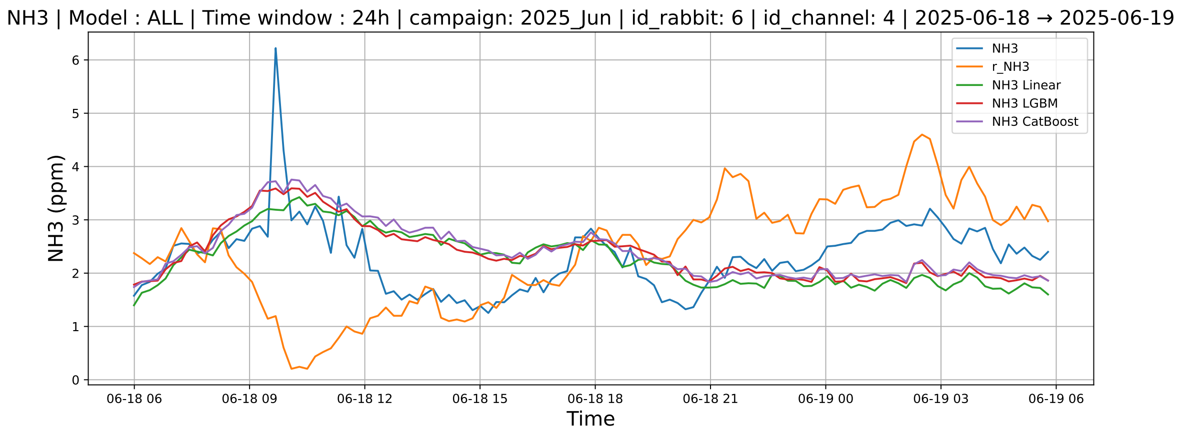
4.2.3.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

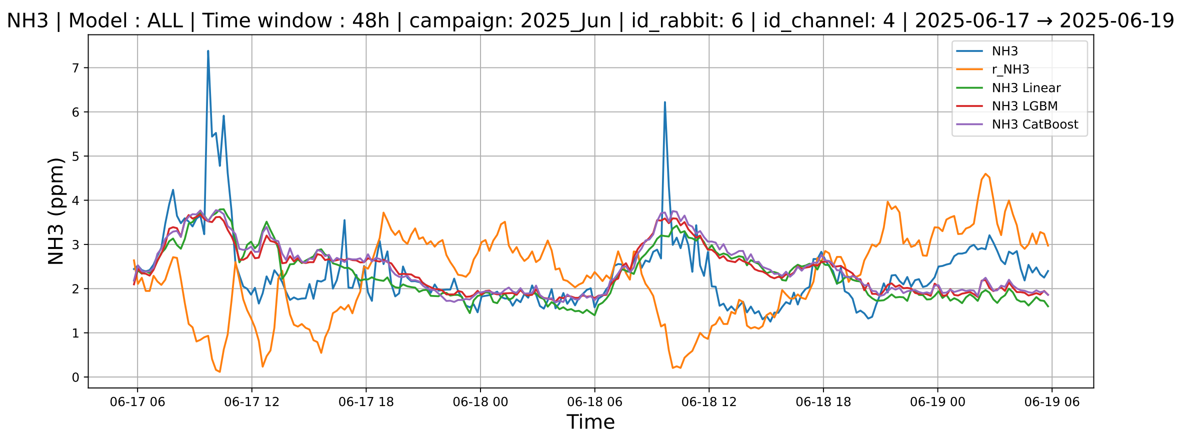


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

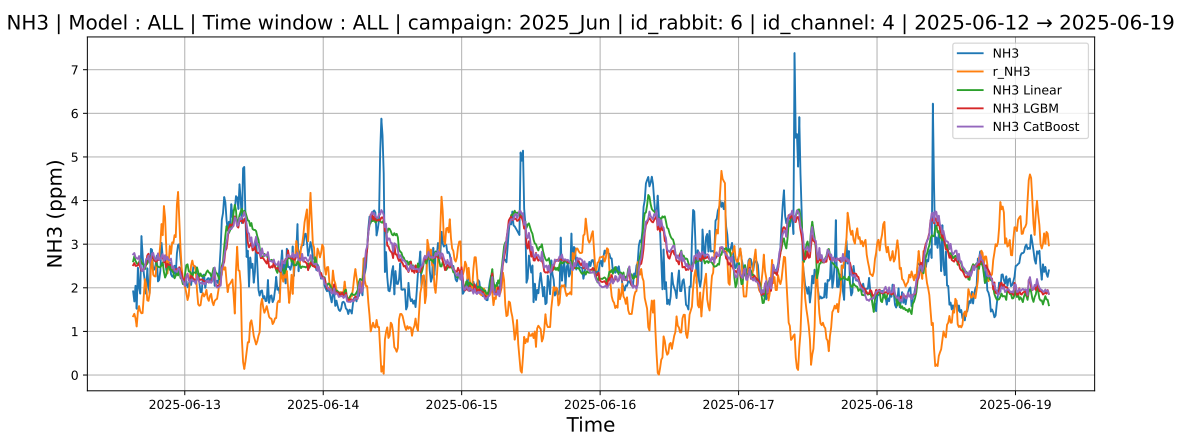
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



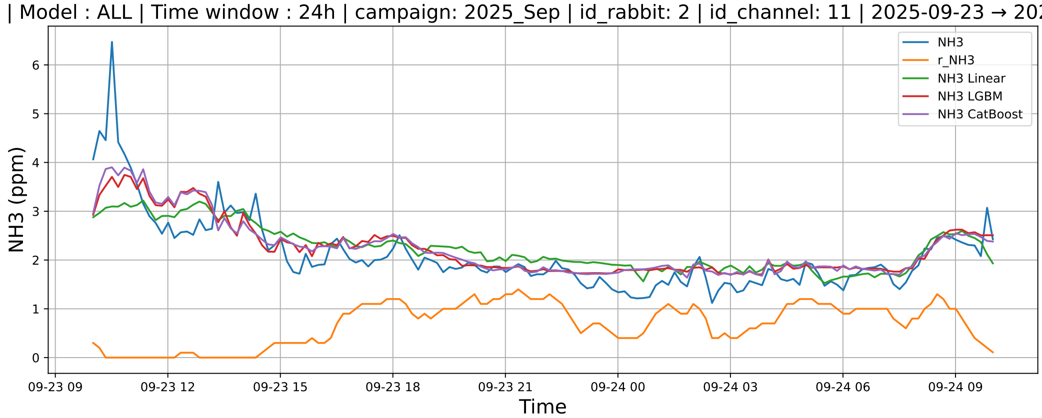
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

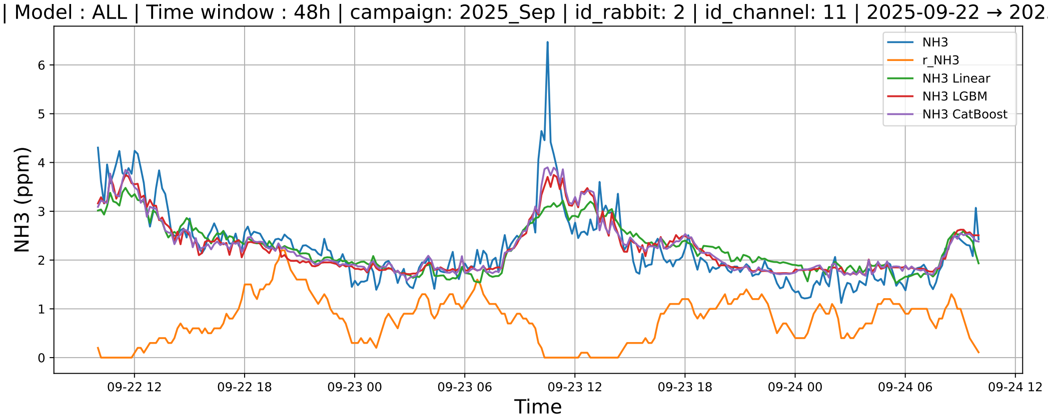
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



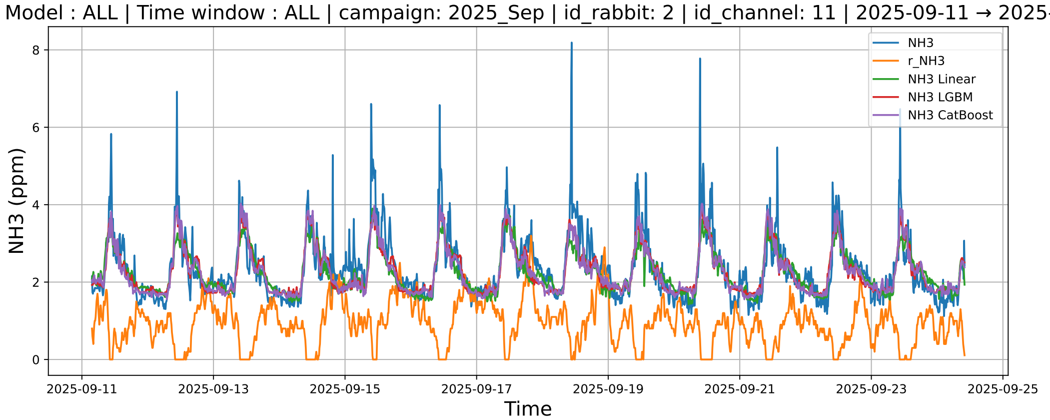
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

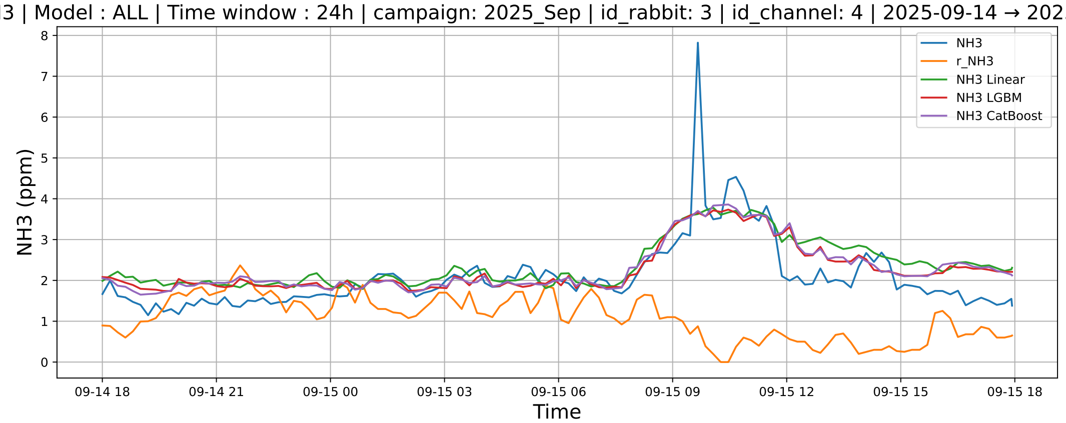
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



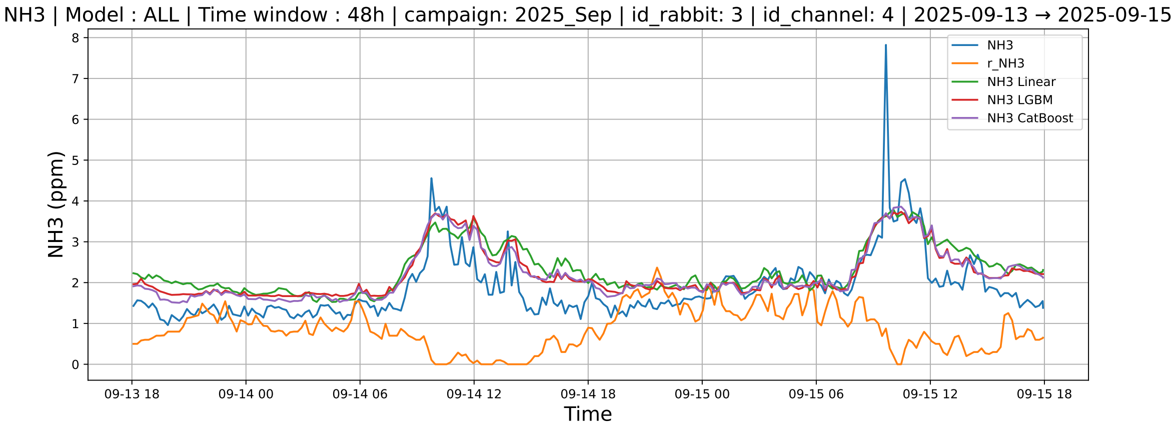
4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.2.6.1 Time window : 24

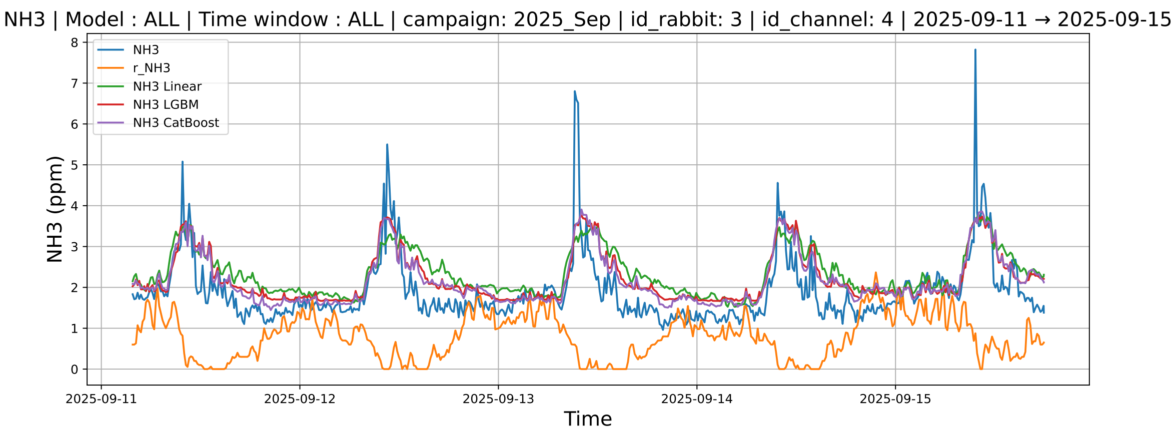
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



4.2.6.2 Time window : 48

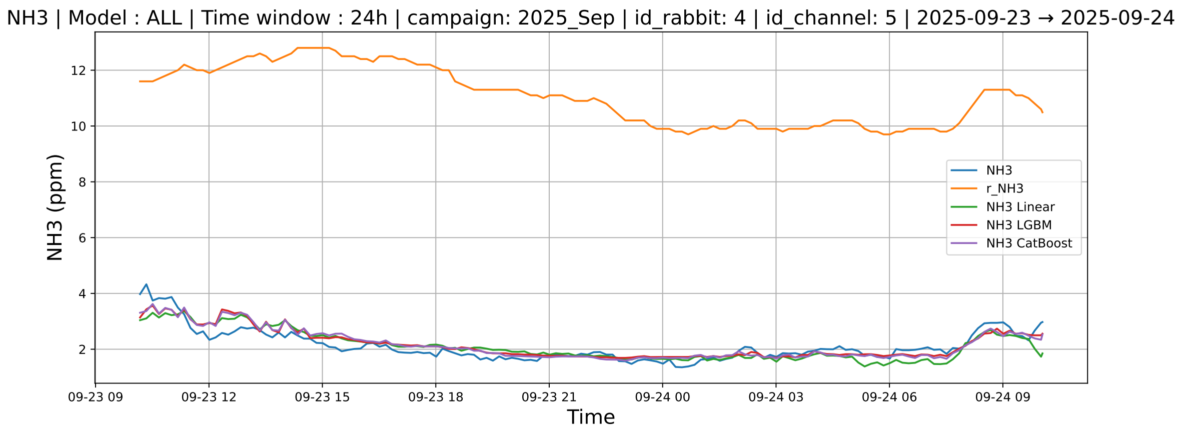


4.2.6.3 Time window : ALL

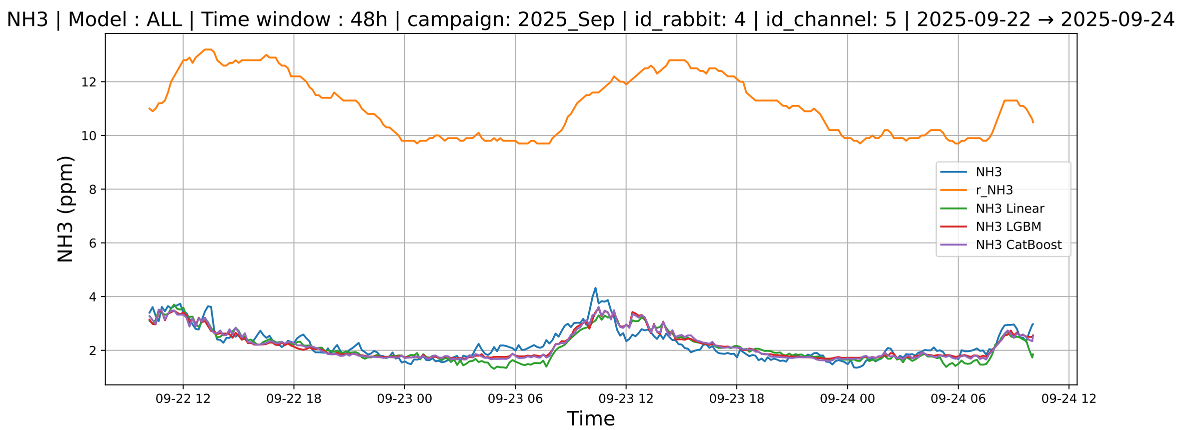


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

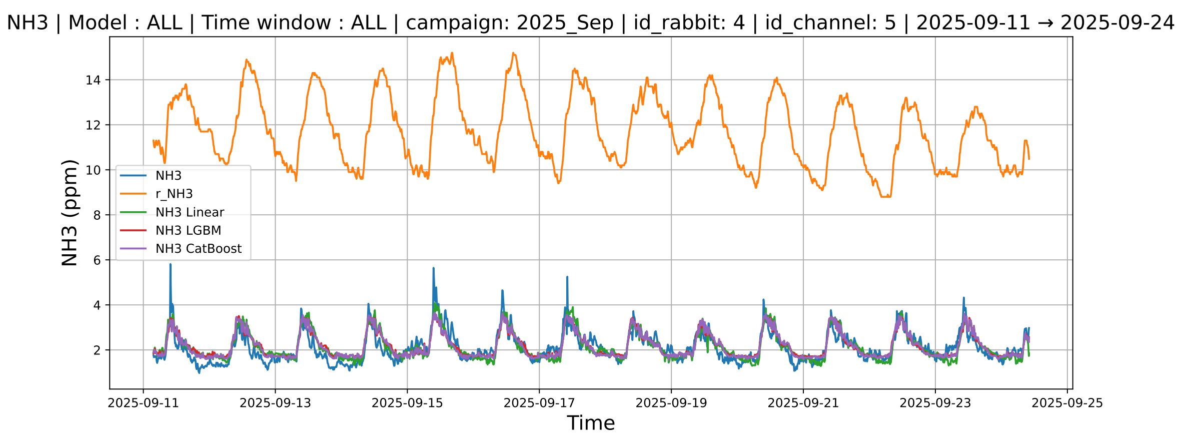
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48

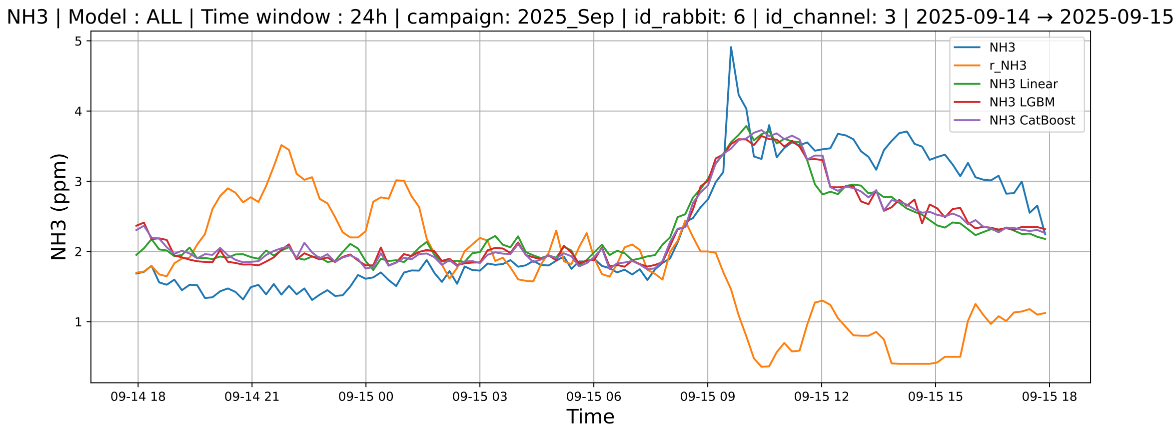


4.2.7.3 Time window : ALL

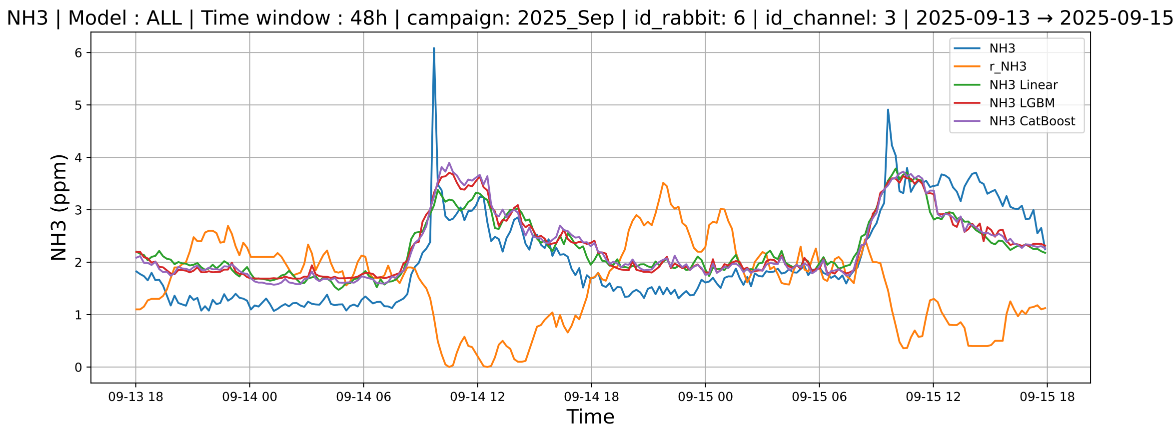


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

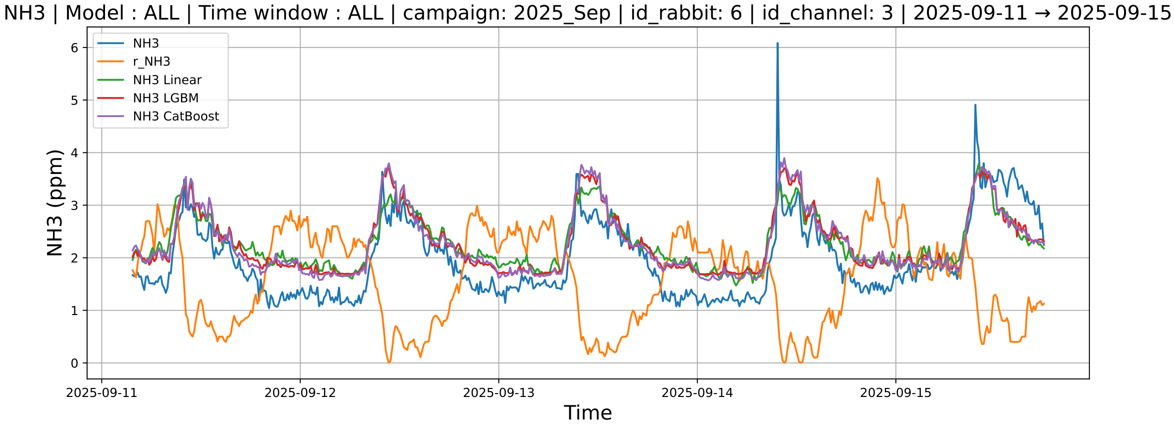
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL



NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
0
1
2
3
4
5
6
2025-09-11
2025-09-12
2025-09-13
2025-09-14
2025-09-15
Time